

PCR-7.5

Structural Stability and Degeneracy-Lifting of the Neutrino Heavy Sector

Symmetry-protected spectral rigidity, carrier-stability margins, and robustness
of the PCR-7 prediction lock

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Scope of the result

The theorem applies to the two source-qualified vectorlike-joint sectors completed by PCR-5E, characterized by PCR-6, and projected to a finite experimental prediction by PCR-7. It classifies finitely presented same-scope perturbations and, on each declared P75-13 trace-transport cell, constructs an explicit positive carrier radius; an uncertified chart remains conditional/NotReady. It proves the nested hierarchy of exact closure, positive-radius rank and aggregate support, and zero-radius symmetry refinements. Positive flavor universality is restricted to the full-rank equal-singular-value stratum, with $\Omega'_H = \delta' I_3$ and $\delta' > 0$. It separates a fibrewise invariant core, maximal relative to a pre-frozen variation registry, from the functional-determination class of the complete covariant structural trace; cross-child agreement exists only for matched ready-child realizations under matched common perturbations and is then the common-source reduct. Finite detector claims require a globally conditioned stratified mass-support/response certificate in one common record space, separation for every core pair outside one declared collision coarsening including the reference core, a frozen sampling-law certificate, and separate record-separator, parameter-inverse, transported-mass, and Gram errors below their own typed thresholds. Under the carrier/reference half-separator guard, the same record retains the true-core inverse and empties the separated wrong-core inverse; on block isometry with $\eta_V \leq \bar{\eta}_V < s_E$, the ideal pair-space gap remains a distinct structural theorem rather than a detector-confidence guard. This source-mass axis is not the physical complex-pole axis, and no universal protocol perturbation radius is claimed. The VJ mass domain is the symbolic source image $H(I_E)$, whose numerical endpoints remain NotReady. Independently, PCR-5D3 and P75-X2A certify an exact source continuum on [28, 30] GeV whose declared response-chart coordinate $c\tau_A^{\text{resp}} = 0.2$ mm is exactly an interior member of the discrete ATLAS lifetime-reweight grid, paired with a narrow on-shell- W energy/boost tag. Its 0.1 mm distance from the lower endpoint of the declared grid span is a placement datum only and supplies no continuous detector-response radius. A frozen HEPData replay places the PCR-5D3 mixing ordinate more than a factor 4.50 below the linearly reconstructed observed Figure 4a threshold across the band under the ATLAS simplified-model convention. This is a source-certified, ATLAS-grid-anchored search benchmark and conditional ordinate screen, not a full-source proper lifetime, full detector-response validation, or source-level ATLAS non-exclusion; only 30 GeV is a direct ATLAS mass anchor inside the band. The supporting records motivate and reproduce the pair but do not derive its endpoints or select the response coordinate uniquely. The ordinate comparison is not a non-exclusion result. The pair has no VJ-specific mass/lifetime authority without the required bridges and carries no uniqueness, optimality, or comparative-priority claim. The marked VJ child obeys the target-qualified bound $0 < \epsilon_{\text{row}} \leq 1/256$. Conditional on applicability of the published light-neutrino response—in particular, closed heavy final states or a channel-aware recast reducing to the same common rescaling—a frozen one-pass HFLAV reconstruction is compatible with flavor universality ($p = 0.1349$) but has exactly zero sensitivity to a common deficit; full VJ physical compatibility remains NotReady. A separate absolute-normalization screen sets the unreported direct/indirect cross-covariance to zero as a screening assumption, without certifying statistical independence, and gives $\epsilon_{95}^{\text{scr}} = 1.62 \times 10^{-3}$; it lacks the mass-, channel-, scheme-, template-, and covariance-complete response needed for promotion. The predeclared stop rule therefore retains 1/256 and records conditional compatibility only. For a fixed child, a future independently covered weak-rate region may still sharpen that source bound by intersection, without selecting the child from a broader class. A numerical VJ lifetime, source-mass inverse, or physical pole-mass projection is conditional on a certified complex-pole continuation atlas, an independent normalization constraint and response record, a pre-record exhaustive route partition, uniform coverage, and the normalization bridge. The separate D3-to-VJ domain bridge can yield a bridge-conditional calibration-derived source-coordinate compatibility/confidence set, but supplies no independent mass-prediction authority, is not part of the core source inverse, and never by itself localizes a physical pole mass. The result neither revises the accepted unperturbed theorems nor extends them to unresolved parent sectors or other fermion kinds.

Abstract

PCR-6 identifies, on two source-qualified vectorlike-joint neutrino sectors, one terminal carrier with three nondegenerate light standing sectors and one rank-three degenerate heavy standing sector. PCR-7 projects that structure to the finite charged-current spectral-tomography packet

$$\Delta_{\text{act}} = \Omega_H = \delta_E I_3, \quad \text{rank } \Omega_H = 3, \quad \Delta_{\text{act}} - \Omega_H = 0,$$

together with one rank-three degenerate structural heavy-mass-support class and a finite source-authorized protocol with carrier and no-vectorlike reference cores in one common record space; under the certified half-separator guard it supports hypothesiswise same-record selection. The present paper determines which parts of that conclusion are structurally rigid, which are protected only by exact heavy-sector symmetry, which persist on a positive perturbation ball, and which fail under arbitrarily small standing symmetry breaking.

The import lock retains PCR-5E, PCR-6, and PCR-7 together with the declared PCR-3/PCR-4, fixed-domain-exhaustion, and SCIC authorities and their frozen transitive dependency closure. Those antecedent results support the source, physicality, update, and carrier-identity interfaces without reopening their targets or importing any unresolved parent conclusion.

A target-relative perturbation theory is formed before any robustness conclusion is inspected. Same-scope perturbations are generated by decorating the frozen source hypergraph with standing-bearing coefficient, matching, continuation, boundary, population, and relation changes; new primitive support is classified as source update, while altered role, witness, or comparison structure is target update. This gives a complete joint perturbation trace. An explicit information-preserving regrouping/factorization map retains the cross-interface incidence and correlation data together with the six displayed interface blocks, rather than replacing the joint trace by independent marginals. The associated fail-closed status calculus separates skin, same-scope tensor deformation, standing symmetry breaking, source update, target update, protocol update, and unresolved status.

On the heavy mass-squared space every Hermitian perturbation decomposes uniquely as

$$\delta K_H = a I_3 + T, \quad \text{tr } T = 0.$$

The commutant of the complete heavy $U(3)_H$ action consists only of scalar operators. Hence exact threefold degeneracy is preserved by all symmetry-preserving perturbations and is lifted by every nonzero standing traceless component. If $\rho_K = \|T\|_2$, the heavy mass-squared spread obeys

$$\frac{3}{2}\rho_K \leq \text{diam Spec}(T) \leq 2\rho_K,$$

with explicit square-root bounds for the structural mass-coordinate spread on the positive heavy domain. A Davis–Kahan estimate proves persistence of the aggregate rank-three heavy subspace whenever the full mass-operator perturbation is smaller than half the light-heavy spectral gap.

For the charged-current block $V'_H = -s_E U + \delta V_H$, singular-value perturbation gives

$$\sigma_{\min}(V'_H) \geq s_E - \|\delta V_H\|_2.$$

Thus rank-three heavy support is stable on the sharp operator-norm ball $\|\delta V_H\|_2 < s_E$. On the full-rank charged-current domain, exact positive flavor universality, $\Omega'_H = \delta' I_3$ with $\delta' > 0$, holds precisely on the equal-singular-value stratum and has zero unrestricted radius against standing anisotropic perturbations. By contrast, whenever the complete light-plus-heavy charged-current map remains isometric,

$$\Delta'_{\text{act}} = I - N' N'^{\dagger} = V'_H V'^{\dagger}_H = \Omega'_H$$

remains exact, independent of mass degeneracy and flavor universality.

Terminal carrier identity is controlled by a finite status-free support/mode skeleton with a finite heavy-partition decoration. On an explicit trace-transport cell, every proper-support skeleton retains a locally constant PCR-6 deletion flag $q_a \equiv 1$, while the full-support $3, 2 + 1$, and $1 + 1 + 1$ decorations map to the same parent carrier orbit. The distance r_{reg} to the cell boundary, the normal-form/update distance d_{nf} , and the explicit nonfrozen guard faces give $r_{\text{car,cert}} > 0$. No finite candidate class, compact candidate fibre, or continuous terminal residual is assumed. Consequently, structural heavy-mass-support splitting changes only the partition decoration and does not by itself split the terminal carrier.

The strongest structural endpoint has two fibrewise levels. Relative to the pre-frozen variation-certified feature registry, the maximal invariant core is

$$\begin{aligned} O_{\text{core}}^{b,X} &= (|\text{TermCar}| = 1, \quad \Delta'_{\text{act}} = \Omega'_H, \quad \text{rank } \Omega'_H = 3, \\ &\quad \lambda_{\min}(\Omega'_H) \geq (s_E - \eta_V)^2 > 0, \quad \text{rank } P'_H = 3, \\ &\quad \text{Spec}(K'_H|_{\lim P'_H}) \subset J_{H,K}^{b,X}). \end{aligned}$$

The maximal lawfully covariant envelope is not identified with that displayed reduct: it is the functional-determination class

$$O_{\text{cov}}^{b,X} = [\pi_{\text{str}}^{b,X} \Theta_{\text{pert}}^{b,X}]_{\sim \text{fd}}$$

of the complete joint structural trace, whose regrouping certificate preserves every cross-interface relation needed for lawful reconstruction, with exact operator laws retained separately. The heavy/CCST tuple is a named reduct. Exact scalar universality and exact threefold heavy degeneracy are excluded from O_{core} , because each is destroyed by arbitrarily small source-authorized standing $U(3)_H$ -breaking perturbations; factors without such a certified displacement remain base data. The cross-child object is formed only afterward, for matched ready-child realizations under matched common perturbations, as the meet of the common-source forgetful images; the decorated child envelopes remain separate. Independently of detector inversion, on the block-isometric subdomain the no-vectorlike reference has the ideal structural pair-space gap

$$\|(\Delta'_{\text{act}}, \Omega'_H)\|_{\text{pair}} \geq \sqrt{6} (s_E - \eta_V)^2, \quad \eta_V = \|\delta V_H\|_2 < s_E.$$

A finite predeclared exposure can shrink only the stochastic part of each confidence radius. Positive acceptance and pointwise Jacobian rank are insufficient: SPIC maps every carrier and reference core into one common ordered record space, requires global lower-Lipschitz response, separates noncoarsened core images, and uses coarser set-valued treatment at structural collisions. The attainable same-record selector is $r_{\text{rec}} < s_{\text{car,ref}}/2$: on true-carrier coverage the carrier inverse contains the truth and the reference inverse is empty, and conversely on true-reference coverage. This is not a distance between two simultaneously nonempty counterfactual regions. The uniform structural mass-support design guard is $r_m < g_{\text{str}}/4$; $g_{\text{str}} > 2r_m$ gives only unique true-support matching, while reported intervals are resolved iff $\widehat{g}_{\text{str}} > 2r_m$. Otherwise the status is `ResolutionNotUniformlyCertified`. The selector, parameter-inverse, transported-mass, and Gram guards remain typed and separate; in particular $2r_{\text{rec}}/\kappa_{\text{car}} < r_{\text{Z,car}}$ and $2r_{\text{rec}}/\kappa_{\text{ref}} < r_{\text{Z,ref}}$, with deterministic-floor attainability, are record-to-parameter guards rather than record separators.

An experimental corollary separates D3 search-benchmark authority from VJ pole authority. The VJ domain is symbolic $H(I_E)$. Independently, PCR-5D3 supplies an exact source continuum on $[28, 30]$ GeV. On its declared response chart, the constructed coefficient maps exactly to the $c\tau_A^{\text{resp}} = 0.2$ mm ATLAS lifetime-reweight-grid point. The pair has status

SourceCertified_ATLASReworkGridAnchored_SearchBenchmark.

Each mass has a common-apex source realization passing the applicable oscillation/LFU/CLFV constraints. ATLAS simulated 30 GeV directly, but not 28 GeV; 0.2 mm is an exact interior member of the frozen discrete lifetime-reweighting grid. Its 0.1 mm distance to the nearer endpoint of the declared grid span is endpoint-distance bookkeeping only, not a continuous detector-response radius. A frozen HEPData replay puts the PCR-5D3 ordinate more than a factor 4.50 below the linearly reconstructed observed Figure 4a limit across the band, conditionally under the ATLAS simplified model. The on-shell- W image gives a near-35 GeV lepton tag and a sub-millimetre response image. Thus the pair is a reproducible, motivated, admissible benchmark, not a derived or uniquely selected endpoint. It is neither a full-source lifetime/response validation nor a VJ pole prediction; source-level ATLAS non-exclusion requires a source-complete recast or equivalent response bridge. No uniqueness, optimality, or comparative priority is claimed. Before the independent record, a D3-to-VJ bridge must freeze the exhaustive split $\mathfrak{T}_{D3} = \text{Compat}_{D3 \rightarrow X}^{\tau} \dot{\cup} \text{Compat}_{D3 \rightarrow X}^R$, exporting covered $C_{D3 \rightarrow X}^{\tau}$ for the lifetime route and same- $\mathcal{Y}^{\text{ind}}$ regions $C_{D3 \rightarrow X,t}^R$ for full-response types, without post-record pruning. The singleton $\{0.2 \text{ mm}\}$ is allowed only after a full-source lifetime identity and exact lifetime bridge. With $\mathcal{E}_{D3} = \mathcal{E}_{\text{spec}} \cap \{\text{true related pair, exhaustive bridge type admission, and route-qualified region membership}\}$, require $\Pr(\mathcal{E}_{\text{spec}} \setminus \mathcal{E}_{D3} \mid \mathcal{G}_{\text{bridge}}) \leq \alpha_{\text{bridge}}$, hence $\Pr(\mathcal{E}_{D3}) \geq 1 - \alpha_{\text{spec}} - \alpha_{\text{bridge}}$. The possibly uninformative route union is a bridge-conditional benchmark-derived source-coordinate compatibility/confidence set, not part of core $\mathcal{I}_M^{\text{src}}$, independent mass-prediction authority, or physical pole-mass localization.

Here $H(I_E)$ is compact Euclidean and the D3–VJ bridge space is a compact Polish product. The lifetime map is continuous/Borel on Ξ_t , $\mathcal{F}_t$ is Borel on $\mathcal{U}_t^R$, and learned bridge relations, finite type graphs, and C^{τ}/C^R

multifunctions obey the corresponding random-closed/Effros–Borel measurability. The $\mathcal{E}_{D3}$ route clauses require true-tuple membership in Ξ_t or $\mathcal{U}_t^R$ before evaluating these partial maps.

The row deficit obeys $\epsilon_{\text{row}} = s_E^2 \leq 1/256$ at marked-child scope. A single frozen public-data pass reconstructs the unconstrained HFLAV Tau-2023 ratio covariance in an algebraically independent basis. Conditional on applicability of the declared light-response model, the universal response has zero common- ϵ Jacobian and its HFLAV statistic is $\chi^2 = 7.0193$ for four degrees of freedom, $p = 0.1349$, giving ConditionalLightResponseCompatibleAt95; full VJ compatibility remains NotReady. A separate direct/indirect m_W linear-response screen, setting the unreported direct/indirect cross-covariance to zero only as a screening assumption, gives $\epsilon_{95}^{\text{scr}} = 1.62 \times 10^{-3}$; this is not an independence certificate, and the finite-mass HNL electroweak and template response needed for covered promotion is absent. The predeclared stop rule therefore retains $1/256$ and records the weak data as conditional compatibility only. X7 uses a certified outer $\mathcal{E}_X^{\text{src}} \ni \epsilon_X$ and the coverage-preserving epsilon projection of a jointly covered weak-record region; its α_{weak} failure is charged inside α_{spec} . The inverse propagator contains one sheet-holomorphic Σ^{ren} ; $\Delta M/\Gamma$ split only on the real/on-shell boundary, while poles and licensed isolated-simple-pole residues use the full self-energy. The Gram and structural weights feed the intrinsic self-energy/pole kernel; if licensed, $\Gamma = \epsilon \mathcal{G}^{\text{self}}$ is boundary-only and C_t^{pole} contains intrinsic factors only. The gauge-complete $\mathcal{F}_t$ uses the full resummed propagator, with a Laurent residue only on a licensed isolated-simple-pole expansion, together with external vertices, interference, nonresonant terms, and detector response; coherent strata retain full- $\mathcal{D}$ inversion or a certified complete local Laurent principal part. On the covered weak event, conditional on the true chart and nuisance tuple, the true lifetime lies in $\mathcal{T}_t(M_X)$. X8 uses the finite-dimensional Euclidean/Borel space $\mathcal{Y}^{\text{ind}} = \text{ran } M_{\text{ind}}$.

The capstone result is a nested robustness closure. One-carrier identity persists on the trace-transport cell and light-heavy closure on the block-isometry class; rank three has a positive operator-norm radius; SPIC-qualified same-record carrier/reference selection has a positive typed half-separator margin only on the declared certified domain; exact flavor universality and heavy degeneracy are zero-radius refinements against unrestricted standing breaking. Detector evaluation still requires instantiated SPIC and held-out data, while X7/X8 inputs remain NotReady.

Keywords. AASC; neutrino carrier; heavy neutral lepton; structural stability; degeneracy lifting; matrix perturbation; terminal carrier identity; charged-current closure; prediction robustness; search localization; trigger strategy; held-out falsification.

Principal result

Nested structural robustness closure

Fix either ready vectorlike-joint child $b \in \mathcal{B}_{7.5}^{\text{ready}}$ and an accepted realization $X \in \text{PhysMod}_{E,VJ}^b$. Let $\mathcal{U}_{\text{str}}^b(X)$ be the connected certified structural component that preserves the frozen source support, child target, one-apex physicality, complete charged-current isometry, positive light-heavy gap, and the P75-13 trace-transport cell with its deletion witnesses. Then:

$$\begin{aligned} \delta \mathcal{S} \in \mathcal{U}_{\text{str}}^b(X) &\implies |\text{TermCar}(X_\delta)| = 1, \\ \Delta'_{\text{act}} &= \Omega'_H, \\ \|\delta V_H\|_2 < s_E &\implies \text{rank } \Omega'_H = 3, \quad \lambda_{\min}(\Omega'_H) \geq (s_E - \|\delta V_H\|_2)^2, \\ \delta K_H &= aI_3 \iff \text{exact threefold heavy degeneracy is preserved,} \\ \delta K_H &= aI_3 + T, \quad T \neq 0 \implies \text{heavy multiplicity } 2 + 1 \text{ or } 1 + 1 + 1, \\ \frac{3}{2} \|T\|_2 &\leq \text{diam Spec}(T) \leq 2 \|T\|_2, \\ \text{rank } V'_H = 3 : \quad \exists \delta' > 0 : \quad \Omega'_H &= \delta' I_3 \iff \text{the three singular values of } V'_H \text{ coincide,} \\ r_{\text{car}}^b(X) &\geq r_{\text{car,cert}}^b(X) > 0, \\ \text{block isometry, } \eta_V < s_E &\implies \|(\Delta'_{\text{act}}, \Omega'_H)\|_{\text{pair}} \geq \sqrt{6} (s_E - \eta_V)^2 > 0 \end{aligned}$$

The last inequality is ideal and structural; it does not compare counterfactual detector regions. On an instantiated carrier-plus-reference SPIC domain, an attainable $r_{\text{rec}} < s_{\text{car,ref}}/2$ instead gives hypothesiswise same-record selection: the true model/core inverse contains the truth and the competing inverse is empty. The invariant

endpoint is fibrewise $\mathcal{O}_{\text{core}}$ relative to its pre-frozen variation registry, while $\mathcal{O}_{\text{cov}}$ retains the complete structural trace modulo functional determination. Exact degeneracy and exact flavor universality have zero radius against certified standing $U(3)_H$ -breaking perturbations, while carrier identity, block-unitary closure, rank three, the ideal pair-space gap, and SPIC-qualified model/core selection retain the feature-specific robustness stated above.

Scope and claim classes

Claim	Status	Scope
One-carrier stability	Established on declared cells	Status-free finite support skeleton, exact proper-support deletion flags, one full-support parent orbit, explicit guard/update faces, and $r_{\text{car,cert}} > 0$. Missing chart witnesses remain conditional/NotReady; compact-family uniformity requires its separately declared common perturbation bundle. Every perturbation preserving complete light-heavy charged-current isometry.
Block-unitary closure	Exact	$\ \delta V_H\ _2 < s_E$; the radius is sharp in operator norm.
$\Delta'_{\text{act}} = \Omega'_H$	Positive-radius	Full mass-operator perturbation smaller than half the light-heavy spectral gap.
Heavy rank three	Positive-radius	Full-rank equal-singular-value stratum, $\Omega'_H = \delta' I_3$ with $\delta' > 0$; zero unrestricted anisotropic radius.
Heavy aggregate subspace	Symmetry-protected	Central heavy mass perturbations; zero unrestricted traceless radius.
Flavor universality	Symmetry-protected	Fibrewise registry-relative $\mathcal{O}_{\text{core}}$; complete-trace $\mathcal{O}_{\text{cov}}$; cross-child common reduct only for matched ready-child realizations under matched common perturbations; decorated envelopes separate.
Threefold heavy degeneracy	Established	Compact carrier/reference SPIC cores in one common record space, global lower-Lipschitz response, collision coarsening, joint nuisance/control coverage, attainable $r_{\text{rec}} < s_{\text{car,ref}}/2$, and separate inverse/mass/Gram guards; the true inverse contains truth and the competing inverse is empty. On block isometry with $\eta_V \leq \bar{\eta}_V < s_E$, the ideal pair-space gap is structural, and no protocol perturbation radius is inferred.
Invariant/covariant maximality	Certified-domain conditional	$H(I_E)$; numerical endpoints are NotReady.
Detector model/core selection	Symbolic source authority	Every $M \in [28, 30]$ GeV has a common-apex source realization with uniform non-direct constraint margins; 30 GeV is a direct ATLAS mass anchor and the public-limit reconstruction gives a conditional ordinate screen. Endpoint derivation and unique coordinate selection are unproved; no full-response or VJ pole authority follows.
VJ mass domain	Predeclared, source-certified, ATLAS-grid-anchored, and admissible	$c\tau_A^{\text{resp}} = 0.2$ mm is an interior member of the declared discrete grid. Its 0.1 mm distance to the nearer endpoint of the $[0.1, 1.0]$ mm grid span is not a continuous detector-response radius. The W -rest-frame response-coordinate image is 0.231–0.252 mm.
D3 search benchmark	Exact discrete ATLAS reweight-grid member; full-source lifetime	$\epsilon_{\text{row}} = \delta_E = s_E^2 \leq 1/256$. Conditional on the declared light-response applicability, HFLAV gives $p = 0.1349$ with zero common-mode sensitivity; full VJ compatibility remains NotReady. The 1.62×10^{-3} absolute screen sets unreported direct/indirect cross-covariance to zero only as a screening assumption; it is neither an independence certificate nor a covered X7 endpoint.
Benchmark response coordinate	NotReady	Requires an independent constraint and a declared electroweak input-scheme response.
Row-normalization bound	Exact at marked-child target scope; public weak pass is conditional compatibility only	Requires a certified complex-pole continuation atlas, independent normalization/response records, exhaustive compatible types, and separate normalization and domain bridges.
Numerical physical weak normalization	NotReady	Published ATLAS/CMS analyses cover the mass scale but do not implement the complete PCR charge/flavour/matrix record quotient.
Lifetime and mass localization	Conditional/NotReady	Future held-out records only.
Existing-search sensitivity	Recast required	Not altered or completed here.
Empirical confirmation	Open	Reserved for the FCGS programme.
Unresolved PCR-5E parent sectors	NotReady/ROOS	
Cross-fermion universality	Not claimed	

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1 Introduction

1.1 From prediction to robustness

PCR-6 and PCR-7 establish a sharply structured result on two completed neutrino realization sectors. Each accepted realization contains one terminal carrier, three nondegenerate light standing sectors, and one rank-three degenerate heavy standing sector. The experimentally projected packet is

$$\Delta_{\text{act}} = \Omega_H = \delta_E I_3, \quad \text{rank } \Omega_H = 3,$$

with a single rank-three degenerate structural heavy-mass-support class and a finite held-out protocol that separates this packet from a no-vectorlike reference [1, 2].

The remaining question is not whether those theorems are exact at their stated scope. It is what happens when the accepted source is changed within that scope. Three superficially similar questions must be distinguished:

1. how small a perturbation can lift the heavy mass degeneracy;
2. how small a perturbation can destroy flavor universality or rank three;
3. how small a perturbation can produce a second independently complete terminal carrier.

They have different answers. Exact degeneracy and exact universality are symmetry-protected and have zero radius against unrestricted standing symmetry breaking. Rank three has a positive and sharp operator-norm radius. Terminal carrier identity is controlled by a separate source/deletion/promotion boundary rather than by the eigenvalue gap alone.

1.2 Structural margins are not confidence intervals

A theorem does not carry a frequentist confidence interval. Its conclusion is exact conditional on the stated premises. The appropriate deterministic analogue is a feature-specific robustness margin: the distance, in a declared perturbation class and metric, from the accepted point to the first locus on which that feature fails. Statistical confidence enters only when the structurally robust packet is transported through a finite experimental protocol.

This paper therefore uses two different kinds of radius:

$$\text{structural radius} \quad \text{and} \quad \text{detector confidence radius}.$$

The first is defined in the source/standing comparison space; the second is defined in reconstructed record space. Neither can substitute for the other.

1.3 Nested robustness hierarchy

The result has the following typed robustness inventory. The vertical placement is expository only: it encodes neither set inclusion nor logical implication.

one terminal carrier declared trace-transport cell	positive same-scope region
light-heavy closure complete block-isometry class	exact on isometry class
rank-three heavy support $\ \delta V_H\ _2 < s_E$	positive norm radius
positive flavor universality full rank; $\Omega'_H = \delta' I_3, \delta' > 0$	coupling-symmetry refinement zero unrestricted anisotropic radius
exact threefold heavy degeneracy central mass perturbations	mass-symmetry refinement zero unrestricted traceless radius

No implication is encoded by the vertical placement. Failure of one feature does not automatically defeat another. A resolved heavy mass splitting, for example, falsifies exact degeneracy but need not falsify rank three, the closure relation, or one-carrier identity. This theorem-localized interpretation is a primary output rather than an editorial afterthought.

1.4 Proof strategy

The proof proceeds in seven stages.

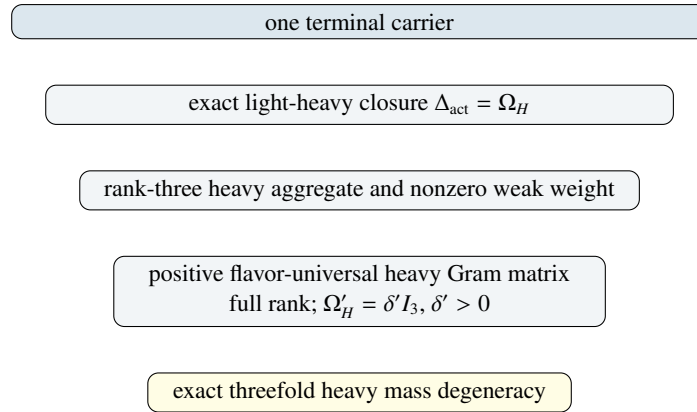
1. Freeze the perturbation target and construct the complete source-first perturbation grammar.
2. Decompose heavy mass perturbations into central and traceless parts and classify exact spectral lifting.
3. Establish heavy-subspace, charged-current-rank, flavor-universality, and block-isometry stability.
4. Construct the terminal-promotion boundary and prove one-carrier stability independently of spectral multiplicity.
5. On domains carrying an explicit stratified structural mass-support/response identifiability certificate in one common record space, extend the PCR-7 protocol and transport confidence regions without replacing injectivity by positive acceptance or identifying source-mass support with physical complex poles.
6. Assemble the feature-specific robustness vector and falsifier hierarchy for PCR-8.
7. Separate the D3 calibration lane from the VJ source domain, derive the marked-child bound $\epsilon_{\text{row}} \leq 1/256$, and state the independent normalization, complex-pole, nuisance, and record gates required for physical lifetime and mass localization.

Nonclaim 1.1 (No retroactive revision). A small perturbation that breaks an exact PCR-6 or PCR-7 refinement does not make the unperturbed theorem approximate. It demonstrates that the exact result is symmetry- or scope-protected. The accepted unperturbed theorem remains unchanged.

1.5 Nested robustness rather than a binary verdict

A structural theorem may be exact at a point and nevertheless have several inequivalent neighborhoods. The accepted heavy sector illustrates the distinction. The mass block is scalar, the charged-current Gram matrix is scalar, the full charged-current block is an isometry, the heavy aggregate has rank three, and the terminal carrier is unique. These statements are related, but none is a synonym for the others. The following status stack displays

them without asserting an implication chain:



The vertical order is mnemonic only and carries no implication arrow. A structural mass-support splitting may preserve rank, closure, and one-carrier identity; an anisotropic Gram matrix may preserve rank and closure; rank loss may still leave one carrier but changes the experimentally locked packet; and a closure failure reaches a deeper constitutive boundary.

1.6 Relation to standard perturbation theory

The matrix estimates used below are standard consequences of Hermitian and singular-value perturbation theory, including Weyl inequalities, Davis–Kahan invariant-subspace control, and distance-to-rank-deficiency results [3–7]. Their role here is deliberately limited. They quantify changes only after PCR-7.5 has established that a proposed variation is a target-relevant perturbation of the accepted source rather than a basis change, source update, or target change. Ordinary matrix perturbation theory supplies the numerical bound; the source and standing analysis supplies the physical authority of the perturbation.

2 Frozen dependencies and starting object

For dependency bookkeeping, the object scope and the proof-authority ledger are kept distinct. The object scope contains only the two ready PCR-5E children. The direct programme-level proof authorities used by the core structural chain P75-00–P75-26 are the PCR-3 finite source-trace and normal-form results, the PCR-4 one-apex physical evaluator, the PCR-5E E-S1/E-S2 joint-incidence and source/update boundaries for the ready children, the accepted PCR-6 one-carrier and heavy-witness results, the accepted PCR-7 prediction packet and protocol, and the fixed-domain and SCIC exhaustion/classifier results. The hypotheses and earlier results on which those named authorities depend are accepted only through their frozen transitive dependency closure; PCR-7.5 does not reopen or strengthen them. The tagged experimental-authority extension P75-X1–P75-X8 plus P75-X2A separately imports the frozen PCR-5D3 source-realization/declared-response-chart certificate and the cited ATLAS, HEPData, CMS, and HFLAV experimental records. It does not import a PCR-5D3 full-source lifetime identity or a complete ATLAS response validation.

Every direct programme-level authority in this ledger is distributed as an exact release-support PDF. Its release path, printed title and date, page count, byte size, semantic version where printed, and SHA-256 identity are fixed in `DIRECT_AUTHORITY_LEDGER_v1.4.4.json`. The package validator checks that ledger against the bundled bytes. This provenance lock identifies the imported statements; it does not enlarge their scopes or certify any transitive dependency as a new direct authority.

Theorem 2.1 (P75-00: ARC insertion, import, and no-backflow lock). *PCR-7.5 is inserted between PCR-7 and PCR-8 with dependency order*

$$\text{PCR-6} \longrightarrow \text{PCR-7} \longrightarrow \text{PCR-7.5} \longrightarrow \text{PCR-8}.$$

Its object domain contains only the two ready child scopes

$$\mathcal{B}_{7.5}^{\text{ready}} = \{b_{\text{acc}}, b_{\text{glob}}\},$$

and its declared direct programme-level proof authorities are

$$\mathfrak{D}_{7.5}^{\text{dir}} = \{\text{PCR-3, PCR-4, PCR-5E, PCR-6, PCR-7, FixedDomain, SCIC}\},$$

restricted to the results and scopes identified above. Their frozen transitive dependency closure is inherited as authority for those results, not promoted to a new direct source or enlarged endpoint. No PCR-7.5 theorem alters an upstream target, source class, equivalence relation, empirical status, or unresolved parent boundary.

Proof. PCR-3 supplies the finite source-facing trace and reconstruction authority used in P75-02/P75-03, while PCR-4 supplies the one-apex physical evaluator used in P75-03 [8, 9]. PCR-5E supplies complete E-H1–E-H10 exports only for the bounded-accidental and exact-global vectorlike-joint children, including the E-S1/E-S2 finite joint-incidence record, and explicitly retains the parent NotReady/ROOS decomposition [10]. PCR-6 characterizes each accepted child realization and proves one terminal bearer with a rank-three heavy sector, but does not collapse the continuum cross-realization quotient [1]. PCR-7 constructs the finite CCST prediction protocol and freezes future validation records as held out [2]. The fixed-domain and SCIC results supply only the invariant/exhaustive classifier authority used in P75-04 [11, 12]. This explicit ledger accounts for every direct upstream programme theorem invoked by P75-02–P75-04; all earlier premises remain inside the cited results’ frozen transitive closure. PCR-7.5 uses these objects only as fixed inputs. Any change to the child target, source support, terminal role, detector record quotient, or held-out split is assigned TargetUpdate, SourceUpdate, or ProtocolUpdate below rather than being treated as an internal perturbation. Thus no backward implication is introduced. $\square$

2.1 Unperturbed source and physical packet

Fix $b \in \mathcal{B}_{7.5}^{\text{ready}}$ and $X \in \text{PhysMod}_{E,VJ}^b$. On the accepted heavy multiplicity space $E_H \cong \mathbb{C}^3$, the unperturbed source has

$$K_H = H_\lambda^2 I_3, \quad V_H = -s_E U, \quad U \in U(3), \quad s_E > 0.$$

The inclusive heavy charged-current weight is

$$\Omega_H = V_H V_H^\dagger = \delta_E I_3, \quad \delta_E = s_E^2.$$

The complete light-plus-heavy block isometry gives

$$NN^\dagger + V_H V_H^\dagger = I_3, \quad \Delta_{\text{act}} = I_3 - NN^\dagger = \Omega_H.$$

PCR-6 proves that the complete heavy witness family descends under the internal $U(3)_H$ action to the aggregate projector P_H , and that the source-connected parent is the unique terminal carrier inside each fixed X [1]. PCR-7 proves the finite source-authorized detector packet and the no-vectorlike reference separation [2].

The coefficient s_E requires a further authority distinction. PCR-5E proves a nondegenerate positive survivor interval in the E-S4A D3-1.3/D5 marginal response system and then uses a marked dyadic witness selected in the proved punctured pass cell before the compact response box is frozen to define the ready-child slice. PCR-7 imports that source-fixed coefficient as a child-source derivation input, so every realization in the accepted child inherits it. Neither theorem proves that a comparison class not already restricted to that child physically selects the same dyadic value. Thus $\delta_E = s_E^2$ is an exact identity on every fixed accepted source, whereas an independently derived numerical physical prediction for δ_E remains a separate selection problem addressed in Section 17.

2.2 Imported strict margins

The ready child source lies in the strict-margin vectorlike product region established by PCR-5E. In particular, source activity is positive, the D3 process response is accepted, matching and continuation are defined, and the heavy threshold remains inside a compact physical interval. These strict inequalities are the local seeds from which the robustness regions below are formed. They are not assumed to be uniform on unresolved parent sectors.

Warning 2.2 (Parent-scope firewall). No perturbation theorem in this paper applies to the D3-1.1 programme, the complete D4 cosmological target, the non-RCF-finite D3-1.3 remainder, the five unresolved D5 comparison rows, or source kinds outside the two E-S5 child categories.

3 Robustness target and source-first perturbation theory

Definition 3.1 (Robustness target). For each ready child b , the robustness target $T_{\text{rob}}^{v,b}$ consists of the accepted child target, the PCR-6 terminal identity target, the PCR-7 observable/protocol target, and the additional work of classifying feature stability under perturbations that preserve the declared source and target ontology.

Theorem 3.2 (P75-01: robustness target formation). *The family $T_{\text{rob}}^{v,b}$, $b \in \mathcal{B}_{7.5}^{\text{ready}}$, can be frozen before perturbation inspection so that it contains no preferred robustness radius, no assumption of exact degeneracy away from the unperturbed point, and no conclusion about carrier splitting. Its target uses are exactly: perturbation typing, source/target update classification, heavy spectral response, charged-current response, terminal carrier reuse, detector response, and theorem-localized failure.*

Proof. Delete from the proposed target all perturbation values, eigenvalue partitions, norms, and desired endpoint labels. What remains is determined by the accepted source, the complete terminal carrier interface, and the accepted detector packet. These objects already fix what must be preserved for same-child comparison: source support and incidence, one-apex physicality, target applicability, terminal role use, and protocol record type. The additional robustness work is therefore definable independently of any perturbation outcome. Any candidate-specific value enters only after the target is frozen. $\square$

Definition 3.3 (Semantic perturbation trace). For a source-authorized perturbation δS of a subobject or relation of X , define

$$\Theta_{\text{pert}}^{b,X}(\delta S) = (\chi_{\text{src}}, \tau_{\text{phys}}, \Sigma_{\text{term}}, O_{\text{CCST}}, \text{Fail}, \text{Update}),$$

where the entries record the activity-reduced source, one-apex physical response, terminal continuation signature, detector-facing invariant packet, first failure, and update boundary. The tuple is a regrouped constrained joint record, not a product of six independent marginal summaries. Its typed codomain and reconstruction maps are fixed in the following lemma.

Lemma 3.4 (Joint trace-regrouping and factorization). *Fix $b \in \mathcal{B}_{7.5}^{\text{ready}}$, $X \in \text{PhysMod}_{E,\text{NJ}}^b$, and the ordered interface set*

$$I_6 = (\text{src}, \text{phys}, \text{term}, \text{CCST}, \text{Fail}, \text{Update}).$$

Let $J_{E-S1}^{b,X}$ be the finite typed coordinate set of the frozen PCR-5E E-S1 joint source/target-incidence trace. For $j \in J_{E-S1}^{b,X}$, let C_j be its codomain. If j is a cross-interface or higher-arity incidence coordinate, then C_j is the codomain of the whole typed relation on its declared incidence legs; it is not replaced by any one-leg marginal. After adjoining the PCR-6 terminal coordinates, the PCR-7 CCST coordinates, and the two fail-closed boundary coordinates, write $J_{\text{aug}}^{b,X}$ for the resulting finite typed coordinate set and

$$\mathcal{T}_{\text{aug}}^{b,X} \subseteq \prod_{j \in J_{\text{aug}}^{b,X}} C_j$$

for the lawful joint-record subspace. All E-S1 incidence equations and cross-coordinate compatibility relations are part of this subspace. Write

$$\widehat{\mathcal{T}}_{\text{aug}}^{b,X} : \text{Dom}(\Theta_{\text{pert}}^{b,X}) \longrightarrow \mathcal{T}_{\text{aug}}^{b,X}$$

for the augmented trace evaluator on the source-authorized perturbations of the preceding definition, and

$$\mathcal{T}_{E-S1}^{b,X} := \text{pr}_{E-S1} \circ \widehat{\mathcal{T}}_{\text{aug}}^{b,X}$$

for its complete imported E-S1 trace component.

There is a fixed ownership map $o : J_{\text{aug}}^{b,X} \rightarrow I_6$ which assigns each entire typed coordinate to its first responsible interface in the declared fail-closed evaluation order. Put

$$C_i^{b,X} := \prod_{o(j)=i} C_j, \quad G^{b,X} : \mathcal{T}_{\text{aug}}^{b,X} \longrightarrow \prod_{i \in I_6} C_i^{b,X},$$

where $G^{b,X}$ merely regroups coordinates according to o , and define the actual six-field codomain by the constrained image

$$C_{\Theta}^{b,X} := G^{b,X}(\mathcal{T}_{\text{aug}}^{b,X}) \subseteq \prod_{i \in I_6} C_i^{b,X}.$$

Then

$$G^{b,X} : \mathcal{T}_{\text{aug}}^{b,X} \xrightarrow{\cong} C_{\Theta}^{b,X}$$

is a typed record isomorphism and

$$\Theta_{\text{pert}}^{b,X} = G^{b,X} \circ \widehat{\mathcal{T}}_{\text{aug}}^{b,X}.$$

The six block projections of this composite, in the order I_6 , are by definition

$$(\chi_{\text{src}}, \tau_{\text{phys}}, \Sigma_{\text{term}}, O_{\text{CCST}}, \text{Fail}, \text{Update}).$$

In particular the complete PCR-5E E-S1 trace is recovered by the lawful map

$$r_{E-S1}^{b,X} := \text{pr}_{E-S1} \circ (G^{b,X})^{-1}, \quad \mathcal{T}_{E-S1}^{b,X} = r_{E-S1}^{b,X} \circ \Theta_{\text{pert}}^{b,X}.$$

Thus every target-authorized use F of the E-S1 joint trace, including a use of a cross-interface or higher-arity correlation, factors as

$$F \circ \mathcal{T}_{E-S1}^{b,X} = (F \circ r_{E-S1}^{b,X}) \circ \Theta_{\text{pert}}^{b,X}.$$

Proof. The E-S1 trace is finite and typed by the imported PCR-5E certificate. The adjoined PCR-6, PCR-7, failure, and update coordinates are likewise finite and typed at the frozen target. The ownership map partitions coordinate labels, so $G^{b,X}$ changes only the bracketing and ordering of a finite record. Its inverse restores every coordinate to its original label. Restricting the codomain to $C_{\Theta}^{b,X} = G^{b,X}(\mathcal{T}_{\text{aug}}^{b,X})$ retains the full lawful joint-record subspace rather than replacing it by the product of its six marginal images. Moreover, an incidence coordinate of arity greater than one is moved as one typed relational object, with all incidence legs intact. Hence neither cross-coordinate compatibility nor cross-interface/higher-arity information is lost. Projection after the inverse regrouping recovers the E-S1 record exactly, and composition gives the displayed factorization. This is a finite record-regrouping argument; it does not add to, or re-prove, the imported PCR-5E completeness theorem. $\square$

Theorem 3.5 (P75-02: semantic perturbation class). *Equality of the complete perturbation trace defines the target-relative semantic perturbation equivalence. Every same-child physical use of a perturbation factors through $\Theta_{\text{pert}}^{b,X}$.*

Proof. PCR-3 proves source-facing trace completeness for finite carrier presentations; PCR-5E proves semantic saturation of joint source/target incidence; PCR-6 proves completeness of the terminal witness family; and PCR-7 proves completeness of the declared CCST record quotient at its protocol scope [1, 2, 8, 10]. By Lemma 3.4, the six displayed fields are an invertible regrouping of that augmented joint record: every E-S1 coordinate, compatibility equation, cross-interface relation, and higher-arity incidence is recoverable, and no independence of the six fields is assumed. A same-child perturbation use can therefore change standing only by changing at least one coordinate of this constrained six-field record, and every such use factors through $\Theta_{\text{pert}}^{b,X}$. Conversely, equal complete traces reconstruct equal augmented joint records and hence agree under every use generated from those interfaces. Extensionality yields the claimed equivalence. $\square$

Construction 3.6 (Perturbation source grammar). Let H_X be the finite activity-reduced source hypergraph of X . A perturbation normal form is obtained by decorating its existing nodes, hyperedges, equations, support maps, matching arrows, continuation arrows, and boundary incidences with typed displacement variables, while retaining the original arities and source responsibilities. A new node, new primitive hyperedge, or new constitutive law is not a normal-form perturbation; it is a source update.

Theorem 3.7 (P75-03: source-first perturbation grammar and completeness). *The decorated source theory $\text{PertSrc}^{v,b}(X)$ is canonically equivalent to the class of finitely presented semantic perturbations that preserve the frozen source ontology, modulo lawful source isomorphism. Every generated normal form has a source-coherent physical interpretation or an exact typed failure; every semantic same-source perturbation has such a normal form.*

Proof. Apply the PCR-3 normal-form reconstruction to the fixed finite source hypergraph and replace each standing-bearing coefficient or relation value by its declared displacement object. Since the underlying node and incidence grammar is fixed, source coherence is the original coherence system evaluated at the perturbed values. A semantic perturbation preserving the ontology changes only those values and therefore reconstructs to a decorated normal form. If it changes the primitive support or formation rules, it is outside the grammar by definition and is recorded as **SourceUpdate**. Physical interpretation and exact failure follow from the PCR-4/PCR-5E one-apex evaluator applied to the decorated source. $\square$

Theorem 3.8 (P75-04: perturbation status and scope classifier). *Every submitted perturbation receives exactly one highest-authority status in*

$$\left\{ \begin{array}{l} \text{Skin, SameScopeTensor, StandingBreakingTensor, SourceUpdate,} \\ \text{TargetUpdate, ProtocolUpdate, NotReady} \end{array} \right\}.$$

The classifier is invariant under lawful redescription and is exhaustive at the fixed perturbation target.

Proof. Order the possible first changed loci as: presentation only; standing value in the fixed source; standing symmetry structure in the fixed source; primitive source formation; target/role/witness/comparison structure; detector protocol/record structure; unresolved typing or theorem. The first changed locus determines the status. Anchor–tensor–skin exhaustion and the fixed-domain no-second-classifier theorem exclude an additional same-scope authority channel [11, 12]. Since lawful redescription does not change the first standing-bearing locus, the classifier is invariant. $\square$

4 Heavy symmetry and exact degeneracy lifting

4.1 The complete heavy action

The unperturbed heavy sector carries the lawful multiplicity action

$$E_H \mapsto WE_H, \quad W \in U(3)_H.$$

PCR-6 proves that the complete target-admitted heavy witness family is equivariant under this action, so individual heavy basis vectors are skin while the aggregate projector P_H has standing [1].

Theorem 4.1 (P75-05: heavy action and central/traceless decomposition). *Every Hermitian perturbation of the heavy mass-squared operator has the unique decomposition*

$$\delta K_H = aI_3 + T, \quad a = \frac{1}{3} \text{tr}(\delta K_H), \quad \text{tr } T = 0.$$

The central part is invariant under the full $U(3)_H$ conjugation action. A Hermitian perturbation commutes with every $W \in U(3)_H$ if and only if its traceless part vanishes.

Proof. The decomposition is the orthogonal decomposition of Hermitian matrices into the scalar subspace and the traceless subspace. If A commutes with every unitary, then in particular it commutes with every rank-one phase and permutation unitary. The phase unitaries force A to be diagonal in every chosen basis, while permutations force equal diagonal entries. Hence $A = aI_3$. The converse is immediate. This is the finite-dimensional commutant form of Schur’s lemma [3, 13]. $\square$

Theorem 4.2 (P75-06: symmetry-preserving heavy rigidity). *Every same-scope heavy mass perturbation preserving the complete $U(3)_H$ action has the form aI_3 . Consequently,*

$$K'_H = K_H + aI_3 = (H_\lambda^2 + a)I_3,$$

so the heavy eigenvalue multiplicity remains exactly three throughout the positive-domain central stratum.

Proof. By Theorem 4.1, complete $U(3)_H$ -preservation forces $T = 0$. Functional calculus then gives one repeated eigenvalue $H_\lambda^2 + a$. Positivity requires $H_\lambda^2 + a > 0$; crossing zero is a support/target boundary rather than a same-scope continuation of the heavy structural support. $\square$

Theorem 4.3 (P75-07: exact degeneracy-lifting classification). *Let $K'_H = (H_\lambda^2 + a)I_3 + T$ with $T = T^\dagger$, $\text{tr } T = 0$. Then:*

1. *$T = 0$ if and only if the heavy mass-squared eigenvalue remains threefold degenerate;*
2. *if $T \neq 0$, exact threefold degeneracy is lost;*
3. *the only remaining multiplicity partitions are $2 + 1$ and $1 + 1 + 1$, according as T has two equal eigenvalues or three distinct eigenvalues.*

These are internal spectral partitions and do not determine terminal carrier cardinality.

Proof. The eigenvalues of K'_H are $H_\lambda^2 + a + t_i$, where t_i are the eigenvalues of T . Threefold equality means $t_1 = t_2 = t_3$. Since their sum is zero, all must vanish, so $T = 0$. A nonzero Hermitian 3×3 matrix has either two or three distinct eigenvalues after excluding threefold equality. The carrier statement is deferred to the terminal-promotion analysis in Section 7. $\square$

Corollary 4.4 (Zero unrestricted degeneracy radius). *For every $\varepsilon > 0$ there exists a standing traceless Hermitian perturbation with $\|T\|_2 < \varepsilon$ that lifts exact threefold degeneracy. For example,*

$$T_\varepsilon = \frac{\varepsilon}{2} \text{diag}(1, -1, 0).$$

Hence exact degeneracy has zero operator-norm radius against unrestricted standing $U(3)_H$ -breaking perturbations.

4.2 Geometric picture of the symmetry decomposition

The Hermitian perturbation space decomposes orthogonally under the Hilbert–Schmidt inner product:

$$\text{Herm}(3) = \mathbb{R}I_3 \oplus \mathfrak{su}(3)_{\text{Herm}}.$$

The first line is tangent to the exact-degeneracy stratum; the second is transverse to it. The accepted point therefore has a large domain of exact degeneracy along the central direction but zero radius in the ambient perturbation space. The two statements are compatible because they refer to different perturbation classes.

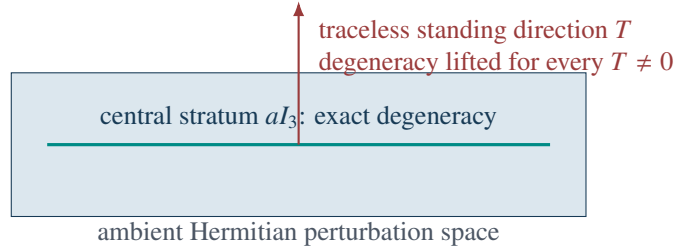


Figure 1: Exact degeneracy is rigid along the protecting central stratum and infinitesimally fragile in transverse standing directions.

4.3 Residual symmetry after splitting

A $2 + 1$ partition has a residual $U(2) \times U(1)$ commutant on the split mass block, while a $1 + 1 + 1$ partition has only the diagonal phase torus (up to permutation of equal labels). These residual groups classify the post-splitting basis freedom. They do not by themselves determine terminal carrier cardinality; that question remains governed by the complete terminal signature and the promotion boundary.

5 Quantitative spectral and subspace stability

Theorem 5.1 (P75-08: quantitative heavy spectral spread). *Let $T = T^\dagger$, $\text{tr } T = 0$, and $\rho_K = \|T\|_2$. Then*

$$\frac{3}{2}\rho_K \leq \text{diam Spec}(T) \leq 2\rho_K.$$

Both constants are sharp.

Proof. Write the ordered eigenvalues as $t_1 \geq t_2 \geq t_3$ with $t_1 + t_2 + t_3 = 0$. The upper bound follows from $|t_1|, |t_3| \leq \rho_K$. Suppose first $\rho_K = t_1$. Set $t_3 = -a\rho_K$. The ordering and trace conditions imply $1/2 \leq a \leq 1$, hence

$$t_1 - t_3 = (1 + a)\rho_K \geq \frac{3}{2}\rho_K.$$

The case $\rho_K = -t_3$ is symmetric. Equality at the lower bound is attained by eigenvalues $(\rho_K, -\rho_K/2, -\rho_K/2)$; equality at the upper bound is attained by $(\rho_K, 0, -\rho_K)$. $\square$

Corollary 5.2 (Structural square-root mass-coordinate spread). *Let $c = H_\lambda^2 + a$ and assume $c > \rho_K$. If $m'_i = \sqrt{c + t_i}$, then*

$$\frac{3\rho_K}{4\sqrt{c + \rho_K}} \leq m'_{\max} - m'_{\min} \leq \frac{\rho_K}{\sqrt{c - \rho_K}}.$$

Proof. Use

$$\sqrt{c + t_1} - \sqrt{c + t_3} = \frac{t_1 - t_3}{\sqrt{c + t_1} + \sqrt{c + t_3}}$$

and combine Theorem 5.1 with

$$2\sqrt{c - \rho_K} \leq \sqrt{c + t_1} + \sqrt{c + t_3} \leq 2\sqrt{c + \rho_K}.$$

$\square$

5.1 Full mass-operator perturbations

Let $\mathcal{K}_0$ denote the complete light-plus-heavy Hermitian mass-squared operator of the accepted realization, and let P_H be the rank-three spectral projector onto the heavy cluster. Define the unperturbed light-heavy gap

$$g_{LH} = \text{dist}(\text{Spec}(\mathcal{K}_0|_{E_H}), \text{Spec}(\mathcal{K}_0|_{E_H^\perp})) > 0.$$

Theorem 5.3 (P75-09: heavy-light subspace and projector stability). *Let $\mathcal{K}' = \mathcal{K}_0 + E$, $E = E^\dagger$, with $\|E\|_2 < g_{LH}/2$. Then the heavy spectral cluster of $\mathcal{K}'$ remains separated from the light spectrum and has rank three. If P'_H is its spectral projector, then*

$$\|P'_H - P_H\|_2 \leq \frac{2\|E\|_2}{g_{LH}}.$$

Proof. Weyl's eigenvalue perturbation theorem moves every eigenvalue by at most $\|E\|_2$, so the two clusters remain disjoint when $2\|E\|_2 < g_{LH}$. Their spectral multiplicities are therefore unchanged. The Davis–Kahan sin-theta theorem bounds the norm of the difference of the associated invariant-subspace projectors by the perturbation norm divided by the reduced gap; using the conservative reduced gap $g_{LH}/2$ yields the displayed estimate [3, 5, 6]. $\square$

Remark 5.4 (Aggregate stability versus internal splitting). Theorem 5.3 preserves the rank-three aggregate heavy subspace. It does not preserve exact degeneracy inside that subspace; Theorem 4.3 governs the internal partition.

6 Charged-current rank, universality, and closure

Let

$$V'_H = -s_E U + E_V, \quad \eta_V = \|E_V\|_2, \quad \Omega'_H = V'_H V'^{\dagger}_H.$$

Theorem 6.1 (P75-10: charged-current rank stability). *The singular values of V'_H satisfy*

$$|\sigma_i(V'_H) - s_E| \leq \eta_V.$$

In particular,

$$\sigma_{\min}(V'_H) \geq s_E - \eta_V.$$

Hence $\eta_V < s_E$ implies

$$\text{rank } \Omega'_H = 3, \quad \lambda_{\min}(\Omega'_H) \geq (s_E - \eta_V)^2.$$

The radius s_E is sharp in operator norm.

Proof. The singular-value perturbation inequality gives $|\sigma_i(A + E) - \sigma_i(A)| \leq \|E\|_2$ [5, 13]. Since all singular values of $-s_E U$ equal s_E , the bound follows. Positive minimum singular value implies full rank, and the eigenvalues of $V'_H V'^{\dagger}_H$ are the squared singular values. Sharpness follows by lawful unitary redescription to $U = I_3$ and choosing

$$E_V = s_E e_1 e_1^\dagger,$$

with sign chosen to cancel the first diagonal entry of $-s_E I_3$. This rank-deficient perturbation has norm s_E . $\square$

Theorem 6.2 (P75-11: flavor-universality and isotropy rigidity). *For a full-rank 3×3 charged-current block V'_H , the following are equivalent:*

1. $\Omega'_H = \delta' I_3$ for some $\delta' > 0$;
2. the three singular values of V'_H are equal;
3. $V'_H = \sqrt{\delta'} W$ for some $W \in U(3)$.

Exact flavor universality is therefore preserved on the isotropic/unitary stratum and has zero unrestricted radius against standing anisotropic perturbations.

Proof. The spectral theorem gives eigenvalues of Ω'_H equal to the squared singular values of V'_H , proving equivalence of the first two statements. Polar decomposition gives $V'_H = \Omega'^{1/2}_H W$ for a unitary W , proving the third. For zero radius, use the lawful basis with $U = I_3$ and perturb by

$$E_\varepsilon = \varepsilon \text{diag}(1, 0, 0).$$

For every nonzero arbitrarily small ε , the singular values are not all equal, while rank remains three for $|\varepsilon| < s_E$. $\square$

Definition 6.3 (Complete block-isometry perturbation). A same-scope charged-current perturbation is block-isometric when the complete light-plus-heavy map satisfies

$$N' N'^\dagger + V'_H V'^\dagger_H = I_3$$

on the retained boundary quotient.

Theorem 6.4 (P75-12: block-unitarity light-heavy closure transport). *For every block-isometric same-scope perturbation,*

$$\Delta'_{\text{act}} = I_3 - N' N'^\dagger = V'_H V'^\dagger_H = \Omega'_H.$$

The equality is independent of heavy mass degeneracy, flavor universality, and heavy-basis choice.

Proof. Rearrange the block-isometry identity. The equality is formulated entirely in the boundary Gram matrices and is therefore invariant under right-unitary rotations of the heavy multiplicity basis. Neither the eigenvalues of K'_H nor scalarity of Ω'_H enters the derivation. $\square$

Corollary 6.5 (Hierarchy of coupling features). *Exact flavor universality implies rank three and, when the complete block is isometric, the closure relation. The converses fail: rank three and $\Delta'_{\text{act}} = \Omega'_H$ allow anisotropic heavy weights and split structural heavy-mass support.*

6.1 Rank boundary and anisotropic examples

The exact rank boundary is visible directly. Let

$$\delta V_H = -s_E u_1 v_1^\dagger,$$

where u_1, v_1 are a singular-vector pair of V_H . Then $\|\delta V_H\|_2 = s_E$ and one singular value reaches zero. Conversely no smaller operator-norm perturbation can make the matrix singular. Thus the radius s_E is sharp.

An arbitrarily small anisotropic perturbation can break universality while preserving rank:

$$V'_H = \text{diag}(s_E + \epsilon, s_E - \epsilon, s_E)U, \quad 0 < |\epsilon| < s_E.$$

Then Ω'_H is positive definite but not scalar. This is the charged-current analogue of the central/traceless mass decomposition.

6.2 Closure as the deeper invariant

The equation

$$\Delta'_{\text{act}} = \Omega'_H$$

expresses conservation of charged-current weight between the light and heavy blocks. It is insensitive to the basis used inside either block and survives both structural mass-support splitting and anisotropic heavy weight. For that reason it is placed above rank, universality, and degeneracy in the robustness hierarchy.

7 Terminal carrier stability under spectral splitting

Spectral perturbation and terminal carrier promotion are different problems. PCR-6 identifies one terminal bearer by independent eligibility formation, deletion-essential support, one-apex compatibility, and complete terminal continuation signatures. A perturbation can split an internal sector without creating a new object that independently carries all of that work.

Definition 7.1 (Status-free finite terminal skeleton atlas). Let $\mathcal{S}_b(X)$ be the finite deletion-essential support and irreducible-relation generator set of the activity-reduced ready-child source. Write $K_{\min}^b(X) \subseteq \mathcal{S}_b(X)$ for the complete deletion-essential generator kernel of the accepted parent: the generators that every terminal-role-complete same-apex candidate must contain. Let

$$\text{Mode} = \{\text{Node, Aggregate, Relation, Mixed, Parent}\}, \quad \Pi_H = \{\perp, 3, 2 + 1, 1 + 1 + 1\}.$$

For a source-authorized candidate A , first define the partition-forgetful support/mode core and then its finite partition decoration by

$$\text{sk}_0(A) = (\text{supp}_{\mathcal{S}}(A), \text{mode}(A)), \quad \text{sk}(A) = (\text{sk}_0(A), \pi_H(A)).$$

The finite decorated image is denoted $\mathcal{A}_b^{\text{dec}}(X)$. It is formed before eligibility, compatibility, terminal status, or carrier cardinality is evaluated. This is the finite support-pattern skeleton induced by PCR-6 Definition 18.1 and P6-04 together with the finite PCR-5E E-S1/E-S2 source and incidence presentation [1, 10]; it is not a claim that the full candidate class or any candidate fibre is finite.

Projector orientations and other lawful continuous presentation data remain inside the fibre over a decorated skeleton and are not used in the finiteness argument. The three heavy rank partitions are included prospectively. Thus splitting changes π_H but not sk_0 . Write

$$\mathcal{P}_X^{\text{dec}} = \{P_{X,\pi} : \pi \in \{3, 2 + 1, 1 + 1 + 1\}\}$$

for the full-support parent decorations that occur on the cell. The carrier projection sends all of them to the same terminal parent orbit, $\varpi_{\text{car}}(P_{X,\pi}) = [P_X]_{\text{term}}$. Define the genuinely proper support skeletons by

$$\mathcal{A}_b^{\text{prop, supp}}(X) = \{a \in \mathcal{A}_b^{\text{dec}}(X) : K_{\min}^b(X) \not\subseteq \text{supp}_{\mathcal{S}}(a)\}.$$

After lawful presentation reduction, any full-support source-faithful candidate belongs to $\mathcal{P}_X^{\text{dec}}$ and hence to the single parent carrier orbit. A full-support object with changed standing-bearing data is a different realization or an explicit update.

Definition 7.2 (Certified trace-transport cell and face slacks). Work in the relative finite-dimensional same-source normal-form chart $\mathfrak{P}^{b,X}$; exact source and target equations are equations of this relative chart rather than small transverse perturbations. Let C_X be a closed certified normal-form/activity cell with $0 \in \text{relint } C_X$ on which:

1. the finite source hypergraph, activity pattern, terminal role, target obligation types, and the boundary, matching, continuation, and common-cone incidences are fixed;
2. the exact obligations remain frozen and every nonfrozen parent definedness, applicability, compatibility, continuation, and failure guard stays in the same strict Pass cell;
3. the heavy aggregate remains a separated rank-three support and the active light-heavy incidence remains nonzero, while all three internal heavy partitions in Π_H are allowed; and
4. no source, target, or normal-form update face is crossed.

Such a cell is obtained by taking a closed relative neighbourhood inside the PCR-5E E-S4A strict response/routing cell and intersecting it with the P75-09 heavy-light gap and P75-10 active-rank neighbourhoods. A chart for which these explicit cell hypotheses are not exhibited is `NotReady` for a positive carrier-radius claim; no upstream continuity of an unspecified terminal residual is inferred.

Set

$$\mathfrak{F}_{\text{reg}}^{b,X} = \mathfrak{P}^{b,X} \setminus \text{relint } C_X, \quad r_{\text{reg}}^b(X) = \text{dist}_{\mathfrak{P}}(0, \mathfrak{F}_{\text{reg}}^{b,X}) > 0,$$

and let $\mathfrak{F}_{\text{nf}}^{b,X}$ be the closed same-source normal-form, source-update, or target-update face, with $d_{\text{nf}}(X) = \text{dist}_{\mathfrak{P}}(0, \mathfrak{F}_{\text{nf}}^{b,X}) > 0$.

For each nonfrozen open parent guard, use its distance-to-complement slack; let $s_P(\delta)$ be the minimum of these finitely many slacks. It is locally Lipschitz with a certified constant L_P , is positive on the declared cell, and is omitted from minima when there is no nonfrozen parent guard. For every proper-support skeleton a , let $\gamma_{a,k}(\delta) > 0$, $k \in \mathcal{G}_a$, be the finite distance-to-complement slacks for any nonfrozen guard needed to keep its chosen deletion witness typed, with Lipschitz constants $L_{a,k}$. These are explicit face distances, not normalized terminal missing-work residuals.

Lemma 7.3 (Finite deletion-witness trace transport). *On relint C_X , the generator identification transports the PCR-6 P6-S2B/G deletion matrix as follows.*

1. Every $a \in \mathcal{A}_b^{\text{prop,supp}}(X)$ omits a fixed deletion-essential generator g_a . The corresponding terminal witness remains defined and changed for every candidate over a , independently of projector orientation.
2. If $q_a(\delta) \in \{0, 1\}$ records whether that transported witness excludes the proper-support skeleton, then

$$q_a(\delta) \equiv 1, \quad L_a^{\text{del}} = 0 \quad \text{on relint } C_X.$$

3. A candidate containing the complete deletion-essential support inherits the parent source values and terminal trace. Its partition-decorated skeleton lies in $\mathcal{P}_X^{\text{dec}}$, and all such decorations map to the same orbit $[P_X]_{\text{term}}$. Changing a standing value moves to another realization or an update.
4. The parent remains independently eligible, terminal-role-complete, and one-apex compatible throughout the cell.

Proof. PCR-6 Definition 18.1 forms the candidate class before eligibility, and P6-04 generates it from the finite source theory. PCR-5E E-S1/E-S2 fixes the finite target-active incidence hypergraph before status evaluation. Hence fixed source incidence identifies the support skeletons throughout C_X .

At the accepted point, P6-S2B and the Appendix G deletion matrix attach a specific lost terminal use to every proper omission. Each such statement is about retained support, activity, boundary, matching, continuation, population, symmetry, or common-apex incidence—precisely the data frozen by the cell. P75-09 transports the rank-three aggregate P_H , and P75-10 keeps its active charged-current support. Refining P_H into a $2 + 1$ or $1 + 1 + 1$ internal spectral partition therefore does not supply the omitted aggregate or any other missing terminal generator. The deletion obstruction is consequently the exact locally constant flag $q_a = 1$, not the positive minimum of a postulated continuous residual over a candidate fibre.

For full support, the proof of PCR-6 Proposition G.2 is a restriction-category argument: a same-apex subobject inherits source parameters, active incidences, boundary data, and continuation laws; it may retain or delete them but may not retune them. The same argument transports through the fixed normal-form cell. The parent construction retains all of those data, and the strict Pass guards exclude parent failure. This proves the four assertions without applying the exact-degenerate P6-S2B theorem pointwise to a standing-breaking perturbation. $\square$

Definition 7.4 (Terminal failure and promotion locus). For $a \in \mathcal{A}_b^{\text{prop,supp}}(X)$ put

$$r_a^{\text{face}}(X) = \min \left(\left\{ \frac{q_a(0)}{L_a^{\text{del}}} \right\} \cup \left\{ \frac{\gamma_{a,k}(0)}{L_{a,k}} : k \in \mathcal{G}_a \right\} \right),$$

where $u/0 = +\infty$ for every positive constant slack u , including a constant parent slack, and a minimum over no nonfrozen guard terms is $+\infty$. Thus the exact obstruction $q_a \equiv 1$, with $L_a^{\text{del}} = 0$, does not manufacture a finite analytic scale.

The explicit terminal failure/promotion locus is

$$\begin{aligned} \partial \mathcal{P}_{\text{term}}^b(X) &= \mathcal{F}_{\text{reg}}^{b,X} \cup \mathcal{F}_{\text{nf}}^{b,X} \cup \{\delta \in \text{relint } C_X : s_P(\delta) \leq 0\} \\ &\cup \bigcup_{a \in \mathcal{A}_b^{\text{prop,supp}}(X)} \left(\left\{ \delta \in \text{relint } C_X : q_a(\delta) = 0 \right\} \cup \bigcup_{k \in \mathcal{G}_a} \left\{ \delta \in \text{relint } C_X : \gamma_{a,k}(\delta) \leq 0 \right\} \right), \end{aligned}$$

with the parent-slack set omitted if there is no nonfrozen parent guard. The regularity-cell boundary is therefore itself an explicit escape/update falsifier. The locus is not the closure of an unspecified promotion set.

Theorem 7.5 (P75-13: finite-skeleton trace transport and carrier radius). *At every accepted ready-child point in the declared strict-margin trace-transport cell,*

$$r_{\text{reg}}^b(X) > 0, \quad d_{\text{nf}}(X) > 0.$$

Every proper-support skeleton has $q_a(0) = 1$, and every displayed nonfrozen guard slack is positive. With the conventions above, set

$$r_{\text{car,cert}}^b(X) = \frac{1}{2} \min \left\{ r_{\text{reg}}^b(X), d_{\text{nf}}(X), \frac{s_P(0)}{L_P}, \min_{a \in \mathcal{A}_{\text{prop, supp}}^b(X)} r_a^{\text{face}}(X) \right\} > 0,$$

where the parent term is omitted when its guard family is empty. Then

$$B_{\mathfrak{P}}(0, r_{\text{car,cert}}^b(X)) \cap \partial \mathcal{P}_{\text{term}}^b(X) = \emptyset, \quad r_{\text{car}}^b(X) := \text{dist}_{\mathfrak{P}}(0, \partial \mathcal{P}_{\text{term}}^b(X)) \geq r_{\text{car,cert}}^b(X) > 0.$$

Hence

$$\mathcal{U}_{\text{car}}^b(X) = \text{Comp}_0(\mathfrak{P}^{b,X} \setminus \partial \mathcal{P}_{\text{term}}^b(X))$$

is a nonempty open same-scope carrier-stability region.

If $K \subset \text{PhysMod}_{E, \mathbb{V}}^b$ is compact and the closed face sets above assemble in a common finite-dimensional perturbation bundle so that the compact zero section Z_K is disjoint from their closed union $\mathfrak{F}_K$, then

$$r_{\text{car,cert}}^b(K) := \frac{1}{2} \text{dist}(Z_K, \mathfrak{F}_K) > 0, \quad \inf_{X \in K} r_{\text{car}}^b(X) \geq r_{\text{car,cert}}^b(K).$$

This compact-family clause uses compact/closed separation of the zero section, not compactness of terminal-candidate fibres.

Proof. The strict ready-child cell contains a relative neighbourhood of the accepted point, so its closed complement and the normal-form/update face have positive distance. Distance-to-complement guard slacks are Lipschitz and positive at the accepted point. For a perturbation smaller than the displayed half-minimum, the regularity and normal-form faces are avoided, the parent guard slacks remain positive, and every nonfrozen witness-typing guard remains positive. The exact proper-support flag has no finite erosion term because $q_a \equiv 1$ and $L_a^{\text{del}} = 0$.

By Lemma 7.3, no proper-support node, aggregate, relation, mixed candidate, or latent split eigensector can enter a compatible complete assignment inside this ball, whereas the parent remains live. Hence the ball is disjoint from every displayed component of $\partial \mathcal{P}_{\text{term}}^b(X)$. For a compact family, the zero section is compact and $\mathfrak{F}_K$ is closed and disjoint from it, so their distance is positive. No candidate-fibre compactness or unproved terminal-residual regularity is used. $\square$

Theorem 7.6 (P75-14: one-terminal-carrier robustness). *For every $\delta S \in \mathcal{U}_{\text{car}}^b(X)$,*

$$|\text{TermCar}_b(X_\delta)| = 1.$$

Proof. By construction of $\partial \mathcal{P}_{\text{term}}^b(X)$, the parent packet remains complete and one-apex compatible, every proper-support skeleton retains its exact deletion obstruction, and every full-support partition decoration maps to the same parent carrier orbit. These are the transported conclusions of Lemma 7.3, not a pointwise application of PCR-6 to a standing-breaking realization. Independent eligibility and Role-Occupancy Closure therefore leave exactly one live assignment orbit. No first-hit, candidate-fibre compactness, or path-continuity assumption is used. $\square$

Theorem 7.7 (P75-15: spectral splitting without carrier promotion). *Within $\mathcal{U}_{\text{car}}^b(X)$, a heavy multiplicity transition*

$$3 \longrightarrow 2 + 1 \longrightarrow 1 + 1 + 1$$

changes the internal standing-sector partition but not terminal carrier cardinality. In particular,

$$\text{heavy spectral splitting} \Rightarrow \text{terminal carrier splitting}.$$

A second terminal carrier requires failure of a transported deletion obstruction, loss of a witness-typing guard, or escape through the declared regularity/update boundary.

Proof. Theorem 4.3 determines the spectral partition from the traceless heavy tensor. All corresponding node and aggregate types are already present in the finite status-free terminal atlas. Spectral splitting changes only the partition decoration of the full-support parent core, and all parent decorations map to the same carrier orbit. Theorem 7.6 shows that every proper-support type retains its exact deletion obstruction throughout $\mathcal{U}_{\text{car}}^b(X)$. Splitting therefore changes the spectral stratum without promoting a terminal bearer. $\square$

7.1 Deletion-essential support matrix

The accepted carrier core requires the following support classes simultaneously:

Support removed	First terminal consequence
one light spectral sector	loss of complete light propagation/boundary reconstruction
heavy aggregate P_H	loss of D3 threshold, heavy response, D5 matching/protection, and PCR-7 structural mass-support/response record
light-heavy incidence	loss of the common charged-current closure and source matching path
charged-current boundary incidence	failure of preparation/detection reuse
D4 population/background support	loss of the admitted joint population/transport record
D5 action/protector support	branch-specific target failure
common-cone relation	no one-apex simultaneous realization
continuation/matching map	failure of identity-preserving reuse across scale/threshold

Each row supplies an exact structural obstruction flag for at least one proper-support skeleton. The matrix also explains why a split heavy eigensector remains internal: receiving a distinct eigenvalue does not supply the other terminal obligations.

7.2 Full-support alternatives

A full-support alternative inside the fixed source point may differ by lawful factorization or basis choice. The trace-transport lemma carries the PCR-6 full-support inheritance argument through the declared cell: these differences collapse at terminal-carrier resolution unless they alter a complete continuation witness. If a full-support candidate changes a standing source value, it is a different realization in the PCR-5E continuum or an update, not a second bearer inside the fixed X_δ .

Remark 7.8 (Positive but target-relative radius). $r_{\text{car},\text{cert}}$ is not asserted to be universal over the unresolved parent neutrino universe. It is positive on the declared ready-child trace-transport cell; uniformity over a compact family requires the explicit closed-face separation in P75-13. Exiting that cell is a typed falsifier or update, not evidence that PCR-6 supplied a larger unproved radius.

8 Heavy mass support and structural resolution

Definition 8.1 (Structural mass-support partition). For a positive heavy spectrum m'_1, m'_2, m'_3 , define

$$\Pi_H^{\text{str}} \in \{\text{Deg3}, \text{Deg2Plus1}, \text{ThreeDistinct}\}$$

from exact eigenvalue equality. This partition is independent of detector resolution.

Definition 8.2 (Experimental mass-support disposition). On a fixed-cardinality fine core with $|S'_H| = |\widehat{S}_{H,p}| = k$, write the unordered observed structural support as $\widehat{S}_{H,p}$. For $A = \{a_1, \dots, a_k\}$ and $B = \{b_1, \dots, b_k\}$, use the bottleneck metric

$$d_B(A, B) = \min_{\sigma \in \mathfrak{S}_k} \max_{1 \leq i \leq k} |a_i - b_{\sigma(i)}|.$$

Equivalently on the real line, sort each support increasingly within that core and take the maximum coordinate difference. Let $r_m(p, \alpha)$ be the predeclared simultaneous radius satisfying

$$\Pr\{d_B(\widehat{S}_{H,p}, S'_H) \leq r_m(p, \alpha)\} \geq 1 - \alpha.$$

Write $\widehat{g}_{\text{str},p}$ for the least gap between distinct points of $\widehat{S}_{H,p}$. A structural splitting is *Resolved* when the reported radius- r_m intervals are disjoint, equivalently $\widehat{g}_{\text{str},p} > 2r_m$; it is *SplitButUnresolved* only when the true structural support points are distinct and the reported intervals actually overlap. Before the record, *ResolutionNotUniformlyCertified* means that the design-level premises do not guarantee disjoint reported

intervals for every covered record; it does not assert that a realized record will overlap. The remaining statuses are `GapMerged` when the heavy cluster is not separated from another target support and `ProtocolUpdate` when the mass-support coordinates leave the protocol support class. At a support-cardinality change the fine-core metric and sorting are not transported across strata: the declared coarser quotient and its compatible adjacent-strata union are used instead.

Definition 8.3 (Distinct structural mass-support gap). On the positive heavy domain define the multiplicity-erased structural mass-coordinate support

$$S'_H = \text{supp Spec}((K'_H)^{1/2}).$$

When $|S'_H| \geq 2$, define

$$g_{\text{str}} = \min\{|x - y| : x, y \in S'_H, x \neq y\}.$$

For $|S'_H| = 1$, the distinct-support gap is not applicable and no distinct-mass-support resolution test is performed. The points of S'_H , their partition, and g_{str} are structural mass-support data of the positive operator $(K'_H)^{1/2}$; they are not complex propagator poles z_i . Physical complex-pole multiplicity and pole continuation belong exclusively to the conditional P75-X7 construction and remain `NotReady` as an empirical VJ record.

Theorem 8.4 (P75-16: heavy mass-support and resolution regimes). *On the positive heavy domain with $\|E\|_2 < g_{LH}/2$, the heavy aggregate remains rank three and its exact structural partition is given by Theorem 4.3. If $|S'_H| \geq 2$ and*

$$g_{\text{str}} > 2r_m(p, \alpha),$$

the bottleneck match to the true support is unique on the simultaneous coverage event. For the reported intervals centered on $\widehat{S}_{H,p}$, the record-dependent condition

$$\widehat{g}_{\text{str},p} > 2r_m(p, \alpha)$$

is necessary and sufficient for pairwise disjointness. The source-level condition

$$g_{\text{str}} > 4r_m(p, \alpha)$$

is a uniform sufficient condition for that reported-interval conclusion, because the coverage event gives $\widehat{g}_{\text{str},p} \geq g_{\text{str}} - 2r_m$. If $0 < g_{\text{str}} \leq 4r_m(p, \alpha)$, the design status is only `ResolutionNotUniformlyCertified`; the realized record is called `SplitButUnresolved` only when its reported intervals overlap. For a $2 + 1$ spectrum g_{str} is the one gap between the multiplicity-erased doublet point and the singlet; for $1 + 1 + 1$ it is the least of three pairwise gaps. On the `Deg3` stratum $|S'_H| = 1$, so no distinct-mass-support test is evaluated. At a cardinality-changing collision the conclusion is replaced by the declared coarse-quotient, compatible-adjacent-strata confidence set; no ordered label is carried through the collision.

Proof. Aggregate support and rank follow from Theorem 5.3. Exact multiplicity here is a property of the source structural spectral operator. On a fixed-cardinality fine core, $2r_m < g_{\text{str}}$ makes the bottleneck association with the true distinct support unique. The reported intervals, however, are centered on the estimated points: they are disjoint exactly when $\widehat{g}_{\text{str},p} > 2r_m$. On the coverage event the reverse triangle inequality gives $\widehat{g}_{\text{str},p} \geq g_{\text{str}} - 2r_m$, so $g_{\text{str}} > 4r_m$ is sufficient uniformly over every covered record. Repeated eigenvalues were erased before the metric was formed, so they do not spuriously force the minimum to zero on the $2 + 1$ stratum. At a cardinality change there is no cross-stratum matching, and the coarse quotient supplies the set-valued result. Overlapping confidence intervals do not imply equality of the underlying structural support points, so experimental nonresolution cannot restore structural degeneracy. No step of this argument computes or counts complex propagator poles. $\square$

Remark 8.5 (Lifetime is separate). Structural mass degeneracy, mass-support resolution, complex-pole structure, and decay width are distinct coordinates. A perturbation can split structural masses without fixing lifetimes, or change widths without lifting structural mass degeneracy, unless an independently licensed bridge links them.

9 Fibrewise invariant core and maximal covariant structural packet

Definition 9.1 (Certified structural robustness domain and typed support intervals). Fix $b \in \mathcal{B}_{7.5}^{\text{ready}}$ and $X \in \text{PhysMod}_{E,VJ}^b$. Write $K_H(X) = \kappa_H(X)I_3$, with $\kappa_H(X) = H_\lambda^2$, and define the structural mass-squared interval

$$J_{H,K}^{b,X} = \left[\kappa_H(X) - \frac{g_{LH}(X)}{2}, \kappa_H(X) + \frac{g_{LH}(X)}{2} \right] \subset (0, \infty).$$

Its structural mass-coordinate image is the separately typed interval

$$J_{H,m}^{b,X} = \{ \sqrt{\kappa} : \kappa \in J_{H,K}^{b,X} \}.$$

Let

$$\mathcal{D}_{\text{str}}^b(X) = \mathcal{U}_{\text{car}}^b(X) \cap \{ \|E\|_2 < g_{LH}(X)/2 \} \cap \{ \eta_V < s_E \} \cap \{ \text{complete block isometry} \},$$

with the source- and target-update loci removed and with the positive heavy chart retained. Weyl's bound gives

$$\text{Spec}(K'_H|_{\lim P'_H}) \subset \text{int } J_{H,K}^{b,X}.$$

This structural interval is defined before detector design. A later protocol may cover its structural mass-coordinate image, but no protocol support interval is used to prove the present section. Neither interval is thereby identified with a physical complex-pole mass.

Definition 9.2 (Complete structural-output trace and maximal covariant information class). Let

$$\Theta_{\text{pert}}^{b,X} = (\chi_{\text{src}}, \tau_{\text{phys}}, \Sigma_{\text{term}}, O_{\text{CCST}}, \text{Fail}, \text{Update})$$

be the complete semantic perturbation trace of [Theorem 3.5](#). Define

$$\mathfrak{U}_{\text{full}}^{b,X} := \pi_{\text{str}}^{b,X} \Theta_{\text{pert}}^{b,X},$$

where $\pi_{\text{str}}^{b,X}$ retains every structural, structural-record, terminal-certificate, and theorem-localization output authorized by the frozen PCR-7.5 target. In particular, it retains the quantitative P75-13 status-free support skeleton, partition decoration, trace-transport cell, exact deletion flags, explicit face slacks, and certified radius, not only its terminal status label.

By [Lemma 3.4](#), this projection acts on the constrained joint-image codomain $C_{\Theta}^{b,X}$, not on a product of independently marginalized fields. Consequently, whenever a frozen structural output uses an E-S1 cross-interface compatibility or higher-arity incidence coordinate, $\pi_{\text{str}}^{b,X}$ retains that whole typed relational coordinate (and its declared legs) in $\mathfrak{U}_{\text{full}}^{b,X}$.

The source identifier, child tag, fixed exact equations, and any source field having no licensed displacement are typed base data. They are not reclassified as varying outputs merely to manufacture a maximality counterexample.

For finite target-authorized packets $\mathcal{A}, \mathcal{B}$, write $\mathcal{A} \leq_{\text{cov}} \mathcal{B}$ when a target-lawful natural map F satisfies $\mathcal{A} = F \circ \mathcal{B}$. Mutual determination defines $\mathcal{A} \sim_{\text{fd}} \mathcal{B}$. The fibrewise maximal covariant information class is

$$O_{\text{cov}}^{b,X} := [\mathfrak{U}_{\text{full}}^{b,X}]_{\sim_{\text{fd}}}.$$

This definition takes maximality from the complete trace itself; no smaller displayed coordinate tuple is silently assumed to reconstruct it.

Definition 9.3 (Lawful quotient and heavy/CCST reduct). The lawful presentation groupoid used below is

$$\mathbf{G}_b^{\text{law}} = \mathbf{G}_{\text{src},b} \times U(3)_H \times G_{\ell,b}^{\text{pres}}.$$

Here $U(3)_H$ acts simultaneously on the heavy mass and coupling coordinates, while $G_{\ell,b}^{\text{pres}}$ contains only permitted active-flavour rephasings and presentation arrows that preserve the physical e, μ, τ labels. An arbitrary physical rotation mixing those labels is a standing-bearing perturbation, not skin.

For $Y = X_{\delta} \in \mathcal{D}_{\text{str}}^b(X)$, define the joint heavy orbit

$$\mathcal{J}_H^{b,X}(Y) = \left[\begin{array}{c} (P_H, K_H|_{\lim P_H}, V_H|_{\lim P_H}); \\ (P'_H, K'_H|_{\lim P'_H}, V'_H|_{\lim P'_H}) \end{array} \right]_{\mathbf{G}_b^{\text{law}}},$$

and

$$R_{\text{cl}}(Y) = \Delta'_{\text{act}}(Y) - \Omega'_H(Y), \quad \Omega'_H(Y) = V'_H(Y)V'_H(Y)^{\dagger}.$$

Let $\Sigma_{\text{term}}^{\text{cert}}$ denote the finite terminal signature together with the P75-13 support-skeleton, parent-orbit, transported deletion-witness, explicit-face, and certified-radius record. Let $\mathcal{D}_b$ be the finitely presented child-qualified D2/D5 coordinate tuple; its values may vary continuously.

The displayed heavy/CCST packet

$$\mathfrak{U}_{H/7}^{b,X} = (\Sigma_{\text{term}}^{\text{cert}}, \mathcal{J}_H^{b,X}, R_{\text{cl}}, \mathcal{D}_b, d(\Delta'_{\text{act}}), q(\Delta'_{\text{act}}), d(\Omega'_H), q(\Omega'_H), \text{StructuralMassSupport})$$

is a lawful reduct

$$\omega_{H/7}^{b,X} : \mathfrak{U}_{\text{full}}^{b,X} \longrightarrow \mathfrak{U}_{H/7}^{b,X}.$$

It is not identified with the complete trace unless a separate reconstruction theorem supplies an inverse.

Definition 9.4 (Distinct-support block coupling weights and collision rule). Let

$$\text{supp Spec}(K'_H|_{\lim P'_H}) = \{\kappa_a : a = 1, \dots, k_\pi\}, \quad k_\pi \in \{1, 2, 3\}.$$

For each distinct spectral value define the canonical block projector and structural coupling weight

$$E'_a = \mathbf{1}_{\{\kappa_a\}}(K'_H), \quad W'_a = V'_H E'_a V'^{\dagger}_H.$$

The packet $\{(\kappa_a, E'_a, W'_a)\}_{a=1}^{k_\pi}$ is unordered and is taken modulo the residual group

$$G_3 = U(3), \quad G_{2+1} = U(2) \times U(1), \quad G_{1+1+1} = U(1)^3.$$

At a structural mass-support collision, the colliding projectors and coupling weights are replaced by their sums. The matrices W'_a are not Laurent residues of a non-Hermitian propagator pole. No ordered eigenvector or rank-one projector inside a degenerate block is transported through the collision, and no physical complex-pole continuation is inferred.

Definition 9.5 (Pre-frozen variation-certified feature registry). Before invariant values are inspected, freeze a finite registry $\mathcal{F}_{\text{var}}^{b,X}$ of finitely presented feature factors. Continuous factor values are allowed. A factor is placed in the varying part of the registry only when it carries a certificate

$$\text{VarCert}_i^{b,X} = (\delta S_i, \text{Status}(\delta S_i), \varepsilon_i, \Theta_{\text{pert}}^{b,X}(\delta S_i))$$

with status `SameScopeTensor` or `StandingBreakingTensor`, and such that every $0 < |\varepsilon| < \varepsilon_i$ remains in $\mathcal{D}_{\text{str}}^b(X)$ along that declared direction.

The registered invariant tests are

$$\text{OneCarrier}, \quad \text{ClosureZero}, \quad \text{RankOmega3}, \quad \text{RankP3}, \quad \text{PositiveHeavy}, \quad \text{SupportIn } J_{H,K}.$$

The possible variation factors are the heavy centroid, traceless heavy spectrum, aggregate projector, Gram scale, Gram anisotropy, physical flavour frame, distinct-support structural coupling-weight packet, terminal decoration, and inherited PCR-7 structural record. A factor without a source-authorized variation certificate remains fixed base data; no predicate is excluded by an unavailable direction. Heavy or active presentation-basis labels have already been removed by $\mathbf{G}_b^{\text{law}}$ and are not feature tests.

For a block-isometric charged-current direction, the certificate must include

$$0 < \Omega'_H < I_3, \quad N' = (I_3 - \Omega'_H)^{1/2} Q, \quad Q \in U(3),$$

and must trace this completion back to a `SameScopeTensor` source-normal-form coordinate. The algebraic square root alone does not establish source authorization.

Definition 9.6 (Exact covariant law packet). Definitional and operator consequences are retained separately from the finite invariant-feature registry:

$$\begin{aligned} \mathcal{L}_{\text{cov}}^{b,X} = \{ & K'_H = K'^{\dagger}_H, \quad \Omega'_H = V'_H V'^{\dagger}_H \geq 0, \quad R_{\text{cl}} = 0, \\ & \sum_a E'_a = P'_H, \quad \sum_a W'_a = \Omega'_H, \\ & \tfrac{3}{2}\rho_K \leq \text{diam Spec}(T) \leq 2\rho_K, \\ & \Omega'_H \text{ is scalar} \iff \text{the singular values of } V'_H \text{ coincide,} \\ & \text{distinct-support block labels are unordered and coarsen by summation at collision} \}. \end{aligned}$$

On the full-rank charged-current domain, this scalarity law is equivalently positive universality: there exists $\delta' > 0$ such that $\Omega'_H = \delta' I_3$, precisely when the three singular values are equal and nonzero. Adjoining a determinant, minimum eigenvalue, spectral moment, anisotropy invariant, structural mass-support partition, or other lawful function of the complete trace does not enlarge $\mathcal{O}_{\text{cov}}^{b,X}$.

Theorem 9.7 (P75-17: fibrewise maximal invariant subpacket and covariant envelope). For every fixed $b \in \mathcal{B}_{7.5}^{\text{ready}}$ and $X \in \text{PhysMod}_{E,V}^b$, the maximal invariant subpacket relative to the pre-frozen variation-certified registry is

$$\boxed{\begin{aligned} \mathcal{O}_{\text{core}}^{b,X} = \{ & |\text{TermCar}_b| = 1, \quad R_{\text{cl}} = 0, \quad \text{rank } \Omega'_H = 3, \\ & \lambda_{\min}(\Omega'_H) \geq (s_E - \eta_V)^2 > 0, \quad \text{rank } P'_H = 3, \\ & \text{Spec}(K'_H|_{\lim P'_H}) \subset J_{H,K}^{b,X} \}. \end{aligned}}$$

Functional consequences of these entries are not duplicated as independent features. The fibrewise maximal target-qualified covariant packet is

$$\mathcal{O}_{\text{cov}}^{b,X} = \left[\pi_{\text{str}}^{b,X} \Theta_{\text{pert}}^{b,X} \right]_{\sim_{\text{fd}}},$$

and it carries the exact relation packet $\mathcal{L}_{\text{cov}}^{b,X}$. The heavy/CCST tuple $\mathfrak{A}_{H/7}^{b,X}$, including the unordered distinct-support structural coupling weights, is a named reduct of this complete envelope.

Proof. One-carrier stability is Theorem 7.6; closure is Theorem 6.4; the Gram rank and minimum-eigenvalue bound are Theorem 6.1; aggregate rank and the Weyl-certified mass-squared support are Theorem 5.3. These conclusions hold pointwise on the declared fibrewise domain.

For covariant maximality, let Q be any target-authorized structural output packet. By P75-02 and the explicit inverse regrouping in Lemma 3.4, Q factors through the complete semantic trace without discarding an E-S1 cross-interface or higher-arity coordinate. Because $\pi_{\text{str}}^{b,X}$ is the target-frozen projection retaining every structural output field, there is a lawful map F_Q with

$$Q = F_Q \circ \mathfrak{A}_{\text{full}}^{b,X}.$$

Hence $Q \leq_{\text{cov}} \mathcal{O}_{\text{cov}}^{b,X}$. No reconstruction from the smaller heavy/CCST reduct is used.

For invariant-feature maximality, consider only factors admitted to $\mathfrak{F}_{\text{var}}^{b,X}$. At fixed b, X , every certified first-difference direction has its own pointwise $\varepsilon_i > 0$:

Certified varying factor	Required source-authorized witness	Core retained for $0 < \varepsilon < \varepsilon_i$
Heavy centroid	Existing common-mass coefficient $\delta K_H = \varepsilon I_3$, traced as SameScopeTensor	Carrier, closure, ranks, positivity, and support containment.
Traceless spectrum	Target-admitted standing comparison $\delta K_H = \varepsilon \text{diag}(1, -1, 0)$, traced as StandingBreakingTensor	The same core, while degeneracy, spread, and partition vary.
Aggregate projector	A certified existing light-heavy mass/matching coordinate whose physical image has a nonzero off-block component	The rank-three aggregate rotates below the half-gap bound. Without this certificate the projector is fixed base data.
Gram scale	A certified isotropic boundary-coupling direction together with the displayed block-isometric N' completion	Closure and rank persist while the Gram scale and $\lambda_{\min}$ vary.
Gram anisotropy or physical flavour frame	A certified anisotropic boundary-coupling direction, not a presentation rotation, together with the SameScopeTensor completion trace	Closure and full rank persist while anisotropy or physical orientation varies.
Terminal decoration	A certified existing matching, continuation, or population coordinate below the P75-13 terminal margin	The complete bearer remains unique. Without such a coordinate the decoration stays in the base.
Distinct-support coupling weights and PCR-7 reduct	The preceding certified mass/coupling directions, followed by the fixed reduct maps	Their values vary lawfully inside $\mathcal{O}_{\text{cov}}^{b,X}$ and do not define additional invariant entries.

The registry contains no other varying factor. Every non-displaceable source field was placed in the typed base before the test, and every presentation basis label was removed by the lawful groupoid. Thus no registered factor can be appended as an additional constant invariant, while all varying values and exact operator laws remain in $\mathcal{O}_{\text{cov}}^{b,X}$ and $\mathcal{L}_{\text{cov}}^{b,X}$, respectively. This is maximality relative to the pre-frozen finite feature registry, not a claim that the algebra of all Boolean functions on a continuous trace is finite. $\square$

Lemma 9.8 (Lawful transport of variation-certified registries). *Let $u : X \rightarrow X'$ be a lawful arrow in the fixed-child groupoid $\mathbf{G}_b^{\text{law}}$. The induced redescription carries the certified structural domain isomorphically,*

$$u_{\#} : \mathcal{D}_{\text{str}}^b(X) \xrightarrow{\cong} \mathcal{D}_{\text{str}}^b(X'),$$

including the transported P75-13 trace cell, its face certificate, and every source-authorized perturbation direction. In particular, every certificate

$$\text{VarCert}_i^{b,X} = (\delta S_i, \text{Status}(\delta S_i), \varepsilon_i, \Theta_{\text{pert}}^{b,X}(\delta S_i))$$

has the transported certificate

$$u_{\#} \text{VarCert}_i^{b,X} = (u_{\#} \delta S_i, \text{Status}(u_{\#} \delta S_i), \varepsilon_i^u, \Theta_{\text{pert}}^{b,X'}(u_{\#} \delta S_i)),$$

where $\varepsilon_i^u > 0$ is the image of the certified interval in the transported normal-form chart (and equals ε_i for the canonical norm-preserving representatives),

$$\text{Status}(u_{\#} \delta S_i) = \text{Status}(\delta S_i), \quad \Theta_{\text{pert}}^{b,X'}(u_{\#} \delta S_i) = u_{\#} \Theta_{\text{pert}}^{b,X}(\delta S_i).$$

Thus a feature factor is variation-admitted at X if and only if its transported factor is variation-admitted at X' . The resulting bijection of frozen registries induces a lattice isomorphism

$$u_{\#}^{\text{inv}} : \text{Pred}_{\text{reg}}(\mathcal{F}_{\text{var}}^{b,X}) \xrightarrow{\cong} \text{Pred}_{\text{reg}}(\mathcal{F}_{\text{var}}^{b,X'}),$$

obtained by transporting each registered predicate and extending to finite conjunctions. Under this isomorphism the maximal invariant cores commute:

$$u_{\#}^{\text{inv}}(\mathcal{O}_{\text{core}}^{b,X}) = \mathcal{O}_{\text{core}}^{b,X'}.$$

Proof. P75-03 transports source-normal-form directions under lawful source isomorphisms, P75-04 makes their status invariant under lawful redescription, and P75-02 makes the complete semantic trace natural. The simultaneous heavy and active presentation action also transports the normed structural chart, the block-isometry class, the spectral-gap and rank inequalities, and the P75-13 terminal certificate. Hence the image of each certified punctured interval remains inside the transported structural domain; the inverse arrow u^{-1} proves the converse and preserves the pre-value-inspection admission rule. This gives the registry bijection and therefore the displayed predicate lattice isomorphism. One-carrier status, closure, ranks, positivity, and support containment are all lawful-invariant predicates, with $J_{H,K}^{b,X}$ transported to $J_{H,K}^{b,X'}$. Transporting the six generators of Theorem 9.7 therefore gives exactly the six generators at X' , proving the final identity. $\square$

Corollary 9.9 (Deepest stable fibrewise relation). *Within each ready-child fibre, $\Delta'_{\text{act}} = \Omega'_H$ together with positive rank-three heavy weight survives perturbations that change structural mass-support multiplicity, Gram anisotropy, or other certified covariant coordinates. The varying joint mass-coupling and terminal-certificate information is retained in $\mathcal{O}_{\text{cov}}^{b,X}$, not discarded from the endpoint.*

Remark 9.10 (Protocol extension does not prove P75-17). P75-17 uses the complete structural-output projection and the inherited PCR-7 structural reduct. Perturbed response kernels, detector mass-support disposition, confidence transport, and same-record carrier/reference core selection are added by P75-19–P75-21 and are not premises of either fibrewise maximality statement.

10 Cross-child common reduct and fibrewise naturality

Let $\mathbf{G}_{\text{VJ}}^{\text{com}}$ denote the groupoid of the vectorlike source, carrier, heavy operator, boundary Gram, and PCR-7 structural-record coordinates shared by the accidental and exact-global ready children. PCR-5E supplies forgetful functors

$$p_b : \mathbf{G}_b^{\text{law}} \longrightarrow \mathbf{G}_{\text{VJ}}^{\text{com}}, \quad b \in \{b_{\text{acc}}, b_{\text{glob}}\},$$

which erase only the child-qualified D2/D5 target decoration and its child-only continuation fields.

Definition 10.1 (Matched ready-child realizations). A pair $(X_{\text{acc}}, X_{\text{glob}})$ is cross-child matched only when

$$p_{\text{acc}}(X_{\text{acc}}) = p_{\text{glob}}(X_{\text{glob}}) = x$$

for one common VJ source point x . Perturbations are compared only when their complete traces have the same image after the common forgetful projection. Arbitrary realizations from the two children are not declared comparable merely because both are ready.

Definition 10.2 (Typed common reduct maps). For a matched pair, let $\mathfrak{A}_{\text{full}}^{\text{com},x}$ be the complete common structural-output trace obtained by retaining precisely the fields typed over the shared VJ source base x . The common structural-output projection is a functor on target-authorized, field-typed representative packets. Its action on the canonical complete-trace representative is

$$F_b^{\text{cov}} : \mathfrak{A}_{\text{full}}^{b,X_b} \longrightarrow \mathfrak{A}_{\text{full}}^{\text{com},x}, \quad b \in \{b_{\text{acc}}, b_{\text{glob}}\}.$$

It retains the common terminal carrier, joint heavy mass-coupling orbit, block-isometry residual, common structural-record reduct, and their exact law packet, while forgetting $\mathcal{D}_b$ and every child-only continuation coordinate.

This typed projection preserves functional-determination equivalence:

$$\mathcal{A} \sim_{\text{fd}} \mathcal{A}' \implies F_b^{\text{cov}}(\mathcal{A}) \sim_{\text{fd}} F_b^{\text{cov}}(\mathcal{A}').$$

Consequently it descends to the well-defined quotient map

$$[F_b^{\text{cov}}] : \mathcal{O}_{\text{cov}}^{b,X_b} \longrightarrow \mathcal{O}_{\text{cov}}^{\text{com},x}, \quad [\mathcal{A}]_{\sim_{\text{fd}}} \mapsto [F_b^{\text{cov}}(\mathcal{A})]_{\sim_{\text{fd}}}.$$

Separately, let $\text{Pred}_{\text{reg}}(\mathfrak{F})$ denote the finite lattice of conjunctions of predicates in a pre-frozen variation-certified feature registry $\mathfrak{F}$. Retaining only predicates well typed over x defines

$$F_b^{\text{inv}} : \text{Pred}_{\text{reg}}(\mathfrak{F}_{\text{var}}^{b,X_b}) \longrightarrow \text{Pred}_{\text{reg}}(\mathfrak{F}_{\text{var}}^{\text{com},x}).$$

On generators, F_b^{inv} sends a common typed predicate to its common counterpart and a child-only predicate to the top predicate; it then extends to finite conjunctions. Thus the common invariant packet is the type-correct logical meet

$$\mathcal{O}_{\text{core}}^{\text{com},x} := F_{\text{acc}}^{\text{inv}}(\mathcal{O}_{\text{core}}^{\text{acc},X_{\text{acc}}}) \wedge F_{\text{glob}}^{\text{inv}}(\mathcal{O}_{\text{core}}^{\text{glob},X_{\text{glob}}})$$

inside the common pre-frozen feature registry.

Theorem 10.3 (P75-18: fibrewise naturality and cross-child common reduct). *The assignments*

$$X_b \mapsto \mathcal{O}_{\text{cov}}^{b,X_b}, \quad X_b \mapsto \mathcal{O}_{\text{core}}^{b,X_b}$$

are natural under lawful arrows within their respective child groupoids. For the invariant assignment, naturality is taken in the predicate lattices through the registry-transport isomorphisms $u_{\#}^{\text{inv}}$ of Lemma 9.8; it is not an identification of independently chosen realization-dependent registries. For matched ready-child realizations and matched common perturbations, the common-reduct cospan

$$\mathcal{O}_{\text{cov}}^{\text{acc},X_{\text{acc}}} \xrightarrow{[F_{\text{acc}}^{\text{cov}}]} \mathcal{O}_{\text{cov}}^{\text{com},x} \xleftarrow{[F_{\text{glob}}^{\text{cov}}]} \mathcal{O}_{\text{cov}}^{\text{glob},X_{\text{glob}}}$$

commutes with every lawful common-source arrow.

The meet $\mathcal{O}_{\text{core}}^{\text{com},x}$ contains exactly the common registered conclusions

$$|\text{TermCar}| = 1, \quad \Delta'_{\text{act}} = \Omega'_H, \quad \text{rank } \Omega'_H = 3, \quad \lambda_{\min}(\Omega'_H) \geq (s_E - \eta_V)^2 > 0, \quad \text{rank } P'_H = 3,$$

together with the child-matched structural mass-squared support containment. The complete child envelopes remain separate functors. There is no cross-child natural map that transports the accidental LNV coordinate into the exact-global protector coordinate or conversely.

Proof. For the covariant envelope, fibrewise naturality follows from P75-02 semantic-trace invariance, P75-03 normal-form naturality, and the fact that the quotient $\mathbf{G}_b^{\text{law}}$ acts simultaneously on every retained source, operator, terminal, and record field. For the invariant core, Lemma 9.8 transports every variation direction together with its positive interval, status, and semantic trace, preserves factor admission, and supplies the predicate-lattice isomorphism under which $\mathcal{O}_{\text{core}}$ commutes. P75-17 supplies the fibrewise maximal packets and is not assumed to have already proved cross-child agreement.

For a matched pair, PCR-5E identifies the common vectorlike source, terminal carrier support, heavy operator class, boundary Gram structure, and PCR-7 structural packet before D2/D5 target decoration is added. The representative projection F_b^{cov} is fieldwise and commutes with lawful source arrows. More explicitly, if $\mathcal{A} = f(\mathcal{A}')$

and $\mathcal{A}' = g(\mathcal{A})$ are lawful mutual-determination maps, their typing over the shared source gives induced common maps $\bar{f}, \bar{g}$ satisfying

$$F_b^{\text{cov}} \circ f = \bar{f} \circ F_b^{\text{cov}}, \quad F_b^{\text{cov}} \circ g = \bar{g} \circ F_b^{\text{cov}}.$$

The two common images therefore mutually determine one another. This proves descent to $[F_b^{\text{cov}}]$, rather than assuming a map on equivalence classes.

Matched common perturbations have equal representative-level common images by definition, so the two descended covariant maps meet at the same common trace object with the same exact operator laws. The invariant map F_b^{inv} has a different domain: it acts only on the pre-frozen registered predicate lattice. The listed invariant conclusions use only common registered predicates and the feature-specific P75-09–P75-14 bounds. Their two images therefore agree, and their logical meet is the displayed core packet.

The accidental nondegenerate LNV amplitude interval and the exact-global zero subfibre with its continuation-stable protector are differently typed target coordinates. Forgetting them is lawful; identifying or transporting them is not. Hence the cospan induced by $[F_b^{\text{cov}}]$ is natural, while no cross-child arrow exists between the complete decorated envelopes. $\square$

Status 10.4 (Cross-child endpoint). The exact status is

$$\text{FibrewiseMaximalPackets} \oplus \text{CrossChildCommonReduct} \oplus \text{TargetDependentD2D5Fibres},$$

not equality or direct transport of the complete child packets.

11 Perturbed CCST protocol and detector confidence transport

11.1 Identifiability-certified protocol domain

Fix compact bounds

$$0 \leq \rho_K \leq \bar{\rho}_K, \quad 0 \leq \eta_V \leq \bar{\eta}_V < s_E, \quad \|E\|_2 \leq \bar{e} < g_{LH}/2,$$

chosen so that all perturbed heavy mass-support coordinates remain inside the tagged parent phase space and all source/analyzer cells remain strictly applicable. Let J_H^{rob} be the resulting compact structural mass-support set. The protocol is complete only at its frozen finite binning, response-kernel class, nuisance family, and control design. Mass-support or coherent-response structure below that declared resolution is not silently absorbed: if it changes the retained record, its status is `ProtocolUpdate`.

For each structural mass-support partition $\pi \in \{3, 2 + 1, 1 + 1 + 1\}$, use the lifted datum and residual quotient Ξ_π defined once in [Appendix E](#). Its independent physical/structural coordinates include the distinct mass-support locations, the active-space mass-support weights $W_a = V_H' P_a V_H'^\dagger$, and the fixed d/q reduct. The nuisance-indexed families $D_\pi^{(j)}(\theta)$ and $V_H' D_\pi^{(j)}(\theta) V_H'^\dagger$ are frozen protocol metadata evaluated inside $\mathcal{R}_{\pi,p}$; they are not coordinates of Ξ_π and their functional form is not reconstructed from held-out data. The fine split-stratum cores obey

$$g_{\text{str}} \geq g_{\pi, \min} > 0, \quad \pi \in \{2 + 1, 1 + 1 + 1\}.$$

If such a core is advertised as uniformly resolved at transported simultaneous mass-coordinate radius r_m , its frozen floor must also obey $g_{\pi, \min} > 4r_m$. The weaker inequality $g_{\text{str}} > 2r_m$ guarantees unique matching to true support points but not uniform disjointness of all reported intervals. Collision neighborhoods are covered instead by the adjacent coarser quotient. Their confidence output is set-valued over the compatible adjacent strata; no ordered eigenvector or mass-support label is transported through a collision.

Let $K_{\pi,p} \subset \Xi_\pi$ be a predeclared compact physical core, $\Theta_{\pi,p}$ its nuisance domain, and $C_{\pi,p}$ its calibration/control domain. Write

$$\mathcal{Z}_{\pi,p} := K_{\pi,p} \times \Theta_{\pi,p} \times C_{\pi,p}, \quad \mathcal{R}_{\pi,p} : \mathcal{Z}_{\pi,p} \longrightarrow \mathbb{R}^{N_{\text{prot}}}.$$

Equip $\mathcal{Z}_{\pi,p}$ with the predeclared product metric $d_{\mathcal{Z}}$. The common ordered protocol record space $\mathbb{R}^{N_{\text{prot}}}$ is frozen once for every core. Its coordinate order contains every declared observed bin, calibrated real and imaginary quadrature, nonnegative rate summary, and calibration/background control summary. Every retained coordinate is a genuinely defined observed summary under every hypothesis, and every response map predicts that same summary. There is no model-predicted padding. A physically observed bin for which one model predicts zero rate remains an ordinary observed coordinate with a zero model prediction.

Freeze one hypothesis-independent availability projection M_{obs} before held-out data and define

$$\mathcal{R}_{\pi,p}^{\text{obs}} := M_{\text{obs}} \mathcal{R}_{\pi,p}.$$

If a coordinate is unavailable, the protocol either assigns it zero weight through M_{obs} or quotients it as unconstrained, identically for every hypothesis. Such a coordinate contributes to neither image distance, lower-Lipschitz conditioning, nor inversion. A mask that records core identity or routing is forbidden. No core label, hypothesis label, structural tag, core-specific shortening, or reordering enters the data vector. All response norms, image distances, and inversions below are taken on the common effective observed-summary space $\text{ran } M_{\text{obs}}$. Positive acceptance guarantees that records can be collected; it does not make the signed/complex response positive and does not prove identifiability.

The no-vectorlike reference is not inferred from the carrier cores. It has an explicit compact augmented core and response

$$\mathcal{Z}_{\text{ref},p} := K_{\text{ref},p} \times \Theta_{\text{ref},p} \times C_{\text{ref},p}, \quad \mathcal{R}_{\text{ref},p} : \mathcal{Z}_{\text{ref},p} \longrightarrow \mathbb{R}^{N_{\text{prot}}},$$

with $\mathcal{R}_{\text{ref},p}^{\text{obs}} := M_{\text{obs}} \mathcal{R}_{\text{ref},p}$. Equip $\mathcal{Z}_{\text{ref},p}$ with its predeclared product metric $d_{\mathcal{Z}_{\text{ref}}}$. Here $K_{\text{ref},p}$ has physical zero heavy amplitude. The reference uses the same observed-summary ordering, calibration/background class, nuisance/control convention, hypothesis-independent availability projection M_{obs} , and held-out split, and it receives its own coverage-certified joint confidence region before physical projection.

A *StratifiedProtocolIdentifiabilityConditioningCertificate* (SPIC) must certify the global lower-Lipschitz inequality

$$\|\mathcal{R}_{\pi,p}^{\text{obs}}(z) - \mathcal{R}_{\pi,p}^{\text{obs}}(z')\|_2 \geq \kappa_{\pi,p} d_{\mathcal{Z}}(z, z'), \quad z, z' \in \mathcal{Z}_{\pi,p}, \quad \kappa_{\pi,p} > 0,$$

on every core. Full rank of a pointwise Jacobian is only a local diagnostic and is not a substitute for this inequality. The reference core independently obeys

$$\|\mathcal{R}_{\text{ref},p}^{\text{obs}}(z) - \mathcal{R}_{\text{ref},p}^{\text{obs}}(z')\|_2 \geq \kappa_{\text{ref},p} d_{\mathcal{Z}_{\text{ref}}}(z, z'), \quad z, z' \in \mathcal{Z}_{\text{ref},p}, \quad \kappa_{\text{ref},p} > 0.$$

Carrier conditioning does not imply this reference bound. For every pair of cores not jointly contained in one explicitly declared collision-coarsening neighborhood, SPIC also records positive image separation in the common record space

$$s_{(\pi,p),(\pi',p')} := \text{dist}(\mathcal{R}_{\pi,p}^{\text{obs}}(\mathcal{Z}_{\pi,p}), \mathcal{R}_{\pi',p'}^{\text{obs}}(\mathcal{Z}_{\pi',p'})) > 0.$$

The exemption is restricted to pieces analyzed together inside that same declared coarser quotient and returned as one compatible adjacent-strata union; generic adjacency does not waive the separator. Positive acceptance without these within-core and inter-core SPIC bounds is insufficient for a complete structural mass-support claim. For every carrier core, SPIC additionally records the explicit positive image separator

$$s_{(\pi,p),(\text{ref},p)} := \text{dist}(\mathcal{R}_{\pi,p}^{\text{obs}}(\mathcal{Z}_{\pi,p}), \mathcal{R}_{\text{ref},p}^{\text{obs}}(\mathcal{Z}_{\text{ref},p})) > 0$$

. A lower-Lipschitz certificate on a disjoint carrier/reference union does not replace this image separator. Thus the aggregate SPIC exported to P75-21 includes explicit reference separation, identifiability, and coverage; it is not inferred from carrier-only cores. SPIC also freezes a sampling-law certificate: either the retained trial summaries are independent under the declared design, or a specified martingale, mixing, or effective-sample-size condition supplies an explicit simultaneous tail bound. Bounded summaries without such a dependence certificate do not license finite-exposure concentration.

For an operator-norm Gram confidence radius ϵ_{op} , define the exact detector threshold

$$\eta_{\text{iso}}^*(\epsilon_{\text{op}}) := \sqrt{\frac{4}{3} s_E^2 - \epsilon_{\text{op}}} - s_E, \quad 0 \leq \epsilon_{\text{op}} < \frac{s_E^2}{3}.$$

Thus $\eta_{\text{iso}}^*(0) = s_E(2/\sqrt{3} - 1)$, and

$$\bar{\eta}_V < \eta_{\text{iso}}^*(\epsilon_{\text{op}}) \iff 2s_E \bar{\eta}_V + \bar{\eta}_V^2 + \epsilon_{\text{op}} < \frac{s_E^2}{3}.$$

This is a uniform detector-certification threshold over the predeclared protocol domain. It is smaller than, and logically independent of, the pointwise structural rank radius $\eta_V < s_E$.

Theorem 11.1 (P75-19: stratified mass-support protocol coverage and identifiability). *On every compact SPIC-certified protocol domain, the finite PCR-7 source/analyzer system extends to a finite protocol covering all declared heavy-mass-support and response bins in J_H^{rob} . The carrier and reference objects use the same calibration, background class, augmented nuisance/control coordinates, common ordered record space $\mathbb{R}^{N_{\text{prot}}}$, record quotient, and held-out split. SPIC exports one common inverse-set construction with hypothesiswise coverage for every declared carrier and reference core, the independent bound $\kappa_{\text{ref},p} > 0$, and an explicit positive image separator for every carrier/reference core pair. On each 3 core and each gap-separated fine $2 + 1$ or $1 + 1 + 1$ core, the protocol reconstructs the target-complete quotient datum Ξ_π jointly with nuisance and control coordinates, and therefore reconstructs the reduct*

$$(d(\Delta'_{\text{act}}), q(\Delta'_{\text{act}}), d(\Omega'_H), q(\Omega'_H), \text{MassSupport}),$$

with finite simultaneous confidence for its jointly inferred coordinates under whichever declared hypothesis is true. The reported physical confidence set is the projection of the joint confidence set, not a fit with nuisance coordinates held fixed.

Within a collision neighborhood the protocol returns the coarser-quotient confidence set together with the compatible adjacent-strata union. Unresolved overlapping mass-support components retain the declared complex coherent response; deleting the interference terms is not licensed as a common-propagator or incoherent-mixture approximation. Each $D_\pi^{(j)}$ is a frozen protocol response kernel, not a computation of the P75-X7 complex propagator poles z_t . Exact structural mass status Deg3 does not imply that the response kernel is common or scalar, and it does not determine physical-pole multiplicity. That multiplicity belongs to conditional P75-X7 and remains NotReady as an empirical VJ record. The theorem is protocol-complete only at the declared binning/response class and is an in-principle finite-record result, not a claim about an existing facility or published analysis.

Let $\epsilon_{\text{rec}}^{\text{det}}$ be the SPIC-certified deterministic record-space floor and let $\epsilon_{\text{rec}}^{\text{stat}}(n) \downarrow 0$ be the stochastic term certified by the frozen sampling law. Thus

$$r_{\text{rec}}(n) = \epsilon_{\text{rec}}^{\text{det}} + \epsilon_{\text{rec}}^{\text{stat}}(n).$$

Freeze the common observed statistic $\widehat{Y}_n \in \text{ran } M_{\text{obs}}$. For each declared carrier or reference core label a , abbreviate its augmented core and observed response by $\mathcal{Z}_a$ and $\mathcal{R}_a^{\text{obs}}$. For each $z_a \in \mathcal{Z}_a$, define

$$\mathcal{E}_a(z_a) := \left\{ \|\widehat{Y}_n - \mathcal{R}_a^{\text{obs}}(z_a)\|_2 \leq r_{\text{rec}}(n) \right\}, \quad \inf_a \inf_{z_a \in \mathcal{Z}_a} \Pr(\mathcal{E}_a(z_a)) \geq 1 - \alpha.$$

This is a uniform hypothesiswise guarantee under the data law indexed by the true (a, z_a) ; it does not assert that mutually exclusive carrier and reference truths hold simultaneously. Define every inverse confidence set from that same record ball:

$$\begin{aligned} C_{\pi,p}^{\text{joint}}(\widehat{Y}_n) &:= \left\{ z \in \mathcal{Z}_{\pi,p} : \|\mathcal{R}_{\pi,p}^{\text{obs}}(z) - \widehat{Y}_n\|_2 \leq r_{\text{rec}}(n) \right\}, \\ C_{\text{ref},p}^{\text{joint}}(\widehat{Y}_n) &:= \left\{ z \in \mathcal{Z}_{\text{ref},p} : \|\mathcal{R}_{\text{ref},p}^{\text{obs}}(z) - \widehat{Y}_n\|_2 \leq r_{\text{rec}}(n) \right\}. \end{aligned}$$

A predeclared record-space target $r_{\text{rec},\star}$ is attainable at finite exposure only when $r_{\text{rec},\star} > \epsilon_{\text{rec}}^{\text{det}}$ and the SPIC sampling-law bound supplies n with $r_{\text{rec}}(n) < r_{\text{rec},\star}$ for every such strict target. This is a radius in $\text{ran } M_{\text{obs}}$, not in parameter space.

Separately freeze carrier and reference parameter-diameter targets $r_{\mathcal{Z},\pi,p} > 0$ and $r_{\mathcal{Z},\text{ref},p} > 0$. Every nonempty inverse set formed from a record ball of radius $r_{\text{rec}}(n)$ obeys its corresponding bound

$$\text{diam}_{d_{\mathcal{Z}}} C_{\pi,p}^{\text{joint}}(\widehat{Y}_n) \leq \frac{2r_{\text{rec}}(n)}{\kappa_{\pi,p}}, \quad \text{diam}_{d_{\mathcal{Z},\text{ref}}} C_{\text{ref},p}^{\text{joint}}(\widehat{Y}_n) \leq \frac{2r_{\text{rec}}(n)}{\kappa_{\text{ref},p}}.$$

Finite-exposure attainment of the parameter targets therefore requires the strict floor guards

$$\frac{2\epsilon_{\text{rec}}^{\text{det}}}{\kappa_{\pi,p}} < r_{\mathcal{Z},\pi,p}, \quad \frac{2\epsilon_{\text{rec}}^{\text{det}}}{\kappa_{\text{ref},p}} < r_{\mathcal{Z},\text{ref},p},$$

under which the sampling-law bound supplies an n satisfying the same two inequalities with $\epsilon_{\text{rec}}^{\text{det}}$ replaced by $r_{\text{rec}}(n)$. For any non-coarsened pair of cores a, b with image separator $s_{a,b}$, write C_a^{joint} and C_b^{joint} for their same-record inverse sets. The attainable guard

$$r_{\text{rec}}(n) < \frac{1}{2} s_{a,b}$$

has the following same-record selection meaning: on $\mathcal{E}_a(z_a^{\text{true}})$, the set $C_a^{\text{joint}}(\widehat{Y}_n)$ contains z_a^{true} and $C_b^{\text{joint}}(\widehat{Y}_n) = \emptyset$; on $\mathcal{E}_b(z_b^{\text{true}})$, the roles reverse. In particular this applies to every carrier/reference pair. It is not a simultaneous comparison of two nonempty counterfactual confidence regions. Uniform reported-interval mass-support resolution separately requires its transported mass radius $r_m(n)$ to obey $r_m(n) < \frac{1}{4} g_{\text{str}}$, and Gram reconstruction has its own operator-norm guard in P75-20. None of these typed quantities may be scalar-compressed into another.

Proof. Strict source activity, positive charged-current singular values, interior parent phase space, and compactness give a finite cover with positive minimum production/analyzer acceptance. That establishes record availability only. The calibrated analyzers may have signed or complex interference response, whose conditioning is supplied separately by SPIC.

On each carrier core the global lower-Lipschitz inequality gives

$$d_Z(z, z') \leq \kappa_{\pi,p}^{-1} \|\mathcal{R}_{\pi,p}^{\text{obs}}(z) - \mathcal{R}_{\pi,p}^{\text{obs}}(z')\|_2,$$

so the inverse is single-valued and Lipschitz there. Two candidates in one record ball differ by at most $2r_{\text{rec}}(n)$, giving the displayed carrier parameter-diameter bound. The independent $\kappa_{\text{ref},p}$ inequality gives the analogous reference bound and is required for reference confidence transport. For separated cores a, b , if both inverse sets were nonempty, points z_a, z_b in them would satisfy

$$s_{a,b} \leq \|\mathcal{R}_a^{\text{obs}}(z_a) - \mathcal{R}_b^{\text{obs}}(z_b)\|_2 \leq 2r_{\text{rec}}(n),$$

contradicting the strict half-separator guard. On the coverage event for the true core, its true point belongs to its inverse set, so that set is nonempty and the incompatible set is empty. This proves the stated hypothesiswise selection claim. At a collision this conclusion is deliberately weakened only for the restricted pieces analyzed in the same coarser quotient: that quotient is used and the compatible adjacent fine-stratum projections are retained set-valuedly. Under the frozen sampling-law certificate, the applicable independent-trial or declared martingale/mixing/effective-sample-size tail bound shrinks $\epsilon_{\text{rec}}^{\text{stat}}(n)$, but it does not shrink the certified deterministic floor $\epsilon_{\text{rec}}^{\text{det}}$. Boundedness alone would not imply this concentration. The strict floor guards are exactly what allow a finite n to put the carrier and reference parameter diameters below their separately declared targets. Projection preserves the physical coverage of whichever hypothesis-indexed joint set is true; it does not manufacture simultaneous coverage of counterfactual models. The d/q packet is a fixed reduct of this complete response. Sub-bin structure outside the frozen response family invokes ProtocolUpdate rather than an unsupported refinement. [2, 14] $\square$

Theorem 11.2 (P75-20: factorized mass-support disposition and confidence transport). *Every certified protocol record carries logically separate coordinates*

MassSupportStatus $\in \{\text{Deg3}, \text{Resolved}, \text{SplitButUnresolved}, \text{GapMerged}, \text{ProtocolOutOfScope}\},$

ResolutionGuaranteeStatus $\in \{\text{UniformlyResolvedCertified}, \text{ResolutionNotUniformlyCertified}\},$

ResponseKernelStatus $\in \{\text{CommonScalarCertified}, \text{NonScalarCertified}, \text{ResponseKernelUnresolved}\},$

and

ReferenceStatus $\in \{\text{ReferenceSeparated}, \text{ReferenceNotCertified}\}.$

In particular, Deg3 is a structural mass-support statement and never supplies CommonScalarCertified or a physical complex-pole multiplicity. The phrase “unresolved but reference-separated” denotes a product status, not a structural mass-support label. Here ReferenceSeparated means that the common-record half-separator guard certifies hypothesiswise selection of the carrier core against its reference (or conversely under reference truth); it does not mean that two counterfactual regions from the same record are both nonempty. Failure of SPIC gives ProtocolOutOfScope, not Deg3 and not a fitted unresolved result.

The three resolution tests are distinct. The true-support condition $g_{\text{str}} > 2r_m$ gives unique support matching only. A realized record is Resolved exactly when its reported-center gap satisfies $\widehat{g}_{\text{str}} > 2r_m$. Uniform disjointness of all reported confidence intervals over the core requires $g_{\text{str}} > 4r_m$, equivalently $r_m < g_{\text{str}}/4$. Below that uniform threshold the guarantee status is ResolutionNotUniformlyCertified; if the realized reported intervals overlap, the record status is SplitButUnresolved. A realized disjoint record does not retroactively strengthen the missing uniform guarantee.

Under a carrier truth, let C_Ω be the Gram-coordinate projection of its selected carrier joint confidence set. On that true carrier hypothesis’s simultaneous coordinate-coverage event

$$\sup_{\widehat{\Omega} \in C_\Omega} \|\widehat{\Omega} - \Omega'_H\|_2 \leq \epsilon_{\text{op}},$$

the phase-insensitive reconstructed Gram set certifies positive rank three by diagonal dominance whenever

$$2s_E \bar{\eta}_V + \bar{\eta}_V^2 + \epsilon_{\text{op}} < \frac{s_E^2}{3}, \quad \text{equivalently} \quad \bar{\eta}_V < \eta_{\text{iso}}^*(\epsilon_{\text{op}}).$$

For every $\eta_V < s_E$, structural rank three remains an unconditional consequence of P75-10, independent of SPIC and detector exposure. The ideal Frobenius and pair-space reference gaps are likewise structural consequences of the rank bound and block isometry. Only finite-record rank certification and same-record hypothesiswise model/core selection are SPIC-conditional.

The detector certificate exports the three guards as the typed vector

$$\left(r_{\text{rec}} < \frac{1}{2} s_{(\pi,p),(\pi',p')}, \quad r_m < \frac{1}{4} g_{\text{str}}, \quad 2s_E \bar{\eta}_V + \bar{\eta}_V^2 + \epsilon_{\text{op}} < \frac{s_E^2}{3} \right).$$

The first entry applies to every pair requiring typed separation, including each carrier/reference pair; r_{rec} must also be attainable above $\epsilon_{\text{rec}}^{\text{det}}$. It certifies the hypothesiswise same-record selection rule of P75-19 and P75-21. The ideal pair-space gap of P75-21 is a separate structural result and supplies no additional detector-radius guard. These guards inhabit different metrics and are not scalar-compressed.

Proof. The structural mass-support coordinate follows from Theorem 8.4; response-kernel scalarity requires its own analyzer certificate because equal structural masses can have noncommuting dispersive, absorptive, or flavor-response structure. The ideal pair-space reference gap is structural, whereas the ReferenceSeparated record status is an independent axis proved from the common-record carrier/reference image separator, its attainable half-separator guard, and hypothesiswise SPIC coverage. Neither this disposition nor the frozen $D_\pi^{(j)}$ kernels compute the X7 complex poles.

For Gram reconstruction, write

$$\Omega'_H - \delta_E I_3 = (-s_E U) E_V^\dagger + E_V (-s_E U)^\dagger + E_V E_V^\dagger,$$

so

$$\|\Omega'_H - \delta_E I_3\|_2 \leq 2s_E \eta_V + \eta_V^2 \leq 2s_E \bar{\eta}_V + \bar{\eta}_V^2.$$

The confidence event and the triangle inequality give, for every $\widehat{\Omega} \in C_\Omega$,

$$\|\widehat{\Omega} - \delta_E I_3\|_2 \leq 2s_E \bar{\eta}_V + \bar{\eta}_V^2 + \epsilon_{\text{op}}.$$

If this uniform bound is below $s_E^2/3$, every diagonal entry is larger than the sum of the two possible off-diagonal magnitudes, so every matrix in the projected confidence set has a positive Gershgorin lower bound [13]. Solving the strict quadratic inequality gives exactly $\bar{\eta}_V < \eta_{\text{iso}}^*(\epsilon_{\text{op}})$. Outside this smaller uniform detector region, P75-10 still proves pointwise structural full rank for every actual $\eta_V < s_E$, while the finite protocol may need increased exposure, a stronger analyzer, or ProtocolUpdate. $\square$

12 Perturbation-stable prediction/reference separation

The no-vectorlike reference retains

$$\Delta_{\text{act}}^0 = 0, \quad \Omega_H^0 = 0, \quad \text{StructuralMassSupport}_H^0 = \emptyset$$

in the same instrument, calibration, nuisance, common ordered record space, and record quotient. This source-mass statement is not a claim about a physical complex-pole multiset.

On the ordered pair of Hermitian matrix coordinates used below, define

$$\|(A, B)\|_{\text{pair}} := (\|A\|_F^2 + \|B\|_F^2)^{1/2},$$

Theorem 12.1 (P75-21: ideal prediction/reference gap and certified-record model selection). *Let $\eta_V < s_E$ and assume block isometry. Then*

$$\|\Omega'_H\|_F \geq \sqrt{3} (s_E - \eta_V)^2$$

and

$$\boxed{\|(\Delta'_{\text{act}}, \Omega'_H)\|_{\text{pair}} \geq \sqrt{6} (s_E - \eta_V)^2.}$$

The boxed inequality is an ideal pair-space gap. It is not itself a detector metric or a confidence-radius guard.

Now let a compact protocol domain satisfy $\eta_V \leq \bar{\eta}_V < s_E$, and fix one declared carrier core $\mathcal{Z}_{\pi,p}$ and its explicit no-vectorlike reference core $\mathcal{Z}_{\text{ref},p}$. Suppose the aggregate response reduct carries the P75-19 SPIC in one common observed-record space, with

$$s_{(\pi,p),(\text{ref},p)} = \text{dist}(\mathcal{R}_{\pi,p}^{\text{obs}}(\mathcal{Z}_{\pi,p}), \mathcal{R}_{\text{ref},p}^{\text{obs}}(\mathcal{Z}_{\text{ref},p})) > 0.$$

Use the same observed statistic $\widehat{Y}_n$ and the two inverse sets $\mathcal{C}_{\pi,p}^{\text{joint}}(\widehat{Y}_n)$ and $\mathcal{C}_{\text{ref},p}^{\text{joint}}(\widehat{Y}_n)$ defined in P75-19. For $h \in \{(\pi, p), (\text{ref}, p)\}$, write $\mathcal{Z}_h$ and $\mathcal{R}_h^{\text{obs}}$ for the corresponding core and response. Their coverage is hypothesiswise:

$$\mathcal{E}_h(z_h) := \{\|\widehat{Y}_n - \mathcal{R}_h^{\text{obs}}(z_h)\|_2 \leq r_{\text{rec}}(n)\}, \quad \inf_h \inf_{z_h \in \mathcal{Z}_h} \Pr(\mathcal{E}_h(z_h)) \geq 1 - \alpha.$$

Whenever

$$r_{\text{rec}}(n) < \frac{1}{2} s_{(\pi,p),(\text{ref},p)},$$

the common record yields the pairwise selector

$$\begin{aligned} \mathcal{E}_{\pi,p}(z_{\pi,p}^{\text{true}}) &\implies z_{\pi,p}^{\text{true}} \in \mathcal{C}_{\pi,p}^{\text{joint}}(\widehat{Y}_n), & \mathcal{C}_{\text{ref},p}^{\text{joint}}(\widehat{Y}_n) &= \emptyset, \\ \mathcal{E}_{\text{ref},p}(z_{\text{ref},p}^{\text{true}}) &\implies z_{\text{ref},p}^{\text{true}} \in \mathcal{C}_{\text{ref},p}^{\text{joint}}(\widehat{Y}_n), & \mathcal{C}_{\pi,p}^{\text{joint}}(\widehat{Y}_n) &= \emptyset. \end{aligned}$$

Thus, under carrier truth the carrier core is retained and the incompatible reference core is rejected with the stated carrier-hypothesis coverage; under reference truth the converse holds with the stated reference-hypothesis coverage. The assertion never requires two counterfactual regions from the same record to be simultaneously nonempty. If

$$\epsilon_{\text{rec}}^{\text{det}} < \frac{1}{2} s_{(\pi,p),(\text{ref},p)},$$

the frozen SPIC sampling-law bound supplies a finite predeclared exposure attaining the record-space half-separator guard. For selection against a finite family of carrier cores, impose the same strict floor and finite-record guard using the minimum applicable carrier/reference separator. Without the aggregate SPIC and explicit reference-core certificate, or when the deterministic record floor reaches that half-separator, the ideal algebraic inequalities remain licensed but detector-record model selection does not follow from positive acceptance alone.

Proof. By Theorem 6.1, all three eigenvalues of Ω'_H are at least $(s_E - \eta_V)^2$. Hence its Frobenius norm is at least $\sqrt{3}$ times that value. By Theorem 6.4, $\Delta'_{\text{act}} = \Omega'_H$, giving the product-space factor $\sqrt{2}$. These ideal-space bounds do not use a detector inverse.

For the finite-record clause, consider carrier truth and its coverage event. The definition of $\mathcal{E}_{\pi,p}$ puts the true point in $\mathcal{C}_{\pi,p}^{\text{joint}}(\widehat{Y}_n)$. If the reference set were nonempty, choose z_{ref} in it. Then

$$s_{(\pi,p),(\text{ref},p)} \leq \|\mathcal{R}_{\pi,p}^{\text{obs}}(z_{\pi,p}^{\text{true}}) - \mathcal{R}_{\text{ref},p}^{\text{obs}}(z_{\text{ref}})\|_2 \leq 2r_{\text{rec}}(n),$$

contradicting the strict half-separator guard. The same argument with carrier and reference interchanged proves the reference-truth implication. The SPIC tail bound shrinks only $\epsilon_{\text{rec}}^{\text{stat}}(n)$; hence finite exposure crosses the half-separator exactly when the deterministic floor lies strictly below it. $\square$

Corollary 12.2 (Conditional robust prediction lock). *On any declared domain lying simultaneously in a P75-13 trace-transport cell, the block-isometric coupling region $\eta_V \leq \bar{\eta}_V < s_E$, and the full carrier-plus-reference SPIC domain, the common-record inverse selects the true carrier core against its no-vectorlike reference, or the reference against that carrier core, with the corresponding hypothesiswise coverage whenever the record-space half-separator guard holds. Independently, the ideal carrier/reference pair-space gap has the boxed structural lower bound. The number s_E is a coupling margin, not a universal protocol radius. The deepest experimentally robust statement is nonzero full-rank heavy charged-current weight exactly accounting for the active deficit, not exact structural mass degeneracy or any uncertified physical complex-pole multiplicity.*

13 Feature-specific robustness radii

Theorem 13.1 (P75-22: robustness-radius, certified-domain, and zero-radius taxonomy). *For each ready child and accepted realization carrying the declared P75-13 trace-transport cell and explicit face certificate, the deterministic robustness vector is*

$$\mathcal{R}_{7.5}(X) = (r_{\text{car}}(X), r_{\text{closure}}, r_{\text{subspace}}, r_{\text{rank}}, r_{\text{univ}}, r_{\text{deg}}, \text{ProtSPIC}),$$

with the following exact dispositions:

$$\begin{aligned}
r_{\text{car}}(X) &= \text{dist}_{\mathbb{P}}(0, \partial \mathcal{P}_{\text{term}}^b(X)) \geq r_{\text{car, cert}}^b(X) > 0 \quad \text{on that declared cell,} \\
r_{\text{closure}} &= \text{ExactOnBlockIsometryClass,} \\
r_{\text{subspace}} &= g_{LH}/2 \quad (\text{conservative operator-norm radius}), \\
r_{\text{rank}} &= s_E \quad (\text{sharp operator-norm radius in } V_H), \\
r_{\text{univ}}^{\text{unrestricted}} &= 0, \\
r_{\text{univ}}^{\text{isotropic}} &= \text{DomainLimitedExact} \quad \text{only on full-rank } \Omega'_H = \delta' I_3, \delta' > 0, \\
r_{\text{deg}}^{\text{unrestricted}} &= 0, \quad r_{\text{deg}}^{\text{central}} = \text{DomainLimitedExact,} \\
\text{Prot}_{\text{SPIC}} &= \text{CertifiedOnDeclaredSPICDomain,}
\end{aligned}$$

where the protocol entry exports the typed guard vector

$$\mathbf{m}_{\text{prot}} = \begin{pmatrix} \{\kappa_{\pi,p}\}, \quad \{\kappa_{\text{ref},p}\}, \\ \{s_{(\pi,p),(\pi',p')}/2 - r_{\text{rec}}(n)\}, \\ \{r_{Z,\pi,p} - 2r_{\text{rec}}(n)/\kappa_{\pi,p}\}, \\ \{r_{Z,\text{ref},p} - 2r_{\text{rec}}(n)/\kappa_{\text{ref},p}\}, \\ \{g_{\text{str},\pi}/4 - r_{m,\pi}\}, \\ s_E^2/3 - (2s_E\bar{\eta}_V + \bar{\eta}_V^2 + \epsilon_{\text{op}}) \end{pmatrix},$$

on the declared common-record-space SPIC domain, together with its positive acceptance, structural-collision coarsening, joint nuisance/control coverage, explicit reference core, and typed deterministic floors. The two r_Z entries are the lower-Lipschitz transports from record-space uncertainty to the carrier and reference parameter metrics; their finite-exposure attainability also requires the corresponding inequalities with $r_{\text{rec}}(n)$ replaced by $\epsilon_{\text{rec}}^{\text{det}}$. Each carrier/reference half-separator component in the record-space entry certifies hypothesiswise same-record selection: under carrier truth and its coverage event the carrier inverse contains truth and the reference inverse is empty, while under reference truth the roles reverse. On the block-isometric subdomain with $\eta_V \leq \bar{\eta}_V < s_E$, the ideal P75-21 pair-space gap is structural, independent of SPIC, and is not an additional detector-radius component. Here each difference is evaluated only in its own declared metric. This is a certified-domain status and margin vector, not a distance in an unspecified perturbation space. No single scalar radius faithfully replaces the typed result.

Proof. The carrier entry is Theorem 7.5; a chart lacking its exhibited trace-transport cell and deletion-witness identification is conditional/NotReady, and compact-family uniformity additionally requires a common perturbation bundle with compact-zero-section/closed-face separation. Closure is an exact class-conditional identity by Theorem 6.4; arbitrary violations of block isometry are target/source changes rather than a norm perturbation inside that class. The subspace entry is Theorem 5.3. The rank radius and sharpness are Theorem 6.1. The zero-radius universality and degeneracy statements follow from the explicit arbitrarily small anisotropic and traceless perturbations in Theorems 4.3 and 6.2, while exactness on the protected strata follows from Theorems 4.2 and 6.2. The protocol entry follows only on the already-declared SPIC domain of Theorems 11.1 and 12.1; those theorems provide typed record-space, record-to-parameter, structural mass-coordinate, and Gram-operator margins together with hypothesiswise same-record carrier/reference selection, but do not define a protocol perturbation metric or prove persistence of SPIC constants on an open perturbation ball. On block isometry with $\eta_V \leq \bar{\eta}_V < s_E$, the ideal P75-21 pair-space gap remains a separate structural inequality. Compactness, positive acceptance, or pointwise Jacobian rank alone is insufficient. Since the entries refer to different metrics, perturbation classes, and certified-domain statuses, scalar compression would erase authority information. $\square$

Status 13.2 (Structural interpretation). The exact heavy degeneracy is not “uncertain.” It is exact on the symmetry-preserving stratum and infinitesimally fragile against standing symmetry breaking. Rank three and carrier identity are more robust features governed by different boundaries.

14 Hierarchical falsifiers and theorem localization

Theorem 14.1 (P75-23: hierarchical falsifier and theorem-localization). *Future observations and source reconstructions admit the following first-failure localization:*

<i>Observed deviation</i>	<i>First theorem layer challenged</i>	<i>Not automatically falsified</i>
<i>The structural heavy-mass support splits</i>	<i>The observation exits and tests the central-stratum hypothesis used by P75-06; P75-07 then classifies the admitted noncentral split</i>	<i>rank three, closure, one carrier; the conditional P75-06 implication itself is not contradicted</i>
<i>Heavy weights become anisotropic</i>	<i>P75-11 flavor-universality layer</i>	<i>rank three, closure, one carrier</i>
<i>Heavy Gram matrix loses rank</i>	<i>P75-10 rank layer or child-source applicability</i>	<i>closure may remain on lower-rank block; carrier requires separate test</i>
$\Delta'_{\text{act}} \neq \Omega'_H$	<i>P75-12 complete block-isometry layer</i>	<i>mass splitting alone</i>
<i>Heavy weight vanishes</i>	<i>ready-child source/rank/nondecoupling layer</i>	<i>unrelated parent sectors</i>
<i>A second complete bearer appears</i>	<i>P75-13–15 terminal-promotion layer</i>	<i>earlier spectral perturbation estimates</i>
<i>The trace-transport cell is left, a proper-support deletion flag is lost, a witness-typing guard fails, or a second full-support orbit appears</i>	<i>P75-13/14 carrier trace-transport layer</i>	<i>PCR-6 closure at the unperturbed point; spectral, rank, and closure estimates whose premises remain valid</i>
<i>In a 2 + 1 spectrum the zero separation inside the doublet is used as the structural support gap</i>	<i>P75-16 distinct-support multiplicity layer</i>	<i>the structural doublet–singlet gap and the 2 + 1 structural classification</i>
<i>A target-authorized structural output cannot be factored through the information-preserving joint-trace regrouping $\pi_{\text{str}} \Theta_{\text{pert}}$, a cross-interface relation is lost by regrouping, or a certified varying factor is absent from the registry</i>	<i>P75-02 joint-trace factorization or P75-17 fibrewise maximality layer, or a target/source update</i>	<i>the already proved core entries whose variation witnesses remain complete</i>
<i>Every source/analyzer cell has positive acceptance but a core lacks a global lower-Lipschitz bound, a required core/reference image separator vanishes in the common record space, or a structural collision is forced into a fine quotient</i>	<i>P75-19 SPIC layer</i>	<i>structural mass-support, rank, closure, and carrier conclusions</i>
<i>Same-record carrier/reference selection fails (the true core is excluded or a separated wrong core survives)</i>	<i>P75-19–21 protocol conditioning, coverage, or image-separator layer</i>	<i>structural source theorem unless the ideal pair-space packet also fails</i>
<i>Result outside ready children</i>	<i>target/update boundary</i>	<i>child-qualified theorem</i>

The weakest failed theorem, not the most dramatic narrative interpretation, determines the scientific revision.

Proof. The implication hierarchy established above is strict: P75-06 is a conditional rigidity theorem on the central perturbation stratum, so an observed split tests or exits that stratum rather than refuting the conditional implication; P75-07 classifies the resulting nonzero traceless branch. Exact structural mass degeneracy is not used in the rank

or closure proofs; flavor universality implies equal singular values but is not used in the rank bound; rank is not used to define terminal eligibility; and detector resolution is downstream of structural multiplicity. P75-13 obtains its positive radius from an explicit trace-transport cell, exact proper-support deletion flags, one full-support parent orbit, and closed guard/update faces; it does not reuse PCR-6 pointwise on a split realization. P75-16 uses separation between distinct structural mass-support values, so the repeated pair in a $2 + 1$ stratum does not create a spurious zero gap and says nothing about physical complex-pole multiplicity. The joint trace-regrouping lemma used by P75-02 supplies an information-preserving factorization of the complete joint trace, including cross-interface relations, and P75-17 derives covariant maximality from that certificate; invariant maximality remains relative only to a pre-frozen variation-certified registry. P75-19 requires a common record space, global within-core lower-Lipschitz response, separation of every core pair outside one declared coarsening including the reference, structural collision coarsening, joint nuisance/control coverage, and typed error guards in addition to positive acceptance. Therefore each listed deviation has a first dependency level. Fixed-domain no-backflow prevents a downstream failure from retroactively defeating an upstream theorem whose premises remain satisfied. $\square$

Warning 14.2 (No all-or-nothing rhetoric). A measured structural heavy-mass-support splitting would be important evidence for standing $U(3)_H$ breaking. It would not, by itself, falsify the one-carrier theorem or the light-heavy strength-accounting relation.

14.1 Extension-layer authority falsifiers

The following rows audit P75-X1–P75-X8 plus P75-X2A and are not additional claims of the core P75-23 theorem. They localize failures of authority, typing, or inference in the later search and lifetime extension.

Attempted promotion	First extension layer challenged	Lawful disposition
An arbitrary pre-frozen open interval is asserted to contain 2^{-n}	P75-X5 dyadic-existence layer	False in general; choose the dyadic in the proved punctured neighborhood before freezing the rational box
The marked-child value or the bound $\epsilon_{\text{row}} \leq 1/256$ is called an independently selected physical value	P75-X5 physical-selection firewall	Retain marked-child/target-qualified authority; independent physical selection remains NotReady
The HFLAV statistic $p = 0.1349$ is called unconditional full-VJ compatibility	P75-X5 light-response-applicability firewall	Retain only conditional light-response compatibility at 95 percent; the ratio block has zero common- ϵ information and full VJ compatibility remains NotReady
The direct/indirect m_W inputs are asserted to be statistically independent for endpoint promotion	P75-X5 absolute-normalization screening firewall	Zero direct/indirect cross-covariance is only the frozen screening assumption, not an independence certificate; uncertified cross-covariance and finite-mass HNL response prevent X7 promotion
The D3 benchmark coefficient is identified with the VJ ϵ_{row}	P75-X6/X7 typed normalization layer	Requires the normalization bridge B_{norm} ; notation matching is insufficient
$c\tau_A^{\text{resp}} = 0.2$ mm is called a predicted D3 or VJ proper lifetime, or its 0.1 mm endpoint distance is called a continuous response radius	P75-X6 benchmark-authority layer	False: it is an exact interior member of the frozen discrete ATLAS reweighting grid. The 0.1 mm number is only its distance to the nearer endpoint of the declared grid span; D3Full SourceProperLifetime and continuous detector-response applicability remain NotReady

Exact heavy mass degeneracy is used to infer a scalar or common width	P75-X7 dispersive–absorptive dynamics layer	Width-eigenvalue interpretation requires the commuting narrow-width reduction; a common width additionally requires equal pole imaginary parts or scalar absorptive response
A W -rest-frame $\beta\gamma c\tau$ benchmark is called the proton–proton laboratory mean	P75-X2/X6 response-frame layer	Requires first the D3 full-source lifetime identity and then the production-spectrum, angle, acceptance, detector, and reconstruction convolution
The D3 [28, 30] GeV benchmark band is called the VJ mass domain or a VJ theory-prioritized slice	P75-X1/X8 domain-authority layer	Its SourceCertified_ATLASR eweightGridAnchored_Sear chBenchmark authority remains established, but only 30 GeV is a direct ATLAS mass anchor; in VJ scope it is only a Predecla redOperationalScanSlice_ NoVJMassAuthority, and transfer requires D3ToVJDomainBridge
An eigenvalue of $\mathbf{\Gamma}$ is called a physical lifetime without the commuting reduction	P75-X7 complex-pole layer	Use chart-local poles of $\det \mathcal{D}^{-1} = 0$; only the separate commuting corollary permits $\gamma_r = 0$ and then gives infinite lifetime
A record is inverted without child, branch, continuation-chart-or-coherent-stratum, response, and electroweak-scheme typing	P75-X8 record-typing layer	Use the exhaustive predeclared compatible typed set
Overlapping poles are assigned separate lifetime intervals without an identifiability certificate	P75-X8 response-identifiability layer	Invert the full coherent-response confidence region
One nuisance point or unresolved type is selected before inversion	P75-X8 coverage-projection layer	Project by union over every compatible nuisance, branch, stratum, and response type under the joint coverage event
The 0.2 mm response-grid point is reused as both benchmark-forming input and an independent validating lifetime	P75-X8 independence firewall	Its response-coordinate authority is unchanged, but reuse supplies compatibility rather than independent full-source lifetime or physical localization; D3Full SourceResponseValidation = NotReady
The linear inverse formula is used without a uniform positive lower bound on $ 5 + \kappa_C $ over the complete envelope	P75-X8 conditioning layer	Use the exact set-valued inverse or a separately certified higher-order expansion

The strongest extension endpoint is therefore the conditional, coverage-preserving complex-pole inversion on $H(I_E)$. A numerical VJ lifetime envelope and numerical VJ mass localization remain NotReady.

15 Empirical status and held-out robustness firewall

Theorem 15.1 (P75-24: empirical-status and held-out robustness firewall). *Before held-out comparison, PCR-7.5 freezes:*

1. the perturbation classes and their norms;
2. the carrier, symmetry, rank, and protocol boundaries;
3. the structural mass-support resolution rule, separated from physical complex-pole authority;
4. the response quotient and the SPIC specification/status gates: one common ordered record space, hypothesiswise uniform coverage, global within-core lower-Lipschitz bounds, all non-coarsened image separators including the explicit reference core, structural collision coarsening, joint nuisance/control coverage, and the separate typed record-separator, carrier/reference parameter-inverse, transported-mass, and Gram-operator floors and thresholds; on block isometry with $\eta_V \leq \bar{\eta}_V < s_E$, the ideal pair-space gap is structural and supplies no detector-radius guard;
5. the feature-specific robustness vector;
6. the falsifier hierarchy;
7. the protocol variants and confidence rule;
8. the specifications and status gates for the VJ normalization and D3-to-VJ domain bridges, including every independence and nuisance-projection condition used for lifetime inversion.

No future signal record may widen a robustness ball, redefine exact symmetry, change the target child, select the norm, or determine which theorem was intended. The status decomposition is

FrozenStructuralPrediction,
DetectorEvaluationConditionalOnInstantiatedSPICAndHeldOutData,
X7X8InputsNotReady.

This freezes what must be supplied and how it may be tested; it does not assert that a SPIC, either bridge, the complex-pole kernel, the independent normalization/response records, or the exhaustive route universe has already been instantiated. Held-out data are therefore not the only outstanding input.

Proof. PCR-7 already partitions derivation and response-forming records from future held-out tomography. PCR-7.5 introduces mathematical/source robustness structure together with specifications and gates for conditional protocol and lifetime extensions. Recording those specifications before data inspection prevents future outcomes from silently redefining them, but does not prove that their NotReady inputs exist. Any post-data change is a new analysis certificate rather than a repair of the current prediction. The D3 pair ([28, 30] GeV, $c\tau_A^{\text{resp}} = 0.2$ mm) retains its SourceCertified_ATLASRweightGridAnchored_SearchBenchmark authority, with 0.2 mm typed as the ATLASRweightGridAnchored_InteriorResponseCoordinate and only 30 GeV a direct ATLAS mass anchor. The declared-linear HEPData Figure 4a screen clears the source curve by a factor greater than 4.50 under the ATLAS simplified-model convention, but is not a source-level non-exclusion or a full likelihood recast. Thus $\text{D3FullSourceProperLifetime} = \text{D3FullSourceResponseValidation} = \text{NotReady}$. The independence firewall further says that the same response-grid anchor may not be reused as an independent VJ lifetime input whose inverse is later reported. $\square$

Nonclaim 15.2 (No current observation). The paper does not claim that exact heavy degeneracy, anisotropy, a structural mass-support splitting, or the robustness radii have been measured. It defines the precomparison theorem and falsifier classes.

16 Theorem-informed search localization and selection strategy

The structural and detector theorems above determine what must be measured, but they do not by themselves select a unique heavy mass. A useful experimental strategy must therefore distinguish three questions:

1. which mass interval is fixed by the accepted source theory;

2. which interval is already known to admit a complete source-certified heavy-state benchmark realization;
3. which trigger, reconstruction, and record quotient are appropriate to the vectorlike Dirac prediction rather than to a different HNL hypothesis.

The present section derives a predeclared search-and-benchmark strategy from those distinctions. It is an experimental corollary of the accepted theorem chain and does not modify the perturbation or robustness endpoint.

16.1 No light-mass-ratio selector

Theorem 16.1 (P75-X1: heavy-scale independence and separated mass-domain authority). *At the current PCR target and source scope, no relation of the form*

$$H = f(m_1, m_2, m_3, \Delta m_{ij}^2)$$

is licensed as a heavy-mass prediction. On the source-qualified vectorlike-joint child, the heavy mass domain is

$$I_{VJ} := H(I_E) = \{H_\lambda = \lambda\Lambda : \lambda \in I_E\}.$$

It is compact and nondegenerate, but the present export does not expose numerical endpoints for I_E , Λ , or $H(I_E)$.

Independently, PCR-5D3 supplies the one-singlet prompt-HNL source-certified search-benchmark band

$$I_{D3} = [28, 30] \text{ GeV}$$

with the declared response-chart anchor $c\tau_A^{\text{resp}} = 0.2 \text{ mm}$ and positive non-direct-constraint and one-apex physicality margins. Therefore

$$\begin{aligned} \text{Status}(H(I_E)) &= \text{SourceQualifiedVJMassDomainSymbolic}, \\ \text{Status}(I_{D3}, 0.2 \text{ mm}) &= \text{SourceCertified_ATLASReweightGridAnchored_SearchBenchmark}, \\ \text{D3ToVJDomainBridge} &= \text{NotReady}. \end{aligned}$$

In particular, the current theorem proves neither $I_{D3} \subseteq H(I_E)$, nor $H(I_E) \subseteq I_{D3}$, nor even numerical overlap. The interval $[28, 30] \text{ GeV}$ may be retained in a predeclared VJ analysis only as an operational comparison slice. Its VJ-scope status is then `PredeclaredOperationalScanSlice_NoVJMassAuthority`, not a VJ-specific theory-motivated interval, a VJ pole-mass prediction, or an exhaustive VJ search domain.

Proof. PCR-5D5 and PCR-5E construct a source-realized family with fixed light singular values and fixed target-retained light charged-current response while $H_\lambda = \lambda\Lambda$ varies on the nondegenerate interval I_E . Hence the retained light packet does not determine the heavy threshold. PCR-5D3 separately constructs a one-singlet source-certified benchmark family on $[28, 30] \text{ GeV}$. These are differently typed source objects. No imported theorem identifies their mass domains. A typed domain bridge is therefore necessary before forming a D3/VJ relation or calibration-matched compatibility set, and even that bridge does not by itself supply physical VJ localization. $\square$

Nonclaim 16.2 (No ratio prediction). The numerical coincidence of any heavy-to-light mass ratio, including a ratio to m_3 , $\sqrt{|\Delta m_{31}^2|}$, the Higgs scale, or the electroweak scale, is not part of the current theorem. Such a relation may be tested as a later target but cannot physically narrow $H(I_E)$ before it is independently derived.

16.2 Source-certified D3 benchmark kinematics and separated VJ search authority

For an on-shell two-body decay $W \rightarrow \ell_\alpha N$ with negligible charged-lepton mass, the rest-frame energies are

$$E_\ell^*(M) = \frac{m_W^2 - M^2}{2m_W}, \quad E_N^*(M) = \frac{m_W^2 + M^2}{2m_W}, \quad \beta\gamma(M) = \frac{E_\ell^*(M)}{M}.$$

Using the ATLAS value $m_W = 80.3665 \text{ GeV}$ [15], the massless-charged-lepton benchmark on $M \in I_{D3}$ gives the rounded intervals

$$E_\ell^*(I_{D3}) \simeq [34.583902, 35.305596] \text{ GeV},$$

$$E_N^*(I_{D3}) \approx [45.060904, 45.782598] \text{ GeV},$$

$$\beta\gamma(I_{D3}) \approx [1.1527967, 1.2609141].$$

For the response-domain-anchored coordinate $c\tau_A^{\text{resp}} = 0.2 \text{ mm}$, the corresponding W -rest-frame response-coordinate image is

$$\beta\gamma c\tau_A^{\text{resp}} \approx [0.2305593, 0.2521828] \text{ mm}.$$

These are rounded response-coordinate images, not physical laboratory mean-flight predictions or strict endpoint inequalities. For a reconstructed tau channel the exact two-body formula with $m_\tau \neq 0$ must be used.

The calibration-equivalent response-width scale is

$$\Gamma_{0.2}^{\text{equiv}} := \frac{\hbar c}{0.2 \text{ mm}} \approx 9.86635 \times 10^{-13} \text{ GeV}.$$

It is the equivalent width associated with the frozen D3 response chart. PCR-5D3 does not identify it with the unrestricted full-source total width. It is not an intrinsic VJ pole width and cannot determine whether a VJ multiplet is experimentally resolved.

Corollary 16.3 (P75-X2: D3 source-certified near-prompt charged-current benchmark tag). *For the D3 source-certified benchmark band, a theorem-consistent search category may be based on*

$$W^{(*)} \rightarrow \ell_\alpha N$$

with the massless-lepton W -rest-frame benchmark above, inclusive mass reconstruction on I_{D3} , and prompt/near-prompt tracking without a hard several-millimetre lower-displacement requirement. This establishes a positive response-grid tag for the D3 benchmark family. It assigns neither I_{D3} nor $c\tau_A^{\text{resp}} = 0.2 \text{ mm}$ to the VJ child. A physical proper lifetime or proton–proton laboratory distribution requires the full-source width, production spectrum, decay angle, lifetime law, acceptance, and reconstruction response.

Proof. The energy and boost intervals follow by monotonicity of the two-body formulas on I_{D3} . Multiplication by the fixed response-grid coordinate gives the displayed sub-millimetre W -rest-frame image. Hence prompt/near-prompt tracking is the appropriate category in which to test this response-grid benchmark, whereas a hard macroscopic-displacement requirement does not target that coordinate. No full-source proper-lifetime, detector-acceptance, VJ mass-domain, or pole-width conclusion is used. $\square$

Corollary 16.4 (P75-X2A: predeclared D3 benchmark-admissibility certificate). *Define the predeclared benchmark pair*

$$\mathfrak{B}_{D3} := ([28, 30] \text{ GeV}) \times \{0.2 \text{ mm}\}.$$

For every $M \in [28, 30] \text{ GeV}$, PCR-5D3 supplies an exact gauge-invariant Weinberg-plus-one-singlet source on the declared response-chart curve

$$\epsilon_{0.2}(M) = \frac{\hbar c}{(0.2 \text{ mm}) \bar{\Gamma}_A^{\text{resp}}(M)}, \quad 0 < \epsilon_{0.2}(M) \leq 2.51 \times 10^{-6}.$$

Thus

$$c\tau_A^{\text{resp}} := \frac{\hbar c}{\Gamma_A^{\text{resp}}} = 0.2 \text{ mm}$$

exactly on that chart. PCR-5D3 expressly does not identify Γ_A^{resp} with the unrestricted full-source total width or a complex-pole width. Every point on the source curve retains the common-apex physicality, oscillation, LFU, CLFV, and matching certificates. The active electron-row norm obeys

$$1 - \epsilon_{0.2}(M) \geq 0.99999749 \geq 0.987;$$

the covariance-complete universality block obeys

$$8.437 < \chi_{\text{LFU}}^2(\epsilon_{0.2}(M)) \leq 8.445 < 11.0704976935;$$

the heavy $\mu \rightarrow e\gamma$ contribution vanishes by $U_{\mu N} = 0$, and the light GIM remainder is below 6×10^{-48} . The entire band lies in the on-shell- W regime and in the published ATLAS prompt-search mass range. ATLAS directly simulated 30 GeV, but not 28 GeV, inside this band. The response coordinate 0.2 mm is an exact member of the

discrete ATLAS lifetime-reweight grid spanning $[0.1, 1.0]$ mm; its distance to the lower endpoint of that declared span is 0.1 mm. This placement datum does not certify response validity at off-grid lifetimes or define a continuous detector-response radius. Together with the narrow kinematic tag of P75-X2, these facts establish

$$\text{Status}(\mathfrak{B}_{D3}) = \text{SourceCertified_ATLASReweightGridAnchored_SearchBenchmark.}$$

The frozen HEPData replay sharpens the numerical comparison: the observed Figure 4a ordinate is 1.1380×10^{-5} at the directly simulated 30 GeV point, a factor greater than 4.53 above the PCR-5D3 uniform upper bound. The package predeclares the straight-line display reconstruction

$$U_{e,95,\text{lin}}^{2,\text{obs}}(M) = 1.0963 \times 10^{-5} + (M - 20 \text{ GeV}) 4.17 \times 10^{-8} \text{ GeV}^{-1}.$$

Hence its value at 28 GeV, and its minimum on $[28, 30]$ GeV, is 1.12966×10^{-5} , preserving a factor greater than 4.50 above the uniform PCR-5D3 upper bound. A log-linear reconstruction, natural for the published logarithmic ordinate, gives 1.129535×10^{-5} at 28 GeV and preserves the same factor-4.50 conclusion. This is a reproducible ordinate-compatibility screen, not a likelihood recast or a certified continuous detector-response morphing in mass. Under the ATLAS single-Majorana, electron-only simplified signal model and either declared interpolation, the PCR-5D3 curve ordinate lies below the published observed upper-limit ordinate. That fact is recorded without promoting it to model-level or source-level non-exclusion. The ATLAS limit couples mixing to the HNL lifetime; the D3 full-source width, branching fractions, and acceptance have not been mapped into that likelihood. Therefore

$$\begin{aligned} \text{ATLASFig4aOrdinateScreen} &= \text{ATLASFig4a_OrdinateBelowObservedLimit} \\ &\quad \text{_ResponseGridCoordinateCovered,} \\ \text{ATLASPointwiseNonExclusion} &= \text{NotReady,} \\ \text{D3FullSourceProperLifetime} &= \text{NotReady,} \\ \text{D3FullSourceResponseValidation} &= \text{NotReady.} \end{aligned}$$

Thus $\mathfrak{B}_{D3}$ is a predeclared, multi-record-anchored admissible search benchmark: it lies at the intersection of an exact source-realized continuum, uniform non-direct-constraint margins, an exact published lifetime-reweight-grid coordinate with a direct 30 GeV mass anchor; a frozen public-ordinate replay, and a deterministic on-shell- W kinematic tag. These facts motivate the benchmark and make it reproducible; they do not derive the numerical endpoints or select 0.2 mm uniquely among admissible grid coordinates. Pointwise ATLAS non-exclusion requires a source-complete likelihood/recast or an equivalent certified response map.

Proof. PCR-5D3 Theorems 10.3 and 10.4 construct the displayed continuum and prove its common-apex physicality, oscillation, LFU, CLFV, and matching records. The ATLAS paper supplies the direct mass grid and lifetime-reweighting grid; the public HEPData table supplies the exact Figure 4a ordinates; and P75-X2 supplies the two-body energy and boost image. The package replay verifies the arithmetic and preserves the model/response firewall. All ingredients are frozen independently of any held-out PCR signal record. [16–19] $\square$

Nonclaim 16.5 (No endpoint-selection or exhaustive-search authority). P75-X2A proves that the predeclared pair is source-certified, ATLAS-reweight-grid anchored, uniformly non-direct-constraint compatible, and kinematically useful. It does not derive the lower or upper mass endpoint, choose 0.2 mm uniquely among grid points, or prove comparative priority or sensitivity optimality. Nor does it prove a full-source physical proper lifetime, complete detector-response validity, or pointwise ATLAS non-exclusion. It also does not prove that $[28, 30]$ GeV exhausts, overlaps, or numerically localizes the VJ pole domain $H(I_E)$.

16.3 Overlap with current searches and the selection mismatch

Current collider searches already cover the mass scale of I_{D3} , but they do not yet implement the complete PCR observable packet. The ATLAS prompt search covers 8–65 GeV and reports its strongest mixing limits in the 15–30 GeV region. It considers only LNV signal samples and requires at least one same-flavour, same-charge lepton pair in every selected event, including the three-lepton category; it also rejects events with three same-flavour signal leptons [18]. Its published response therefore does not directly test the LNC/vectorlike PCR signal, and an experiment-specific recast is required. The CMS trilepton search covers 10 GeV–1.5 TeV, includes both Dirac and Majorana interpretations and hadronic tau categories, and reports trigger efficiencies above 90% for the three-lepton sample, but its published limits are organized around exclusive coupling to one Standard-Model generation at a time [20]. The most recent ATLAS displaced-HNL analysis reaches masses below 14.5 GeV and therefore does not directly cover the 28–30 GeV benchmark band [21].

Status 16.6 (Published-search overlap). The current experimental status of the PCR search packet is

PublishedMassCoverage $\oplus$ SelectionIncompleteForPCRPacket $\oplus$ ExistingDataRecastRequired.

Published mass coverage is not a proof that existing analyses already have the required sensitivity. A dedicated recast must preserve the actual trigger, acceptance, lifetime, flavour, covariance, and background structure before a sensitivity claim can be made.

The selection mismatch is physically important. The exact-global PCR child is vectorlike and $B - L$ preserving; therefore lepton-number violation is not a universal signal requirement. A same-sign selection can retain the accidental child but can suppress the exact-global child. Likewise, a single-flavour interpretation does not test the PCR-7 prediction

$$\Delta_{\text{act}} = \Omega_H$$

as a flavour-complete matrix relation.

16.4 Theorem-consistent trigger and selection strategy

Theorem 16.7 (P75-X3: source-certified D3 benchmark and separated VJ search architecture). *A theorem-consistent experimental programme separates the following lanes and record obligations.*

1. **D3 benchmark lane.** *The D3 lane contains an exact source continuum on $I_{D3} = [28, 30]$ GeV, paired with the exact $c\tau_A^{\text{resp}} = 0.2$ mm ATLAS reweight-grid anchor. Prompt/near-prompt categories retain the energy and boost support of Corollaries 16.3 and 16.4. This establishes a predeclared, grid-anchored admissible D3 search target, not endpoint-selection authority, a full-source proper lifetime, guaranteed detector sensitivity, unconditional ATLAS non-exclusion, or a VJ mass/lifetime prediction.*
2. **VJ mass-domain lane.** *The source-qualified VJ domain is $H(I_E)$. Its numerical endpoints are not exposed at the current certificate. If $[28, 30]$ GeV is retained in a VJ analysis, its status is only PredeclaredOperationalScanSlice_NoVJMassAuthority. A typed domain bridge can at most support a D3/VJ relation or benchmark-matched compatibility set; it does not itself make the slice a VJ-specific theory-motivated interval or a physical VJ localization.*
3. **Production tag.** *A charged-current tag for*

$$pp \rightarrow W^{(*)} + X, \quad W^{(*)} \rightarrow \ell_a N_H,$$

with reconstructed support declared for the chosen VJ scan slice; the D3 lane separately contains the benchmark of Corollary 16.3.

4. **Lepton-number-inclusive selection.** *Both lepton-number-conserving and lepton-number-violating final states are retained. Same-sign charge is a category, not a universal requirement; same-flavour trilepton topologies are not vetoed merely because they are background-rich.*
5. **Lifetime scan.** *Until P75-X7's pole-width certificate is supplied, lifetime remains a branch-indexed response-category scan coordinate rather than a physical pole prediction. Conditional on that certificate, labels are local to certified simple-pole continuation charts, while collision, threshold, exceptional-point, and unresolved-overlap regions retain the unordered pole multiset and full coherent response rather than a global mode label. No prompt, near-prompt, or displaced category is declared universally complete for VJ at the current certificate.*
6. **Flavour-complete response.** *Electron, muon, and tau boundary categories are represented at the record level. Electron and muon channels may provide the discovery sample, but the tau channel remains part of the closure test rather than an optional reinterpretation.*
7. **Inclusive heavy reconstruction.** *Leptonic and semileptonic heavy-decay categories are retained, including opposite-sign channels, so that a common reconstructed heavy mass-support cluster can be tested without assuming Majorana character. A physical complex-pole cluster is reported only under the separate P75-X7 certificate.*
8. **Joint normalization.** *Heavy and light charged-current records are evaluated in one declared electroweak input scheme, permitting reconstruction of Ω_H , Δ_{act} , and $\Delta_{\text{act}} = \Omega_H$ without interpreting a source normalization coefficient as a scheme-free measured rate deficit.*

9. **Hierarchical mass-support test.** Aggregate nonzero rank-three heavy support and the light-heavy sum rule are the primary selection objects. One unresolved heavy peak is not evidence of structural Deg3 or of a common physical pole. Multiple nearby reconstructed support features are retained and tested against the PCR-7.5 structural mass-support atlas rather than rejected automatically.

A claim that existing data exclude or detect the VJ packet requires a frozen, experiment-specific recast certificate and a declared covered subset of $H(I_E)$.

Proof. The first two items follow from the separated source authorities of Theorem 16.1 and corollaries 16.3 and 16.4. The remaining detector-facing items are the minimum content needed to reconstruct the accepted PCR-7 packet rather than a single raw rate. The vectorlike exact-global child invalidates an LNV-only universal selection, while the flavour-complete Gram packet invalidates a one-flavour-only final interpretation. Aggregate support, rank, and light-heavy closure persist on a positive perturbation region, whereas exact degeneracy has zero unrestricted symmetry-breaking radius. None of these facts supplies numerical endpoints for $H(I_E)$ or a VJ lifetime. $\square$

16.5 Resolution-adaptive window tightening

Let $\sigma_M(M)$ denote the predeclared reconstructed heavy-mass resolution of a particular detector category and let $\kappa \geq 1$ be the chosen simultaneous sideband factor. Define

$$\sigma_{M,D3}^{\max} = \sup_{M \in I_{D3}} \sigma_M(M)$$

and, whenever $2\kappa\sigma_{M,D3}^{\max} < 2 \text{ GeV}$, the sideband-safe core

$$I_{\text{core},D3}^{(\kappa)} = [28 + \kappa\sigma_{M,D3}^{\max}, 30 - \kappa\sigma_{M,D3}^{\max}].$$

A complete D3 resolution-adaptive benchmark scan is a finite overlapping cover

$$\mathcal{W}_{\text{scan},D3} = \{W_k = [M_k - \kappa\sigma_M(M_k), M_k + \kappa\sigma_M(M_k)]\}_{k=1}^N$$

with

$$I_{D3} \subseteq \bigcup_{k=1}^N W_k.$$

The core windows carry two-sided local sidebands; the edge windows carry one-sided sidebands and retain explicit edge nuisance parameters. This construction tightens the D3 benchmark analysis resolution without changing or supplying VJ mass-domain authority.

Corollary 16.8 (P75-X4: strongest licensed numerical refinement). *Before held-out signal inspection, detector resolution may refine the D3 operational calibration programme as*

$$I_{D3} \longrightarrow (I_{\text{core},D3}^{(\kappa)}, \mathcal{W}_{\text{scan},D3}, I_{E_t^*,D3}),$$

where $I_{E_t^*,D3}$ is the rounded interval of Corollary 16.3, enlarged only by the predeclared kinematic resolution. This is a partition and prioritization of the source-certified D3 benchmark band, not a physical narrowing of the VJ mass domain.

At VJ scope the strongest current source authority remains

$$M \in H(I_E),$$

with numerical endpoints NotReady. A narrower numerical object may acquire either OperationalScanPriority, from a predeclared detector optimization using calibration or anchor records disjoint from the held-out signal set, or PhysicalMassLocalization, only from a new source theorem or an independent physical constraint acting inside $H(I_E)$. Transporting a D3-derived interval to VJ additionally requires a D3ToVJDomainBridge.

Proof. Detector resolution and sideband geometry can partition and prioritize the D3 benchmark range without discarding any D3 source point. The two-body kinematic tag is a deterministic consequence of that benchmark interval and measured W mass. It has no VJ-domain implication without a typed bridge. A physical VJ narrowing requires a new source theorem or independent physical constraint, whereas a detector optimization creates only an operational priority. The empirical firewall of Theorem 15.1 forbids using held-out outcomes to create either object retrospectively. $\square$

Remark 16.9 (Conditional yield-ordering heuristic). For run planning only, one may use the leading lifetime-calibration and on-shell W phase-space benchmark

$$\tilde{w}(M) = M^{-5} \left(1 - \frac{M^2}{m_W^2} \right)^2 \left(1 + \frac{M^2}{2m_W^2} \right).$$

It decreases on $[28, 30]$ GeV. Under flat instrumental efficiency, the lower half $[28, 29]$ GeV contains approximately 54.7% of the normalized benchmark weight, and the median benchmark mass is approximately 28.91 GeV. This is a secondary scheduling heuristic within the source-certified D3 benchmark, not a predicted VJ mass distribution and not a licence to discard $[29, 30]$ GeV.

Warning 16.10 (Trigger and selection authority). The paper identifies a theorem-consistent search architecture, a symbolic VJ source domain $H(I_E)$, and a separate source-certified D3 benchmark band. It does not claim that an existing trigger menu, reconstruction chain, or published analysis already satisfies the complete PCR-7 record quotient. Such a statement requires an experiment-specific recast or a new analysis certificate with actual efficiencies, covariances, backgrounds, and systematics.

17 Row-normalization exposure and complex-pole lifetime localization

The PCR-5D3 prompt family establishes a source-certified, ATLAS-reweight-grid-anchored search benchmark on its own mass band. It does not promote the D3 response-chart coordinate to a full-source physical lifetime or convert it into a property of the PCR-7 vectorlike-joint (VJ) carrier. This section separates the algebraic row-normalization identity, the positive authority of the D3 source/grid construction, the physical VJ pole problem, and the independent lifetime inversion.

Throughout this section

$$\epsilon_{\text{row}} := 1 - (NN^\dagger)_{ee}$$

denotes the electron-row light charged-current normalization deficit of the source model. It is distinct from the protocol record-space radius r_{rec} , the Gram-operator radius ϵ_{op} , and the D3 calibration coefficient $\epsilon_{0.2}(M)$. Its interpretation as a measured weak-rate deficit requires a declared electroweak input-scheme response certificate.

17.1 Structural identity, dyadic existence, and source-qualified bound

Theorem 17.1 (P75-X5: row-normalization identity, dyadic-existence repair, and marked-child bound). *On the unperturbed flavor-universal VJ stratum,*

$$N = c_E U, \quad V_H = -s_E U, \quad U \in U(3),$$

and therefore, for every active flavor α and heavy index i ,

$$1 - (NN^\dagger)_{\alpha\alpha} = (V_H V_H^\dagger)_{\alpha\alpha} = (V_H^\dagger V_H)_{ii} = \delta_E = s_E^2.$$

The PCR-5E E-S4A proof supplies a uniform punctured strict-pass neighborhood of $s = 0$ together with a finite operational guard registry. The marked-child witness is made well-defined in the following order:

1. choose n_{mark} sufficiently large that $s_E = 2^{-n_{\text{mark}}}$ lies in that punctured strict-pass neighborhood and away from every exceptional guard frontier;
2. choose a threshold point in the corresponding open guard cell;
3. only then freeze compact rational intervals I_E and I_{s_E} whose product lies in that cell and with $s_E \in \text{int } I_{s_E}$.

This comparison-certificate repair uses $2^{-n} \rightarrow 0$; it does not assume that reciprocal powers of two meet an arbitrary interval already frozen elsewhere.

The retained PCR-5E tester gives

$$1 - s_E^2 > 0.987,$$

hence $0 < \epsilon_{\text{row}} = s_E^2 < 0.013$. Dyadicity then forces $n_{\text{mark}} \geq 4$, and therefore

$$0 < \epsilon_{\text{row}} = 4^{-n_{\text{mark}}} \leq \frac{1}{256} = 3.90625 \times 10^{-3}.$$

The identity and upper bound are exact at the marked-child and retained-target scope. They do not expose the exact integer n_{mark} , provide a target-independent lower bound, or independently select the marked child as the physical realization.

Proof. The matrix identities follow from $NN^\dagger = c_E^2 I_3$, $V_H V_H^\dagger = V_H^\dagger V_H = s_E^2 I_3$, and $c_E^2 + s_E^2 = 1$.

In the E-S4A construction, every target-active direct-state response converges uniformly to its accepted zero response as $s \downarrow 0$; the nontrivial U response is continuous in s^2 ; the retained tester is $1 - s^2$; and the L response has a uniform inherited margin. The finite guard registry is constant on a sufficiently small punctured cell after its exceptional frontiers are removed. Since $2^{-n} \rightarrow 0$, a dyadic point lies in that cell. Openness and continuity then supply a compact rational product box around the selected point. Reversing only this witness-selection order changes neither the raw source class, the target, the accepted response predicates, nor the later quotient.

Finally, $4^{-3} = 1/64 > 0.013$, while $4^{-4} = 1/256 < 0.013$. Thus every accepted dyadic witness has $n_{\text{mark}} \geq 4$. [2, 10] $\square$

Nonclaim 17.2 (No scheme-free observed weak deficit). The algebraic coordinate ϵ_{row} is not automatically a measured rate deficit. A physical extraction must distinguish the Lagrangian parameter

$$G_F^{(0)} = \frac{g_2^2}{4\sqrt{2}m_W^2}$$

from the muon-decay input G_μ , account for normalization changes in the muon channel, include any kinematically open heavy final states, and propagate the complete production and decay response. Until an `EWInputSchemeResponseCertificate` is supplied,

ObservableWeakRateDeficit = NotReady.

Criterion 17.3 (Two independent row-normalization promotions). A numerical value or confidence set for ϵ_{row} may be promoted to a *fixed-child conditional physical normalization constraint* if an independent weak-rate record, together with a declared electroweak input-scheme response, constrains the coordinate conditional on the chosen child. That record must be independent of the D3 prompt-lifetime calibration and of the later lifetime/response record used for mass inversion. This is the level required by P75-X7, and it does not select the child as the realized target.

A stronger *target-independent child or realization selection* requires a new theorem or record acting on a comparison class not already restricted to the marked child and supplying a standing-bearing selection across that class. Evaluating the dyadic rule or its $1/256$ upper bound satisfies neither promotion.

17.2 Bounded public weak-record reconstruction

One public-data pass was frozen before its numerical disposition: one look at the HFLAV Tau-2023 unconstrained coupling-ratio block, one separately sourced absolute electroweak-normalization screen, no within-release retry, and the decision rule

$$\begin{array}{ll} \epsilon_{95}^{\text{weak}} < 1/256 & \implies \text{tightened source-weak intersection,} \\ \text{otherwise} & \implies \text{retain } 1/256; \text{ compatibility only.} \end{array}$$

Here $\epsilon_{95}^{\text{weak}}$ means a *certified*, jointly covered physical endpoint; a leading-response screening number is not substituted for it.

In the HFLAV order

$$\mathbf{r} = (A, B, C, D, E) = ((g_\tau/g_\mu)_\tau, (g_\tau/g_e)_\tau, (g_\mu/g_e)_\tau, (g_\tau/g_\mu)_\pi, (g_\tau/g_\mu)_K),$$

the frozen public values and standard uncertainties are

$$\widehat{\mathbf{r}} = (1.0016, 1.0018, 1.0002, 0.996, 0.986), \quad \boldsymbol{\sigma} = (0.0014, 0.0014, 0.0011, 0.004, 0.008),$$

with printed correlation matrix

$$R = \begin{pmatrix} 1 & 0.67 & -0.41 & 0.18 & 0.12 \\ 0.67 & 1 & 0.40 & 0.19 & 0.12 \\ -0.41 & 0.40 & 1 & 0.01 & -0.01 \\ 0.18 & 0.19 & 0.01 & 1 & 0.07 \\ 0.12 & 0.12 & -0.01 & 0.07 & 1 \end{pmatrix}, \quad V = \text{diag}(\boldsymbol{\sigma})R \text{diag}(\boldsymbol{\sigma}).$$

The exact ratio identity $B = AC$ makes one coordinate algebraically redundant. The frozen test therefore uses the independent basis (A, C, D, E) , rather than inverting the rounded five-coordinate near-null mode or applying a post hoc covariance repair. [22]

On the flavor-universal VJ stratum of P75-X5,

$$g_e^{\text{eff}} = g_\mu^{\text{eff}} = g_\tau^{\text{eff}} = g_2 \sqrt{1 - \epsilon_{\text{row}}}.$$

Consequently every component of the ratio prediction is one for every common ϵ_{row} , and

$$\frac{\partial \mathbf{r}}{\partial \epsilon_{\text{row}}} = 0, \quad \mathcal{I}_\epsilon = \left(\frac{\partial \mathbf{r}}{\partial \epsilon_{\text{row}}} \right)^T V^+ \left(\frac{\partial \mathbf{r}}{\partial \epsilon_{\text{row}}} \right) = 0.$$

The independent-basis universality statistic is

$$\chi_{\text{HFLAV}}^2 = 7.019300995, \quad \nu = 4, \quad p = 0.134871765,$$

below $\chi_{4,0.95}^2 = 9.487729037$. This calculation has the canonical status

ConditionalLightResponseCompatibleAt95.

The condition is that the published light-neutrino response map applies to the evaluated child—in particular, that heavy-neutrino final states affecting the HFLAV inputs are kinematically closed and that no mass-dependent acceptance change invalidates the unit-ratio map—or that a channel-aware recast certifies that same map. Conditional on that applicability predicate, the HFLAV record does not reject the universal response at 95 percent, but it cannot determine the magnitude of a common deficit. Its inverse image in the common- ϵ direction is the whole declared domain when unity is accepted and empty when unity is rejected. Without the applicability predicate or a channel-aware recast,

FullVJPhysicalCompatibility = NotReady.

Using the alternate independent basis (B, C, D, E) as a printed-rounding audit gives $\chi^2 = 7.0403$ and $p = 0.1338$; the conditional compatibility disposition is unchanged.

For the separately sourced absolute-normalization screen, freeze the CMS direct result

$$m_W^{\text{dir}} = 80.3602 \pm 0.0099 \text{ GeV}$$

and the Standard-Model comparison value quoted with that result,

$$m_W^{\text{ind}} = 80.353 \pm 0.006 \text{ GeV}.$$

As an explicit screening assumption, set the unreported direct/indirect cross-covariance to zero for this calculation. This treatment is not a certificate of statistical independence. Holding the radiative inputs fixed and using $G_\mu = G_F^{(0)}(1 - \epsilon_{\text{row}})$ then gives the on-shell linear response

$$K_W := \frac{\partial m_W}{\partial \epsilon_{\text{row}}} = \frac{m_W}{2} \frac{s_W^2}{c_W^2 - s_W^2} \Big|_{m_W=m_W^{\text{ind}}} = 16.239776 \text{ GeV}, \quad s_W^2 = 1 - \frac{m_W^2}{m_Z^2}, \quad m_Z = 91.1876 \text{ GeV}.$$

The corresponding one-sided 95-percent screening calculation is

$$\widehat{\epsilon}_{\text{scr}} = \frac{m_W^{\text{dir}} - m_W^{\text{ind}}}{K_W} = 4.4336 \times 10^{-4},$$

$$\epsilon_{95}^{\text{scr}} = \widehat{\epsilon}_{\text{scr}} + 1.6448536 \frac{\sqrt{0.0099^2 + 0.006^2}}{K_W} = 1.6159 \times 10^{-3} < \frac{1}{256}.$$

The exact nonlinear on-shell inversion changes the endpoint only to 1.6176×10^{-3} , which is the stored screening value. This screen has real numerical leverage: its endpoint is about a factor 2.4 below the source ceiling. It is not promoted to $\epsilon_{95}^{\text{weak}}$. The quoted indirect value is not a supplied finite-mass HNL likelihood; the numerical endpoints of $H(I_E)$ are unexposed; heavy-neutrino loop and open-channel effects are not jointly propagated; and the direct m_W template, acceptance, electroweak-scheme, theory-covariance, and direct/indirect cross-covariance responses have not been certified uniformly on the VJ source domain. [23]

Frozen one-pass disposition

The HFLAV ratio block gives ConditionalLightResponseCompatibleAt95 with $p = 0.1349$ and exactly zero common-mode ϵ information; full VJ physical compatibility remains NotReady unless the published light-response applicability condition or a channel-aware recast is supplied. The separately sourced absolute-normalization calculation gives only $\epsilon_{95}^{\text{scr}} = 1.62 \times 10^{-3}$, with status ScreeningOnly_NotCertifiedForX7. Therefore

$$\epsilon_{95}^{\text{weak}} = \text{NotReady}, \quad 0 < \epsilon_{\text{row}} \leq \frac{1}{256}$$

remains the published marked-child bound. The stop rule takes its “otherwise” branch: HFLAV is recorded as an external, light-response-conditional flavor-universality compatibility check. Because its underlying observables overlap the PCR-5D3 LFU block, no statistical independence from that block is claimed. Nor is direct/indirect statistical independence certified for the absolute-normalization screen: zero cross-covariance is only its declared screening assumption. Meanwhile, $\mathcal{E}_X^{\text{weak}}$, $\mathcal{E}_X^{\text{adm}}$, child selection, lifetime, and mass promotion remain NotReady.

17.3 Response-grid authority of $c\tau_A^{\text{resp}} = 0.2$ mm

Theorem 17.4 (P75-X6: ATLAS-reweight-grid membership and D3 response-chart authority). *On the PCR-5D3 prompt response chart, write*

$$\Gamma_A^{\text{resp}}(M, \epsilon) = \epsilon \bar{\Gamma}_A^{\text{resp}}(M), \quad \bar{\Gamma}_A^{\text{resp}}(M) > 0.$$

The D3 calibration curve is

$$\epsilon_{0.2}(M) = \frac{\hbar c}{(0.2 \text{ mm}) \bar{\Gamma}_A^{\text{resp}}(M)}.$$

Thus $c\tau_A^{\text{resp}} := \hbar c / \Gamma_A^{\text{resp}} = 0.2$ mm holds exactly on the declared response chart. The coefficient is generally mass-dependent and is nonconstant whenever $\bar{\Gamma}_A^{\text{resp}}(M)$ is nonconstant; nonconstancy is not inferred merely from notation. PCR-5D3 does not identify Γ_A^{resp} with the unrestricted full-source total width or a complex-pole width.

The value 0.2 mm is not an arbitrary continuous-coordinate insertion. It is an exact point of the frozen ATLAS lifetime-reweighting grid. Across $M \in [28, 30]$ GeV, the PCR-5D3 source construction realizes that response-chart coordinate with

$$0 < \epsilon_{0.2}(M) \leq 2.51 \times 10^{-6},$$

while retaining the positive oscillation, LFU, CLFV, matching, and common-apex margins collected in P75-X2A. The frozen HEPData replay supplies the independently sourced statement that the uniform upper bound is a factor greater than 4.50 below both the declared linear and log-linear reconstructions of the observed Figure 4a ordinate on $[28, 30]$ GeV, conditionally within the ATLAS simplified model; it is not a full-source likelihood or response certificate.

The frozen ATLAS lifetime-reweighting coordinates form the discrete set

$$\mathcal{G}_\tau^{\text{ATLAS}} = \{0.10, 0.15, 0.20, 0.25, 0.30, 0.35, 0.40, 0.45, 0.50, 0.60, 0.70, 0.80, 0.90, 1.00\} \text{ mm}.$$

Hence $0.2 \text{ mm} \in \mathcal{G}_\tau^{\text{ATLAS}}$ exactly and is not an endpoint of that grid. Its endpoint-placement distance inside the numerical span of the grid is

$$d_{\tau, \text{end}} := \min\{0.2 - 0.1, 1.0 - 0.2\} \text{ mm} = 0.1 \text{ mm}.$$

This number certifies only interior placement relative to the two endpoint coordinates. The discrete grid does not by itself license interpolation, an open lifetime neighborhood, or continuous detector-response validity away from its listed coordinates. The exact equality $c\tau_A^{\text{resp}} = 0.2$ mm has the separate PCR-5D3 source response-chart authority established above. Moreover,

D3ResponseGridAuthority = ATLASReweightGridAnchored_InteriorResponseCoordinate,
D3FullSourceProperLifetime = NotReady,
D3FullSourceResponseValidation = NotReady,
ContinuousResponseNeighborhood = NotReady,
PredictedVJLifetime = NotReady.

The associated $\beta\gamma c\tau_A^{\text{resp}}$ interval is only the massless-lepton W -rest-frame response-coordinate image of Corollary 16.3; it is not a physical mean flight length or a proton–proton laboratory mean.

Proof. Substitution gives

$$\Gamma_A^{\text{resp}} = \frac{\hbar c}{0.2 \text{ mm}}$$

identically, so 0.2 mm is the predeclared response-chart coordinate defining the curve, not an independently predicted full-source lifetime. PCR-5D3 proves the displayed mixing bound and simultaneous positive non-direct constraint margins. Direct enumeration of the ATLAS reweighting coordinates proves exact membership and gives the 0.1 mm endpoint-placement distance; it supplies no off-grid neighborhood. P75-X2 maps the coordinate to a 0.231–0.252 mm W -rest-frame response image. Nothing in this construction identifies the one-singlet D3 response width, mass domain, or coordinate with an unrestricted physical total width or with the rank-three VJ source. [16–19] $\square$

Nonclaim 17.5 (No optimal-lifetime claim). The theorem proves neither that 0.2 mm is the full-source physical proper lifetime nor that the D3 source has been validated by the complete ATLAS detector response. It also does not establish source-level pointwise ATLAS non-exclusion from the ordinate screen. It proves an exact source-coherent continuum mapped to an interior ATLAS reweight-grid coordinate, with positive non-direct constraint and kinematic motivation; uniqueness, sensitivity optimality, and any VJ lifetime remain unproved.

17.4 Complex-pole lifetime transport and bridge factorization

Definition 17.6 (Typed VJ pole-response datum). A typed pole-response datum is a tuple

$$\mathbf{t} = (X, b, \sigma_z, r, \mathcal{R}, s_{\text{EW}}),$$

where X is a fixed accepted VJ child, b is a source-qualified Dirac branch, $\mathcal{R}$ is the frozen production–propagation–decay response class, and s_{EW} is the declared electroweak input scheme. On a simple-pole continuation chart, σ_z is a fixed physical complex-pole stratum and r is a chart-local pole index. For a coherent response at a collision or across a stratum frontier, $(\sigma_z, r) = (\star, \star)$ denotes the stratified unordered multiset of every source-connected relevant-sheet pole. The index σ_z is not the structural source-mass partition Π_H^{str} and cannot be inferred from it without the target-complete pole kernel. An ordered label is never transported through a collision, exceptional point, channel-threshold frontier, or stratum change.

Throughout P75-X7/X8, freeze one physical length unit ℓ_0 (for example, $\ell_0 = \text{mm}$) and write

$$\overline{\mathbb{L}}_{+, \ell_0} := \{\lambda \ell_0 : \lambda \in [0, \infty]\}$$

for the extended nonnegative length-coordinate space, equipped with the order topology and Borel sigma-field transported from $[0, \infty]$ by $\lambda \mapsto \lambda \ell_0$. Thus every displayed $c\tau$, including 0.2 mm, is a typed element of one frozen space.

Definition 17.7 (Noninterchangeable normalization and domain bridges). Write $\mathbf{B}_{\text{norm}}$ for the typed VJ-packet-to-pole-width normalization bridge. It maps the PCR-7 charged-current packet of X , in scheme s_{EW} , to the coupling normalization of the complete VJ pole kernel. If a factorized absorptive boundary form is licensed, it proves on the declared real/on-shell boundary

$$\Gamma_{X,b}(E; M, \epsilon_{\text{row}}, \theta) = \epsilon_{\text{row}} \mathcal{G}_{X,b}^{\text{self}}(E; M, \epsilon_{\text{row}}, \theta; s_{\text{EW}}),$$

rather than inferring that identity from matching notation or separately analytically continuing dispersive and absorptive pieces. Off the exact flavor-universal stratum, the bridge must transport the full heavy Gram and the structural mass-support coupling weights $W_a = V_H E_a V_H^\dagger$ into the self-energy/pole kernel. These Hermitian weights are not Laurent residues of the continued propagator. A scalar factorization by ϵ_{row} is licensed only when every remaining self-energy channel, flavor, mode, threshold, and energy-dependence factor is retained in $\mathcal{G}_{X,b}^{\text{self}}$; the electron-row coordinate alone is insufficient. Separately, the target-complete *observable-response* certificate uses the certified Laurent residue only on an isolated simple-pole chart. At a multiple pole, collision, exceptional point, or coherent stratum, it instead retains the full continued resummed matrix propagator, or a certified complete local Laurent principal part including every required higher-order term. It then combines that propagator information consistently with production and decay vertices, interference, all required nonresonant terms, and detector response. Those external-response factors shape $\mathcal{F}_i$; they are not inserted into the intrinsic pole-width kernel. The retained

observable response, rather than any field-propagator coefficient in isolation, must be gauge-parameter independent through the declared order.

Give the compact set $H(I_E)$ its Euclidean subspace topology and Borel sigma-field, and define the compact Polish bridge parameter space

$$\mathfrak{Z}_{D3,X} := [28, 30] \text{ GeV} \times H(I_E)$$

with its product topology and product Borel sigma-field. The separate `D3ToVJDomainBridge` is a typed relation

$$\mathfrak{D}_{D3 \rightarrow X} \subseteq \mathfrak{Z}_{D3,X}$$

which, if supplied, certifies a common mass coordinate and the stated response compatibility between the source-certified D3 benchmark family and the VJ child. The bridge declares whether this relation is exact or coverage-qualified and exports, for every related pair, the compatible VJ child, response class, mass convention, uncertainty envelope, and a predeclared lifetime-coordinate benchmark region

$$C_{D3 \rightarrow X}^\tau(M_D, M_X) \subseteq \bar{\mathbb{L}}_{+, \ell_0}.$$

Any bridge coverage failure is carried by its declared benchmark-compatibility qualification. The singleton $\{0.2 \text{ mm}\}$ is permitted only after both a full-source lifetime identity and an exact lifetime-coordinate bridge are certified; a coverage-qualified bridge exports its covered region instead. The bridge neither identifies the D3 benchmark coefficient with ϵ_{row} nor contributes to VJ width normalization. It is the typed transport required before the already source-certified D3 benchmark authority can enter a VJ-domain statement, but it does not by itself predict or localize a VJ mass; the P75-X7/X8 normalization, kernel, independent-record, coverage, and inversion premises remain necessary.

Theorem 17.8 (P75-X7: conditional complex-pole lifetime envelope). *Fix a typed VJ child and Dirac branch. Before any lifetime record is opened, suppose a successor certificate supplies:*

1. *a nonempty closed certified outer source-admissible normalization set $\mathcal{E}_X^{\text{src}} \subset (0, 1)$, with closedness understood relative to $(0, 1)$, containing the true fixed-child coordinate*

$$\epsilon_X := s_E(X)^2 \in \mathcal{E}_X^{\text{src}}.$$

For the marked child of P75-X5,

$$\mathcal{E}_X^{\text{src}} \subseteq (0, 1/256],$$

with the singleton $\{s_E(X)^2\}$ admissible only if that exact source coordinate is exposed. Let $v_X^{\text{weak}} \in \mathcal{N}_X^{\text{weak}}$ collect every weak-extraction coordinate other than ϵ , including the compatible source mass, branch, electroweak input scheme, muon normalization, open-channel, response, and theory nuisances. Independently, a weak-rate record W_X and declared response supply a compact jointly covered region

$$C_X^{\text{weak}}(W_X) \subseteq (0, 1) \times \mathcal{N}_X^{\text{weak}}, \quad \Pr\left[(\epsilon_X, v_X^{\text{weak}}) \in C_X^{\text{weak}}(W_X)\right] \geq 1 - \alpha_{\text{weak}},$$

with the guarantee holding almost surely conditional on any design data used to form it. Define the coverage-preserving projection

$$\mathcal{E}_X^{\text{weak}} := \text{pr}_\epsilon C_X^{\text{weak}}(W_X),$$

rather than profiling the other coordinates at best-fit values. The weak record is independent of the records used to establish the source-certified D3 0.2 mm benchmark and of the later lifetime/response record. Freeze the source-compatible compact intersection

$$\boxed{\mathcal{E}_X^{\text{adm}} := \mathcal{E}_X^{\text{src}} \cap \mathcal{E}_X^{\text{weak}} \neq \emptyset.}$$

A point normalization is the singleton special case. If the intersection is empty, report

`FixedChildNormalizationIncompatibility,`

not a VJ lifetime curve. The weak-region failure α_{weak} is charged inside the combined α_{spec} budget of P75-X8 rather than added again as an independent term;

2. *a compact nuisance/theory set $\Theta_{X,b}$ and, after the declared Lorentz/chiral reduction, a target-complete renormalized inverse propagator on the heavy subspace in the rest-frame complex-energy coordinate z ,*

$$\mathcal{D}_{X,b}^{-1}(z; M, \epsilon, \theta) = zI_{E_H} - M_X^{(0)}(M) - \Sigma_{X,b}^{\text{ren}}(z; M, \epsilon, \theta),$$

where $\Sigma_{X,b}^{\text{ren}}$ is one renormalized sheet-holomorphic self-energy on the declared continuation domain. Only on its declared real/on-shell boundary is it decomposed as

$$\Sigma_{X,b}^{\text{ren}}(E + i0) = \Delta M_{X,b}(E) - \frac{i}{2} \Gamma_{X,b}(E), \quad \Delta M_{X,b}(E) = \Delta M_{X,b}(E)^\dagger, \quad \Gamma_{X,b}(E) = \Gamma_{X,b}(E)^\dagger \geq 0.$$

The pole equation and propagator residue use the full Σ^{ren} ; no separate complex continuation of ΔM or Γ is asserted. If the resummed determinant is instead formulated in the invariant $s = p^2$ plane, the certificate first defines $z_r = \sqrt{s_r}$ on the declared physical branch; the pole width is never obtained as $-2\Im s_r$. The certificate fixes the renormalization prescription, counterterms, chiral wave-function factors, analytic continuation and pole sheet, proves gauge-parameter independence of the pole positions through the declared perturbative or EFT order, and encloses residual scheme and truncation dependence in $\Theta_{X,b}$. Every source-authorized self-energy channel, threshold, charged-particle mass, CKM and color factor, finite-propagator effect, QCD/electroweak/radiative correction, scalar or Higgs contribution, and intrinsic pole-normalization uncertainty is included or enclosed. The separate observable-response kernel includes the matched external production/decay vertices, coherent propagation and interference, nonresonant terms, and detector-response uncertainties;

3. the bridge $\mathbf{B}_{\text{norm}}$ of Definition 17.7, including the electroweak input-scheme map. The kernel uses $G_F^{(0)}$, or its explicitly proved image in $\mathfrak{s}_{\text{EW}}$, rather than silently identifying it with G_μ .

The physical heavy-pole positions are the lower-half-plane solutions on the declared analytically continued sheet that are continuously connected to the source heavy sector,

$$\det \mathcal{D}_{X,b}^{-1}(z; M, \epsilon, \theta) = 0.$$

Write $\mathcal{Z}_{X,b}(M; \epsilon, \theta)$ for the resulting unordered multiset of source-connected relevant-sheet poles. At each isolated simple root z_t , let v_t and u_t be right and left null vectors of $\mathcal{D}_{X,b}^{-1}(z_t)$. The target-complete certificate must compute or enclose the propagator Laurent residue in the declared reduction and scheme,

$$\text{Res}_{z=z_t} \mathcal{D}_{X,b}(z) = \frac{v_t u_t^\dagger}{u_t^\dagger [\partial_z \mathcal{D}_{X,b}^{-1}](z_t) v_t},$$

with the corresponding declared chiral generalization; the nonzero denominator is part of the simple-root certificate. That residue is not asserted to be separately physical: the response certificate must combine it with the corresponding vertices and required nonresonant contributions and prove gauge-parameter independence of the retained observable response through the declared order. For a typed simple-pole chart $t = (X, b, \sigma_z, r, \mathcal{R}, \mathfrak{s}_{\text{EW}})$, define its certified continuation domain

$$\Xi_t := \left\{ (M, \epsilon, \theta) : \begin{array}{l} M \in H(I_E), \epsilon \in \mathcal{E}_X^{\text{adm}}, \theta \in \Theta_{X,b}, \\ z_t \in \mathcal{Z}_{X,b}(M; \epsilon, \theta) \text{ exists as an isolated simple root and } -2\Im z_t > 0 \end{array} \right\}.$$

For every $(M, \epsilon, \theta) \in \Xi_t$, define

$$\Gamma_t^{\text{pole}}(M; \epsilon, \theta) := -2 \Im z_t(M; \epsilon, \theta) > 0,$$

and

$$c\tau_t(M; \epsilon, \theta) = \frac{\hbar c}{\Gamma_t^{\text{pole}}(M; \epsilon, \theta)}.$$

The licensed polewise lifetime output is the chart-qualified set-valued envelope

$$\mathcal{T}_t(M) = \left\{ \frac{\hbar c}{-2 \Im z_t(M; \epsilon, \theta)} : (M, \epsilon, \theta) \in \Xi_t \right\}.$$

On the jointly covered weak-record event, source containment gives $\epsilon_X \in \mathcal{E}_X^{\text{adm}}$. Therefore, for any fixed true tuple $(M_X, \epsilon_X, \theta_X) \in \Xi_t$,

$$c\tau_t(M_X; \epsilon_X, \theta_X) \in \mathcal{T}_t(M_X).$$

The weak-record failure probability is at most α_{weak} ; chart, theory, and nuisance-domain failures retain their separately declared qualification and are combined with it in α_{spec} when P75-X8 is invoked. If the allowed parameter envelope crosses a collision, exceptional point, channel threshold, or stratum frontier, the pole packet is transported as the unordered multiset $\mathcal{Z}_{X,b}$ and covered by separate continuation charts; no global r is asserted. When poles overlap within response resolution, the experimental record is generated by the full gauge-complete

coherent production–propagation–decay observable response built from $\mathcal{D}_{X,b}$, matched vertices and interference, and all required nonresonant terms; it is not replaced by a mixture of independently decaying exponentials or by independently inferred pole lifetimes.

If the bridge licenses the usual sub- W factorization of the intrinsic pole width obtained from the complete self-energy/pole kernel, one may write on a certified simple-pole chart

$$\Gamma_i^{\text{pole}}(M; \epsilon, \theta) = \epsilon \frac{(G_F^{(0)})^2 M^5}{192\pi^3} C_i^{\text{pole}}(M, \epsilon, \theta; s_{EW}),$$

where M is the frozen source mass coordinate used only to extract the conventional dimensional factor M^5 . Dependence on the self-energy channels, shifted pole, thresholds, finite propagators, flavor/mode structure, full Gram/structural-weight packet, pole-derivative effects, and ϵ_{row} remains in C_i^{pole} . External production/decay vertices and nonresonant response interference belong to $\mathcal{F}_i$, not C_i^{pole} . Thus even a factorized pole width need not be strictly linear in ϵ_{row} .

The current statuses are

FixedChildIndependentPhysicalNormalizationConstraint = NotReady,
 NonemptySourceWeakNormalizationIntersection = NotReady,
 TargetIndependentChildSelectionFromWeakNormalization = NotReady,
 TargetCompleteComplexPoleKernel = NotReady,
 VJPacketToPoleWidthNormalizationBridge = NotReady,
 NumericalVJPoleLifetimeEnvelope = NotReady.

The separate $D3\text{ToVJDomainBridge} = \text{NotReady}$ obstructs only transfer of $D3$ band authority; it does not obstruct the formal VJ construction on $H(I_E)$.

Proof. The relevant-sheet zeros of the renormalized inverse propagator, rather than eigenvalues of an absorptive matrix in isolation, define the basis-independent unstable-state poles. For an isolated simple energy pole $z = m - i\Gamma^{\text{pole}}/2$, the proper-lifetime coordinate follows. The implicit-function theorem gives a continuous local continuation on each Ξ_i . At a loss of simplicity or a stratum frontier, the unordered pole multiset and the full coherent response replace the chart-local label.

The normalization bridge is required because the PCR-7 Gram packet, a decay-kernel mixing normalization, and a measured weak rate are distinct typed objects. In a nonunitary light sector, converting $G_F^{(0)}$ to G_μ can depend on the input scheme, muon-decay normalization, and accessible heavy channels. Exact degeneracy of $M_X^{(0)} = MI_{E_H}$ does not force either $\Delta M_{X,b}$ or $\Gamma_{X,b}$ to be scalar or mutually commuting. Tree-level degeneracy therefore cannot license a common lifetime or width-eigenvalue interpretation. [24, 25] $\square$

Corollary 17.9 (Commuting narrow-width reduction). *Assume, at the declared accuracy, an energy-independent Wigner–Weisskopf or narrow-width reduction*

$$H_{X,b}^{\text{eff}} = A_{X,b} - \frac{i}{2}\Gamma_{X,b}, \quad A_{X,b} = M_X^{(0)} + \Delta M_{X,b},$$

with $A_{X,b}$ and $\Gamma_{X,b}$ Hermitian, $\Gamma_{X,b} \geq 0$, and $[A_{X,b}, \Gamma_{X,b}] = 0$. Here $\Gamma_{X,b}$ is defined and frozen only in the declared real-energy/on-shell reduction; no separate complex continuation of it is used. Then a common unitary basis gives

$$U^\dagger A_{X,b} U = \text{diag}(m_r), \quad U^\dagger \Gamma_{X,b} U = \text{diag}(\gamma_r), \quad \gamma_r \geq 0,$$

and

$$z_r = m_r - \frac{i}{2}\gamma_r, \quad c\tau_r = \begin{cases} \hbar c/\gamma_r, & \gamma_r > 0, \\ +\infty, & \gamma_r = 0. \end{cases}$$

The displayed hypotheses therefore guarantee that the width-matrix eigenvalues equal the pole widths. Without an equivalent pole-level certificate, diagonalizing $\Gamma_{X,b}$ separately does not in general determine pole lifetimes.

17.5 Coverage-preserving source-parameter inversion and pole-mass projection

Theorem 17.10 (P75-X8: typed nuisance-envelope inversion and physical pole-mass projection). *Under Theorem 17.8, freeze the VJ child, branch set, physical complex-pole strata σ_z , response class, electroweak input scheme, normalization set, complex-pole kernel, nuisance envelopes, typing rule, inversion rule, and exhaustive finite route-assigned type universes before opening the independent response record.*

Let $R_{\text{raw}}^{\text{ind}} \in \mathcal{R}_{\text{raw}}^{\text{ind}}$ be a raw response record formed from data disjoint from the records used to establish the source-certified D3 benchmark, from the weak-rate record used to construct $\mathcal{E}_X^{\text{weak}}$, and from every record used to construct the pole kernel or set its nuisance envelope. Freeze one hypothesis-independent availability projection M_{ind} and the finite-dimensional effective observed-response space

$$\mathcal{Y}^{\text{ind}} := \text{ran } M_{\text{ind}} \subseteq \mathbb{R}^{N_{\text{ind}}},$$

with its Euclidean norm, induced topology, and Borel sigma-field, together with one ordered coordinate list and a measurable, type-independent statistic

$$S^{\text{ind}} : \mathcal{R}_{\text{raw}}^{\text{ind}} \longrightarrow \mathcal{Y}^{\text{ind}}, \quad R^{\text{ind}} := S^{\text{ind}}(R_{\text{raw}}^{\text{ind}}).$$

Every retained coordinate is a genuinely defined observed summary under every admitted hypothesis, and each model predicts that same summary; a physically observed bin with predicted zero rate is an ordinary model prediction, not padding. If a coordinate is unavailable, M_{ind} gives it zero weight (equivalently quotients it out) for every type, and it contributes to neither distance, conditioning, nor inversion. All response maps, confidence regions, bridge-exported response regions, distances, conditioning statements, and inversions below use this same effective normed space and topology. No support, missingness pattern, mask, or type-dependent rebinning may reveal or select the type label. Before $R_{\text{raw}}^{\text{ind}}$ is opened, freeze a finite admitted type universe $\mathfrak{T}_{\text{adm}}$ and a deterministic route map

$$q : \mathfrak{T}_{\text{adm}} \longrightarrow \{\tau, R\}.$$

Write $\mathfrak{T}_{\tau} = q^{-1}(\tau)$ and $\mathfrak{T}_R = q^{-1}(R)$, so that

$$\mathfrak{T}_{\text{adm}} = \mathfrak{T}_{\tau} \cup \mathfrak{T}_R.$$

For either $S = \overline{\mathbb{L}}_{+, \ell_0}$ or the finite-dimensional Polish space $S = \mathcal{Y}^{\text{ind}}$, write $\text{Cl}(S)$ for the hyperspace of closed subsets equipped with its Effros–Borel structure. The two confidence procedures below are measurable into these declared hyperspaces. The two routes have the following pre-record qualifications:

1. $\mathfrak{T}_{\tau}$, containing only simple-pole types for which the response certificate proves that the isolated pole lifetime is identifiable uniformly on the entire routed domain Ξ_t , the map $c\tau_t : \Xi_t \rightarrow \overline{\mathbb{L}}_{+, \ell_0}$ is continuous (hence Borel measurable), and a predeclared measurable confidence procedure $I_{\tau, t}^{\text{ind}} : \mathcal{Y}^{\text{ind}} \rightarrow \text{Cl}(\overline{\mathbb{L}}_{+, \ell_0})$ is supplied and evaluated as $I_{\tau, t}^{\text{ind}}(R^{\text{ind}})$;
2. $\mathfrak{T}_R$, containing every remaining admitted type, including collision/coherent types with $(\sigma_z, r) = (\star, \star)$ and simple-pole types whose lifetime is not identifiable. Every such type must have a Borel-measurable full gauge-complete production–propagation–decay observable-response procedure

$$\mathcal{F}_t : \mathcal{U}_t^R \longrightarrow \mathcal{Y}^{\text{ind}},$$

on a certified validity domain

$$\mathcal{U}_t^R \subseteq H(I_E) \times \mathcal{E}_X^{\text{adm}} \times \Theta_{X, b(t)}$$

and a predeclared measurable confidence procedure $C_{R, t}^{\text{ind}} : \mathcal{Y}^{\text{ind}} \rightarrow \text{Cl}(\mathcal{Y}^{\text{ind}})$, evaluated as the response-space region $C_{R, t}^{\text{ind}}(R^{\text{ind}}) \subseteq \mathcal{Y}^{\text{ind}}$.

For every $t \in \mathfrak{T}_R$, $\mathcal{F}_t$ is constructed from the full resummed matrix propagator. On a licensed isolated simple-pole chart it may use the certified Laurent residue as one component of that construction. At a multiple pole, collision, exceptional point, or coherent stratum, no simple-pole residue is presumed: the response and its inversion retain the full matrix propagator. In every case the gauge-complete response also contains the appropriate external vertices, interference, required nonresonant terms, and detector response. Positive acceptance or the existence of pole coordinates does not by itself place a type in $\mathfrak{T}_{\tau}$; polewise route assignment requires the stated domain-uniform lifetime-identifiability certificate. If identifiability holds only on a parameter subset, that subset must be frozen as a separate pre-record subtype and its complement routed to $\mathfrak{T}_R$; otherwise the entire type is routed to $\mathfrak{T}_{\tau}$. If any admitted type has neither the uniformly certified lifetime route nor a full-response procedure, the premises

of P75-X8 fail before the record is opened and the corresponding localization is **NotReady**. No type is removed from either frozen route after R^{ind} is observed; a confidence procedure may return the empty set, but that outcome counts against its stated coverage.

The polewise and full-response source-parameter inverse sets are

$$\mathcal{I}_{M,\tau}^{\text{src}}(R^{\text{ind}}) := \{M \in H(I_E) : \exists t \in \mathfrak{T}_\tau, \\ \exists \epsilon \in \mathcal{E}_X^{\text{adm}}, \exists \theta \in \Theta_{X,b(t)}, \\ (M, \epsilon, \theta) \in \Xi_t, \quad c\tau_t(M; \epsilon, \theta) \in I_{\tau,t}^{\text{ind}}(R^{\text{ind}})\}.$$

$$\mathcal{I}_{M,R}^{\text{src}}(R^{\text{ind}}) := \{M \in H(I_E) : \exists t \in \mathfrak{T}_R, \\ \exists \epsilon \in \mathcal{E}_X^{\text{adm}}, \exists \theta \in \Theta_{X,b(t)}, \\ (M, \epsilon, \theta) \in \mathcal{U}_t^R, \quad \mathcal{F}_t(M, \epsilon, \theta) \in C_{R,t}^{\text{ind}}(R^{\text{ind}})\},$$

and

$$\mathcal{I}_M^{\text{src}}(R^{\text{ind}}) = \mathcal{I}_{M,\tau}^{\text{src}}(R^{\text{ind}}) \cup \mathcal{I}_{M,R}^{\text{src}}(R^{\text{ind}}).$$

Here M is the frozen source/renormalized mass coordinate in $H(I_E)$, not a physical pole mass. Nuisance coordinates and unresolved branch or pole-mode labels are projected by union, not fixed at a best-fit value. If a unique branch, mode, response, and scheme are independently assigned, the corresponding singleton type may be used.

The distinct physical output is obtained by projecting the accepted witnesses through the complex-pole kernel. Define the route witness sets $\mathcal{A}_\tau(R^{\text{ind}})$ and $\mathcal{A}_R(R^{\text{ind}})$ to be the tuples (t, M, ϵ, θ) satisfying, respectively, the braces in the definitions of $\mathcal{I}_{M,\tau}^{\text{src}}$ and $\mathcal{I}_{M,R}^{\text{src}}$. Then the coverage-preserving physical pole-mass set is

$$\mathcal{I}_{m_{\text{pole}}}^{\text{phys}}(R^{\text{ind}}) := \left\{ \mathfrak{R}_{z_t}(M; \epsilon, \theta) : (t, M, \epsilon, \theta) \in \mathcal{A}_\tau(R^{\text{ind}}) \right\} \\ \cup \bigcup_{(t,M,\epsilon,\theta) \in \mathcal{A}_R(R^{\text{ind}})} \{ \mathfrak{R}_z : z \in \mathcal{Z}_{X,b(t)}(M; \epsilon, \theta) \}.$$

The second line is a multiplicity-erased projection of the unordered physical pole multiset; it does not invent global pole labels across collisions.

Core inverse measurability. For the coverage statements below, require every routed domain Ξ_t and $\mathcal{U}_t^R$ to be Borel in its declared Polish parameter product, the chart pole map z_t to be Borel on Ξ_t (as already follows from its stated continuity), and the multiplicity-erased graph of $(M, \epsilon, \theta) \mapsto \mathcal{Z}_{X,b(t)}(M; \epsilon, \theta)$ to be Borel on every full-response domain. If a normalization set, nuisance domain, routed domain, pole graph, type universe, route map, or confidence procedure depends on frozen design data ω , its corresponding graph must be jointly measurable in ω and the displayed record and parameter variables. All probabilities are taken in the completions of the induced design and record laws.

Let

$$G_M := \{(y, M) : M \in \mathcal{I}_M^{\text{src}}(y)\}, \quad G_{\text{pole}} := \{(y, m) : m \in \mathcal{I}_{m_{\text{pole}}}^{\text{phys}}(y)\}.$$

Because $\mathfrak{T}_{\text{adm}}$ is finite, the confidence-set membership graphs are Borel, and the routed model maps are Borel, the route-witness relations are Borel. Therefore G_M and G_{pole} are analytic projections of Borel sets and hence universally measurable. In particular, every fixed-target inclusion event and every open-set hit event used below is measurable under the completed law. No closedness, connectedness, interval form, or numerical localization is inferred; any such property requires a separate theorem.

For each nonempty frozen route, assume parameter-uniform coverage,

$$\inf_{\substack{t \in \mathfrak{T}_\tau \\ (M, \epsilon, \theta) \in \Xi_t}} \Pr_{t, M, \epsilon, \theta} [c\tau_t(M; \epsilon, \theta) \in I_{\tau,t}^{\text{ind}}(R^{\text{ind}})] \geq 1 - \alpha_{\text{rec}},$$

and

$$\inf_{\substack{t \in \mathfrak{T}_R \\ (M, \epsilon, \theta) \in \mathcal{U}_t^R}} \Pr_{t, M, \epsilon, \theta} [\mathcal{F}_t(M, \epsilon, \theta) \in C_{R,t}^{\text{ind}}(R^{\text{ind}})] \geq 1 - \alpha_{\text{rec}},$$

with an equivalent simultaneous-coverage construction also admissible. If the weak-record construction, $\mathcal{E}_X^{\text{weak}}$, $\mathcal{E}_X^{\text{adm}}$, routed domains Ξ_t and $\mathcal{U}_t^R$, certificate, type universe, route map, or nuisance sets are outputs of design

data, all displayed probabilities and infima are required to hold almost surely conditional on the frozen design sigma-field; mere disjointness of the records is not a substitute for conditional validity. Marginal coverage followed by record-dependent type pruning is not admissible. Define the combined model/type/domain well-specification event

$$\mathcal{E}_{\text{spec}} := \left\{ \begin{array}{l} t_{\text{true}} \in \mathcal{T}_{\text{adm}}, \text{ and either} \\ q(t_{\text{true}}) = \tau, (M_{\text{true}}, \epsilon_{\text{true}}, \theta_{\text{true}}) \in \Xi_{t_{\text{true}}}, \text{ or} \\ q(t_{\text{true}}) = R, (M_{\text{true}}, \epsilon_{\text{true}}, \theta_{\text{true}}) \in \mathcal{U}_{t_{\text{true}}}^R \end{array} \right\}.$$

Membership in Ξ_t or $\mathcal{U}_t^R$ already includes membership of ϵ_{true} and θ_{true} in their frozen normalization and nuisance envelopes. If $\Pr(\mathcal{E}_{\text{spec}}) \geq 1 - \alpha_{\text{spec}}$, where α_{spec} accounts for every type-admission, routed-domain, normalization-envelope, and nuisance-envelope failure, then, for that predeclared true inferential target, $\mathcal{I}_M^{\text{src}}$ contains the true source coordinate with probability at least

$$1 - \alpha_{\text{rec}} - \alpha_{\text{spec}},$$

without any independence assumption. On the lifetime route the same bound applies to inclusion of the one type-qualified pole mass $\mathcal{R}_{\tau_t}^{\text{true}}$ in $\mathcal{I}_{m_{\text{pole}}}^{\text{phys}}$; on the full-response route it applies to inclusion of the complete multiplicity-erased source-connected pole-mass support. If an entire simple-pole packet is to be covered through several chart-local lifetime targets, the confidence construction must be simultaneous over all of those targets (or its familywise error must be budgeted explicitly); marginal type-uniform coverage alone does not license a whole-packet statement. Take $\alpha_{\text{spec}} = 0$ only when type admission, routed-domain validity, normalization membership, and nuisance membership are all deterministically certified.

If, for a fixed type, ϵ , and θ , the map $M \mapsto c\tau_t(M; \epsilon, \theta)$ is continuous and strictly monotone, the inverse image of a lifetime interval is an interval and the inverse of an exact lifetime is single-valued. The union over branches, modes, schemes, or nuisance coordinates need not be connected and is not called an interval without a separate connectedness theorem.

The D3 band already has the source-certified benchmark authority proved by PCR-5D3 and P75-X6. Transport of that authority into the VJ source-coordinate inversion requires a D3ToVJDomainBridge supplying $\mathcal{D}_{D3 \rightarrow X} \subseteq \mathcal{Z}_{D3, X}$. Before $\mathcal{R}_{\text{raw}}^{\text{ind}}$ is opened, that bridge must freeze its relation, its set-valued procedures and, for every related (M_D, M_X) , a bridge-admitted type set

$$\mathcal{T}_{D3}(M_D, M_X) \subseteq \mathcal{T}_{\text{adm}}.$$

This set exhausts every true type that the bridge declares compatible with the related mass pair. Define without post-record pruning

$$\text{Compat}_{D3 \rightarrow X}^{\tau}(M_D, M_X) := \mathcal{T}_{D3}(M_D, M_X) \cap \mathcal{T}_{\tau},$$

$$\text{Compat}_{D3 \rightarrow X}^R(M_D, M_X) := \mathcal{T}_{D3}(M_D, M_X) \cap \mathcal{T}_R,$$

so that, at every related pair,

$$\mathcal{T}_{D3}(M_D, M_X) = \text{Compat}_{D3 \rightarrow X}^{\tau}(M_D, M_X) \dot{\cup} \text{Compat}_{D3 \rightarrow X}^R(M_D, M_X).$$

In addition to $\mathcal{C}_{D3 \rightarrow X}^{\tau}$, the bridge exports for every full-response type a region

$$\mathcal{C}_{D3 \rightarrow X, t}^R(M_D, M_X) \subseteq \mathcal{Y}^{\text{ind}}$$

constructed in the same effective space from the same type-independent statistic and availability projection. The two route-specific source-coordinate compatibility sets are

$$\begin{aligned} \mathcal{I}_{M, \tau}^{D3\text{-cal,src}} &:= \{M_X \in H(I_E) : \exists M_D \in [28, 30] \text{ GeV}, (M_D, M_X) \in \mathcal{D}_{D3 \rightarrow X}, \\ &\quad \exists t \in \text{Compat}_{D3 \rightarrow X}^{\tau}(M_D, M_X), \\ &\quad \exists \epsilon \in \mathcal{E}_X^{\text{adm}}, \exists \theta \in \Theta_{X, b(t)}, \\ &\quad (M_X, \epsilon, \theta) \in \Xi_t, \quad c\tau_t(M_X; \epsilon, \theta) \in \mathcal{C}_{D3 \rightarrow X}^{\tau}(M_D, M_X)\}. \end{aligned}$$

and

$$\begin{aligned} \mathcal{I}_{M, R}^{D3\text{-cal,src}} &:= \{M_X \in H(I_E) : \exists M_D \in [28, 30] \text{ GeV}, (M_D, M_X) \in \mathcal{D}_{D3 \rightarrow X}, \\ &\quad \exists t \in \text{Compat}_{D3 \rightarrow X}^R(M_D, M_X), \\ &\quad \exists \epsilon \in \mathcal{E}_X^{\text{adm}}, \exists \theta \in \Theta_{X, b(t)}, \\ &\quad (M_X, \epsilon, \theta) \in \mathcal{U}_t^R, \quad \mathcal{F}_t(M_X, \epsilon, \theta) \in \mathcal{C}_{D3 \rightarrow X, t}^R(M_D, M_X)\}, \end{aligned}$$

with $\mathcal{I}_M^{D3\text{-cal,src}} := \mathcal{I}_{M, \tau}^{D3\text{-cal,src}} \cup \mathcal{I}_{M, R}^{D3\text{-cal,src}}$. P75-X6 certifies 0.2 mm only as an exact ATLAS-reweight-grid response coordinate on the D3 chart. Replacing $\mathcal{C}_{D3 \rightarrow X}^{\tau}$ by its singleton $\{0.2 \text{ mm}\}$ requires both a full-source lifetime identity

and an exact lifetime-coordinate bridge to the VJ chart. Define the bridge-qualified event

$$\mathcal{E}_{D3} := \mathcal{E}_{\text{spec}} \cap \left\{ \begin{array}{l} (M_{D,\text{true}}, M_{X,\text{true}}) \in \mathcal{D}_{D3 \rightarrow X}, \\ t_{\text{true}} \in \mathcal{T}_{D3}(M_{D,\text{true}}, M_{X,\text{true}}), \text{ and either} \\ q(t_{\text{true}}) = \tau, \quad (M_{X,\text{true}}, \epsilon_{\text{true}}, \theta_{\text{true}}) \in \Xi_{t_{\text{true}}}, \quad c\tau_{t_{\text{true}}}(M_{X,\text{true}}; \epsilon_{\text{true}}, \theta_{\text{true}}) \in C_{D3 \rightarrow X}^{\tau}(M_{D,\text{true}}, M_{X,\text{true}}), \text{ or} \\ q(t_{\text{true}}) = R, \quad (M_{X,\text{true}}, \epsilon_{\text{true}}, \theta_{\text{true}}) \in \mathcal{U}_{t_{\text{true}}}^R, \quad \mathcal{F}_{t_{\text{true}}}(M_{X,\text{true}}; \epsilon_{\text{true}}, \theta_{\text{true}}) \in C_{D3 \rightarrow X, t_{\text{true}}}^R(M_{D,\text{true}}, M_{X,\text{true}}) \end{array} \right\}.$$

If the relation, bridge-admitted type multifunction, or either benchmark-response-region multifunction is learned from benchmark/design data ω , the bridge certificate must impose the following joint measurability conditions before the independent record is opened:

1. $\omega \mapsto \mathcal{D}_{D3 \rightarrow X}(\omega)$ is an Effros–Borel random closed subset of $\mathcal{Z}_{D3,X}$, with measurable graph in $\mathcal{G}_{\text{bridge}} \otimes \mathcal{B}(\mathcal{Z}_{D3,X})$;
2. $(\omega, M_D, M_X) \mapsto \mathcal{T}_{D3}(\omega; M_D, M_X)$ is jointly measurable into the finite power set $2^{\mathcal{T}_{\text{adm}}}$ with its discrete sigma-field (equivalently, every type-membership graph is measurable);
3. $(\omega, M_D, M_X) \mapsto C_{D3 \rightarrow X}^{\tau}(\omega; M_D, M_X)$ and, for each $t \in \mathcal{T}_R$, $(\omega, M_D, M_X) \mapsto C_{D3 \rightarrow X, t}^R(\omega; M_D, M_X)$ are jointly Effros–Borel measurable closed-valued multifunctions into $\text{Cl}(\overline{\mathcal{L}}_{+, \ell_0})$ and $\text{Cl}(\mathcal{Y}^{\text{ind}})$, respectively.

Together with the measurable model maps above, the bridge certificate must verify that $\mathcal{E}_{D3}$ and the two displayed existential source-coordinate projections are measurable random sets; this follows, for example, from compact parameter sections and closed graphs, and is otherwise an explicit bridge premise. The bridge-specific loss of the core well-specification event must obey

$$\Pr(\mathcal{E}_{\text{spec}} \setminus \mathcal{E}_{D3} \mid \mathcal{G}_{\text{bridge}}) \leq \alpha_{\text{bridge}} \quad \text{almost surely}$$

for the frozen bridge-design sigma-field $\mathcal{G}_{\text{bridge}}$. On $\mathcal{E}_{D3}$, the true VJ source coordinate belongs to $\mathcal{I}_M^{D3\text{-cal,src}}$; therefore its benchmark-compatibility coverage is at least

$$1 - \alpha_{\text{spec}} - \alpha_{\text{bridge}}$$

without an independence assumption. That failure budget is carried by the CalibrationMatchedCompatibilitySet qualification. This union is the coverage-qualified VJ transport of the source-certified D3 benchmark and may be broad if the bridge response is weak. It is not an independent mass-prediction authority, is not part of the core $\mathcal{I}_M^{\text{src}}$, and never by itself localizes a physical pole mass. Without the bridge its VJ transport is NotReady, while the underlying D3 benchmark authority remains established and the D3 interval does not appear in the core VJ inversion.

At the current certificate,

$\begin{aligned} \text{VJMassDomain} &= H(I_E), \\ \text{NumericalEndpointsOf } H(I_E) &= \text{NotReady}, \\ \text{D3ToVJDomainBridge} &= \text{NotReady}, \\ \text{NumericalSourceMassParameterLocalization} &= \text{NotReady}, \\ \text{NumericalPhysicalPoleMassLocalization} &= \text{NotReady}. \end{aligned}$

Proof. Fix a predeclared true inferential target. On $\mathcal{E}_{\text{spec}}$ its type is admitted, its route was assigned before the record, and its true tuple—including the normalization and nuisance coordinates—lies in the corresponding routed domain. Because no record-dependent pruning is allowed and the record guarantee is uniform over every nonempty route domain, it applies at the true parameter point. On the lifetime route, the target’s true proper lifetime lies in the corresponding interval, so the true source coordinate is in $\mathcal{I}_{M,\tau}^{\text{src}}$ and the real part of that chart-local simple physical pole is in $\mathcal{I}_{m_{\text{pole}}}^{\text{phys}}$. This is a marginal target-qualified statement unless the procedure supplies simultaneous coverage over the complete simple-pole packet. On the full-response route, the true response lies in its confidence region, so the true source coordinate is in $\mathcal{I}_{M,R}^{\text{src}}$, and every real part in the true source-connected pole support is included by the unordered-multiset projection. The stated bound follows by a union bound and does not require statistical independence. Conditional validity given a random frozen design yields the same unconditional bound after averaging.

The existential projection is necessary for coverage: choosing one unresolved branch, mode, response model, or nuisance best fit can remove the true realization. Monotonicity applies only after those coordinates are fixed.

For the bridge transport of the source-certified D3 benchmark, $\mathcal{E}_{D3} \subseteq \mathcal{E}_{\text{spec}}$ supplies the true related mass pair, an exhaustive bridge-admitted true type, and the corresponding lifetime- or response-region membership. The true normalization and nuisance tuple is available from $\mathcal{E}_{\text{spec}}$, so it is an admissible witness in the applicable route-specific existential set. Hence $M_{X,\text{true}} \in \mathcal{I}_M^{D3\text{-cal,src}}$ on $\mathcal{E}_{D3}$. Moreover,

$$\mathcal{E}_{D3}^c = \mathcal{E}_{\text{spec}}^c \cup (\mathcal{E}_{\text{spec}} \setminus \mathcal{E}_{D3}),$$

and the conditional bridge bound, averaged over $\mathcal{G}_{\text{bridge}}$, gives the stated $1 - \alpha_{\text{spec}} - \alpha_{\text{bridge}}$ lower coverage without independence. Finally, 0.2 mm remains source-certified evidence for the D3 response-grid benchmark, not for a full-source physical proper lifetime. Because that same value generated the D3 benchmark curve, it is not a second, independent weak or lifetime-validation record. The domain bridge and its covered route regions can transport the benchmark into the stated bridge-conditional source-coordinate confidence/compatibility set, but cannot turn the reused value into independent VJ mass-prediction authority or a VJ physical pole-mass determination. $\square$

Corollary 17.11 (Local scalar-limit source-coordinate conditioning). *Assume the commuting reduction of Corollary 17.9, a fixed typed simple pole, and*

$$\Gamma_t^{\text{pole}}(M, \epsilon, \theta) = \epsilon \frac{(G_F^{(0)})^2 M^5}{192\pi^3} C_{D,t}^{\text{pole}}(M, \epsilon, \theta), \quad C_{D,t}^{\text{pole}} > 0.$$

Define

$$\kappa_C = \frac{\partial \log C_{D,t}^{\text{pole}}}{\partial \log M}, \quad \kappa_\epsilon = \frac{\partial \log C_{D,t}^{\text{pole}}}{\partial \log \epsilon},$$

with the remaining typed coordinates held fixed. A first-order inverse condition number is licensed only on a complete local allowed envelope $\mathcal{U}$ for which

$$\inf_{(M, \epsilon, \theta) \in \mathcal{U}} |5 + \kappa_C(M, \epsilon, \theta)| \geq \kappa_{\min} > 0.$$

Let $\sigma_{\ln N_{\text{res}}}$ be the small residual fractional uncertainty in the complete multiplicative normalization

$$N_{\text{res}} = (G_F^{(0)})^2 C_{D,t}^{\text{pole}}$$

at fixed M and ϵ , after removing the M - and ϵ -induced variation already represented by κ_C and κ_ϵ . Thus an independently varied $G_F^{(0)}$ contributes $2 \delta \log G_F^{(0)}$ to $\delta \log N_{\text{res}}$. For small mutually independent uncertainties,

$$\frac{\sigma_M}{M} \simeq \frac{1}{|5 + \kappa_C|} \left[(1 + \kappa_\epsilon)^2 \left(\frac{\sigma_\epsilon}{\epsilon} \right)^2 + \left(\frac{\sigma_\tau}{c\tau} \right)^2 + \sigma_{\ln N_{\text{res}}}^2 \right]^{1/2}.$$

Here σ_M/M conditions the source/renormalized coordinate M ; it is not directly an uncertainty on $\Re z$. For correlated inputs the full covariance matrix, including input-scheme correlations, must be retained. If $5 + \kappa_C = 0$, or an allowed nuisance envelope crosses that locus, the linearized inverse is not licensed; one must use the exact set-valued inverse of Theorem 17.10 or a separately certified higher-order expansion. Discrete branch and mode ambiguity remains a union of typed sets, and physical pole-mass uncertainty must be propagated through the complex-pole kernel.

Remark 17.12 (Leading-scaling illustration only). If $C_{D,t}^{\text{pole}}$ is held locally constant, then $c\tau \propto M^{-5}$. Evaluating this scaling on the numerical coordinate [28, 30] GeV illustrates how a fixed normalization differs from the source-certified constant-response-coordinate D3 benchmark family, but does not place the VJ child in that band. Such a VJ interpretation requires the D3ToVJDomainBridge.

Nonclaim 17.13 (No dyadic branch inference from benchmark reuse). The D3 one-channel upper bound and selected 0.2 mm coordinate establish a source-certified, ATLAS-reweight-grid-anchored benchmark with positive non-direct margins. They do not establish a full-source physical lifetime, determine n_{mark} , select a physical dyadic branch, or supply an independent ϵ_{row} . Any such inference requires the promotion, input-scheme, pole-kernel, and normalization-bridge certificates above.

Current row-normalization/lifetime endpoint

PCR-7.5 proves the exact algebraic row-normalization identity, the corrected existence of the marked dyadic witness, the target-qualified bound $\epsilon_{\text{row}} \leq 1/256$, and the source-certified, ATLAS-reweight-grid-anchored authority of $c\tau_A^{\text{resp}} = 0.2$ mm across the D3 [28, 30] GeV family, together with exact discrete grid membership,

interior endpoint placement, and positive non-direct constraint margins. It does not infer an off-grid detector-response neighborhood from that membership. It also proves the correct complex-pole transport rule, the commuting-width corollary, and a coverage-preserving conditional source-parameter inversion with a separate physical pole-mass projection. The response-grid benchmark is not a full-source proper-lifetime prediction, a full detector-response validation, an independent weak-validation record, or a VJ pole-lifetime prediction. The manuscript does not currently derive an independently selected numerical physical weak normalization, a target-complete VJ complex-pole kernel, a predicted VJ lifetime envelope, a D3-to-VJ domain transfer, a numerical source-mass constraint, or a numerical physical pole-mass localization.

18 Nested structural robustness closure

Theorem 18.1 (P75-25: nested structural robustness capstone). *For each $b \in \mathcal{B}_{7.5}^{\text{ready}}$ and $X \in \text{PhysMod}_{E,VJ}^b$, the accepted PCR-6/PCR-7 endpoint admits the following exact robustness classification.*

1. **Carrier stability.** *On the declared trace-transport cell, the finite status-free support skeleton has a finite heavy-partition decoration; each proper-support skeleton carries an exact transported deletion flag $q_a \equiv 1$, and all full-support partition decorations map to one parent carrier orbit. The regularity radius r_{reg} , normal-form/update distance d_{nf} , and explicit nonfrozen guard faces give $r_{\text{car}}^b(X) \geq r_{\text{car,cert}}^b(X) > 0$. Hence on the nonempty open same-scope region $\mathcal{U}_{\text{car}}^b(X)$,*

$$|\text{TermCar}_b(X_\delta)| = 1.$$

2. **Closure stability.** *On the complete block-isometry class,*

$$\Delta'_{\text{act}} = \Omega'_H$$

exactly.

3. **Rank and aggregate support.** *If $\eta_V < s_E$ and the full mass perturbation is smaller than half the light-heavy gap, then*

$$\text{rank } \Omega'_H = 3, \quad \text{rank } P'_H = 3, \quad \lambda_{\min}(\Omega'_H) \geq (s_E - \eta_V)^2.$$

4. **Symmetry refinements.** *On the full-rank charged-current domain, exact positive flavor universality holds precisely on the equal-singular-value stratum, equivalently $\Omega'_H = \delta' I_3$ with $\delta' > 0$. Exact threefold heavy degeneracy holds precisely on the central heavy mass-perturbation stratum. Both have zero radius against unrestricted standing $U(3)_H$ breaking.*
5. **Spectral lifting and resolution.** *A nonzero traceless mass perturbation gives heavy partition $2 + 1$ or $1 + 1 + 1$ with*

$$\frac{3}{2}\rho_K \leq \text{diam Spec}(T) \leq 2\rho_K,$$

*but does not change terminal carrier cardinality inside $\mathcal{U}_{\text{car}}^b(X)$. For structural mass support, $g_{\text{str}} > 2r_m$ only identifies the true-support matching; reported intervals are resolved iff $\widehat{g}_{\text{str}} > 2r_m$, and a uniform design guarantee requires $r_m < g_{\text{str}}/4$. Otherwise the status is **ResolutionNotUniformlyCertified**. This is not a physical-pole statement.*

6. **Two-level fibrewise maximality.** *Relative to the pre-frozen variation-certified feature registry, $\mathcal{O}_{\text{core}}^{b,X}$ of Theorem 9.7 is the maximal invariant core on the certified fibrewise structural domain. The joint trace-regrouping lemma used by P75-02 first supplies an information-preserving factorization of the complete joint trace, retaining the cross-interface incidence and correlation data as well as the six displayed blocks. The functional-determination class*

$$\mathcal{O}_{\text{cov}}^{b,X} = [\pi_{\text{str}}^{b,X} \Theta_{\text{pert}}^{b,X}]_{\sim \text{fd}}$$

is the maximal target-qualified covariant structural envelope. Exact laws form a separate packet, and the displayed heavy/CCST coordinates are only a reduct.

7. **Conditional same-record model/core selection.** *On every compact carrier-plus-reference SPIC-certified subdomain, global joint signal/nuisance/control conditioning, one common ordered record space, an explicit coverage-certified reference core, and an attainable*

$$r_{\text{rec}} < \frac{1}{2} s_{\text{car,ref}},$$

the observed record selects hypotheses: on true-carrier coverage the carrier inverse contains the truth and the reference inverse is empty, and conversely on true-reference coverage. No theorem compares two simultaneously nonempty counterfactual confidence regions. On the block-isometric subdomain with $\eta_V \leq \bar{\eta}_V < s_E$, the ideal structural inequality $\|(\Delta'_{\text{act}}, \Omega'_H)\|_{\text{pair}} \geq \sqrt{6}(s_E - \bar{\eta}_V)^2$ holds; it remains independent of SPIC and is not a detector guard. The lower-Lipschitz inverses additionally require

$$\frac{2r_{\text{rec}}}{\kappa_{\text{car}}} < r_{Z,\text{car}}, \quad \frac{2r_{\text{rec}}}{\kappa_{\text{ref}}} < r_{Z,\text{ref}},$$

with r_{rec} attainable strictly above its deterministic floor. These typed record-to-parameter transports are distinct from the same-record selector, structural-mass, and Gram guards. This is a certified-domain intersection, not a universal protocol perturbation ball.

8. **Cross-child status.** For matched ready-child realizations and matched common perturbations, the meet of the two forgetful images gives the common robust core and structural reduct. The complete decorated envelopes remain separate functors; D2/D5 coordinates are not transported across children.
9. **Source-certified benchmark and separated VJ authority.** The D3 pair ([28, 30] GeV, $c\tau_A^{\text{resp}} = 0.2$ mm) is a SourceCertified_ATLASRweightGridAnchored_SearchBenchmark: every mass in the band has an explicit source-coherent physical realization and the complete non-direct LFU, oscillation, CLFV, and matching records pass. The supporting records establish a motivated, reproducible, admissible benchmark; they do not derive its endpoints or select 0.2 mm uniquely. The response coordinate is an exact interior member of the frozen discrete ATLAS lifetime-reweighting grid, and 30 GeV is a direct ATLAS mass anchor. Its 0.1 mm distance to the nearer endpoint of the declared grid span does not establish a continuous detector-response neighborhood. The frozen HEPData replay places the uniform PCR-5D3 mixing bound more than a factor 4.50 below the declared linear reconstruction of the observed Figure 4a ordinate across the band; under the ATLAS simplified-model convention this is an ordinate-compatibility screen, not a non-exclusion result. It is not a full-source proper lifetime, full detector-response validation, or unconditional source-level ATLAS non-exclusion. Separately, the VJ source domain is $H(I_E)$, with numerical endpoints NotReady; the benchmark has no numerical VJ pole-mass or pole-lifetime authority and is not asserted unique, sensitivity-optimal, or comparatively high-priority. Before the independent response record, a D3-to-VJ bridge must freeze $\mathfrak{T}_{D3} = \text{Compat}_{D3 \rightarrow X}^{\tau} \cup \text{Compat}_{D3 \rightarrow X}^R$, export covered $C_{D3 \rightarrow X}^{\tau}$ for lifetime types and same- $\mathcal{Y}^{\text{ind}} C_{D3 \rightarrow X, t}^R$ for full-response types, and forbid post-record pruning. The singleton {0.2 mm} is licensed only after a full-source lifetime identity and exact lifetime bridge. Let $\mathcal{E}_{D3} = \mathcal{E}_{\text{spec}} \cap \{\text{bridge pair/type/route membership}\}$; the conditional bridge bound $\Pr(\mathcal{E}_{\text{spec}} \setminus \mathcal{E}_{D3} \mid \mathcal{G}_{\text{bridge}}) \leq \alpha_{\text{bridge}}$ gives $\Pr(\mathcal{E}_{D3}) \geq 1 - \alpha_{\text{spec}} - \alpha_{\text{bridge}}$. The possibly uninformative union is a bridge-conditional benchmark-derived source-coordinate compatibility/confidence set outside core $\mathcal{I}_M^{\text{src}}$, not independent mass-prediction authority or physical pole-mass localization. The bridge relation lives in a compact Polish product over compact Euclidean $H(I_E)$; learned relations, finite type graphs, and C^{τ}/C^R multifunctions obey random-closed/Effros–Borel measurability. The lifetime map is continuous/Borel on Ξ_t , $\mathcal{F}_t$ is Borel on $\mathcal{U}_t^R$, and $\mathcal{E}_{D3}$ requires true-tuple domain membership before either partial map is evaluated.

10. **Marked-child row normalization.** The corrected dyadic-first witness construction gives

$$0 < \epsilon_{\text{row}} = \delta_E = s_E^2 \leq \frac{1}{256},$$

as a target-qualified marked-child bound, not an independently selected physical weak-rate value. The frozen HFLAV covariance reconstruction gives $\chi_4^2 = 7.0193$ and $p = 0.1349$ only conditional on applicability of the declared light-response model, with status ConditionalLightResponseCompatibleAt95; its common- ϵ Jacobian is zero and full VJ compatibility remains NotReady. The separate m_W calculation sets the unreported direct/indirect cross-covariance to zero only as a screening assumption and yields $\epsilon_{95}^{\text{scr}} = 1.62 \times 10^{-3}$; it is neither an independence certificate nor a covered X7 endpoint. The stop rule therefore retains 1/256, while $\epsilon_{95}^{\text{weak}}$, $\mathcal{E}_X^{\text{weak}}$, and $\mathcal{E}_X^{\text{adm}}$ remain NotReady.

11. **Conditional pole lifetime and localization.** The inverse propagator uses one sheet-holomorphic $\Sigma^{\text{ren}}(z)$; $\Delta M/\Gamma$ split only on the real/on-shell boundary, while poles and licensed isolated-simple-pole residues use the full self-energy. The certified outer $\mathcal{E}_X^{\text{src}} \ni \epsilon_X$ intersects the coverage-preserving epsilon projection $\mathcal{E}_X^{\text{weak}}$ of a jointly covered weak region to form nonempty $\mathcal{E}_X^{\text{adm}}$; α_{weak} is charged inside α_{spec} . The normalization bridge transports the Gram and structural weights into the intrinsic self-energy/pole kernel; if licensed, $\Gamma = \epsilon \mathcal{G}^{\text{self}}$ is boundary-only and C_t^{pole} contains no external response factor. The gauge-complete $\mathcal{F}_t$ is built from the full resummed propagator, using a Laurent residue only on a licensed isolated-simple-pole chart, together with external vertices, interference, nonresonant terms, and detector response; collision/coherent strata retain

full- $\mathcal{D}$ inversion or a certified complete local Laurent principal part. On the weak covered event, conditional on the true chart and nuisance tuple, the true lifetime lies in $\mathcal{T}_1(M_X)$. P75-X8 freezes exhaustive routes in finite-dimensional Euclidean/Borel $\mathcal{Y}^{\text{ind}} = \text{ran } M_{\text{ind}}$; every non-lifetime type uses $\mathcal{F}_1$, not the propagator alone. It yields $\mathcal{I}_M^{\text{src}}$ then $\mathcal{I}_{m_{\text{pole}}}^{\text{phys}}$, with targeted-pole versus full-support/familywise coverage as declared. Numerical outputs remain NotReady. Here the bridge-free $\mathcal{E}_{\text{spec}}$ combines true-type and routed-domain validity, $\mathcal{E}_X^{\text{adm}}$, the jointly covered weak-region failure α_{weak} , and nuisance coverage. The resulting core level is $1 - \alpha_{\text{rec}} - \alpha_{\text{spec}}$; optional D3-to-VJ benchmark compatibility carries its separate α_{bridge} .

The final status is

$$\begin{aligned}
 & \text{NestedRobustnessClosure}_{\text{VJ}} \\
 & \oplus \text{RobustPredictionLock}_{\text{CCST}}^{\text{SPIC}} \\
 & \oplus \text{TargetQualifiedRowBound}_{1/256} \\
 & \oplus \text{ConditionalComplexPoleEnvelope}_{H(I_E)} \\
 & \oplus \text{FrozenStructuralPrediction} \\
 & \oplus \text{DetectorEvaluationConditional}_{\text{InstantiatedSPIC+HeldOutData}} \\
 & \oplus \text{X7X8InputsNotReady}
 \end{aligned}$$

DetectorEvaluationConditional denotes same-record carrier/reference model/core selection and requires both an instantiated SPIC certificate and held-out data; held-out data alone are not the missing step. The pole/domain bridges, target-complete kernel, nonempty source/weak normalization intersection $\mathcal{E}_X^{\text{adm}}$, route universe, and coverage procedures remain specification/status gates rather than supplied certificates.

Proof. Items 1–11 are collectively supplied by Theorems 4.2, 4.3, 5.1, 5.3, 6.1, 6.2, 6.4, 7.5 to 7.7, 8.4, 9.7, 10.3, 11.1, 11.2, 12.1, 16.1, 16.7, 17.1, 17.4, 17.8 and 17.10 and corollaries 16.3, 16.4 and 16.8. Source-authorized variation witnesses prove registry-relative invariant maximality; the information-preserving complete-joint-trace regrouping/factorization certificate proves covariant maximality without discarding cross-interface relations; global SPIC bounds and collision coarsening supply detector identifiability; and the search/lifetime clauses preserve the source, benchmark, input-scheme, continuation, bridge, and held-out-data firewalls. The import ledger includes the declared PCR-3/PCR-4, fixed-domain, and SCIC authorities and their frozen transitive dependency closure, but no clause imports unresolved parent data or changes an upstream theorem. $\square$

Interpretive summary

The deepest perturbation-stable prediction is not exact degeneracy. It is the fibrewise invariant core $\mathcal{O}_{\text{core}}$: one carrier with nonzero rank-three heavy support whose inclusive charged-current weight exactly accounts for the active deficit. The complete-joint-trace covariant envelope $\mathcal{O}_{\text{cov}}$, supported by the information-preserving regrouping certificate, retains lawful variation and cross-interface relations discarded by an invariant-only summary; cross-child agreement is only the common reduct for matched ready-child realizations under matched common perturbations. Exact heavy degeneracy and full-rank positive flavor universality are stronger symmetry-protected specializations. Detector separation is globally SPIC-qualified, while numerical VJ mass and lifetime conclusions remain conditional on their explicitly named continuation, bridge, and record certificates.

19 PCR-8 robustness handoff

Theorem 19.1 (P75-26: PCR-8 robustness handoff). *PCR-7.5 exports the following status-preserving target-qualified package to PCR-8:*

- P75-H1 : ready child tags and upstream no-backflow boundary,
- P75-H2 : semantic/source perturbation category and status classifier,
- P75-H3 : heavy symmetry action, central/traceless decomposition, and multiplicity atlas,
- P75-H4 : spectral/structural mass-coordinate spread and heavy-projector stability bounds,
- P75-H5 : rank radius; full-rank $\Omega'_H = \delta' I_3$, $\delta' > 0$; and closure identity,
- P75-H6 : declared-cell trace transport, exact flags/faces, carrier radius, and nonpromotion,
- P75-H7 : structural mass support and separate mass/response/reference/physical-pole dispositions,
- P75-H8 : fixed-child fibrewise core/envelope; matched-realization/common-perturbation reduct only,
- P75-H9 : SPIC same-record selection, sampling law, floors/attainability, and typed guards,
- P75-H10 : robustness vector, falsifier hierarchy, held-out firewall, and parent boundary.

P75-H9 includes the distinct typed inverse-radius guards

$$2r_{\text{rec}}/\kappa_{\text{car}} < r_{\mathcal{Z},\text{car}}, \quad 2r_{\text{rec}}/\kappa_{\text{ref}} < r_{\mathcal{Z},\text{ref}},$$

in addition to the structural-mass and Gram guards. Same-record carrier/reference selection requires the separately attainable $r_{\text{rec}} < s_{\text{car},\text{ref}}/2$: on true-carrier coverage the carrier inverse contains truth and the reference inverse is empty, and conversely for true-reference coverage. On block isometry with $\eta_V \leq \bar{\eta}_V < s_E$, the ideal pair-space gap remains structural and is not a detector guard. The record radius must be attainable strictly above its deterministic floor. H6 applies only on declared trace-transport cells; missing chart witnesses remain conditional/NotReady. H8 remains fixed-child/fibrewise, with cross-child output only for matched ready-child realizations under matched common perturbations and no identification of decorated envelopes. The structural mass guard is $r_m < g_{\text{str}}/4$ for uniform resolution; $g_{\text{str}} > 2r_m$ alone only matches true supports, and observed intervals resolve iff $\widehat{g_{\text{str}}} > 2r_m$. Otherwise the handoff status is ResolutionNotUniformlyCertified. The tagged experimental-authority extension is:

1. P75-X1, P75-X2, P75-X2A, P75-X3, and P75-X4 export $([28, 30] \text{ GeV}, c\tau_A^{\text{resp}} = 0.2 \text{ mm})$ as the SourceCertified_ATLASRweightGridAnchored_SearchBenchmark and separately preserve $H(I_E)$ as the VJ source domain. The export includes the exact PCR-5D3 source continuum, non-direct constraint margins, the exact ATLAS lifetime-reweight-grid coordinate, the direct 30 GeV mass anchor, and the conditional HEPData Figure 4a ordinate screen. It explicitly excludes a full-source proper lifetime, full detector-response validation, and source-level ATLAS non-exclusion. If the band is retained at VJ scope, it is only a PredeclaredOperationalScanSlice_NoVJMassAuthority, not a VJ-specific theory-motivated or optimized interval. Before the independent response record, a D3 bridge freezes $\mathfrak{T}_{D3} = \text{Compat}_{D3 \rightarrow X}^r \dot{\cup} \text{Compat}_{D3 \rightarrow X}^{\text{ind}}$, exports covered $C_{D3 \rightarrow X}^r$ and same- $\mathcal{Y}^{\text{ind}}$ $C_{D3 \rightarrow X, \text{t}}^R$, and permits no post-record pruning. The singleton $\{0.2 \text{ mm}\}$ is allowed only after a full-source lifetime identity and exact lifetime bridge. With $\mathcal{E}_{D3} = \mathcal{E}_{\text{spec}} \cap \{\text{bridge pair/type/route membership}\}$, the conditional bound $\Pr(\mathcal{E}_{\text{spec}} \setminus \mathcal{E}_{D3} \mid \mathcal{G}_{\text{bridge}}) \leq \alpha_{\text{bridge}}$ gives $\Pr(\mathcal{E}_{D3}) \geq 1 - \alpha_{\text{spec}} - \alpha_{\text{bridge}}$. The possibly uninformative union is a bridge-conditional benchmark-derived source-coordinate compatibility/confidence set outside core $\mathcal{I}_M^{\text{src}}$, not independent mass-prediction authority or physical pole-mass localization. The bridge relation lives in a compact Polish product over compact Euclidean $H(I_E)$; learned relations, finite type graphs, and C^r/C^R multifunctions obey random-closed/Effros–Borel measurability. The lifetime map is continuous/Borel on Ξ_t , $\mathcal{F}_t$ is Borel on $\mathcal{U}_t^R$, and $\mathcal{E}_{D3}$ requires true-tuple domain membership before either partial map is evaluated.
2. P75-X5 proves the row-normalization identity, repaired dyadic existence, and marked-child upper bound, while retaining the independent-selection and electroweak-scheme firewalls. P75-X6 certifies $c\tau_A^{\text{resp}} = 0.2 \text{ mm}$ as an ATLASRweightGridAnchored_InteriorResponseCoordinate; it records $D3\text{FullSourceProperLifetime} = \text{NotReady}$, $D3\text{FullSourceResponseValidation} = \text{NotReady}$, and separately denies promotion to a predicted VJ pole lifetime.
3. P75-X7 uses a certified outer $\mathcal{E}_X^{\text{src}} \ni \epsilon_X$ and the coverage-preserving epsilon projection $\mathcal{E}_X^{\text{weak}}$ of a jointly covered weak-record region over mass, branch, EW scheme, muon/open-channel, response, and theory nuisances. It requires nonempty $\mathcal{E}_X^{\text{adm}}$; empty is fixed-child incompatibility, and α_{weak} is folded into α_{spec} . The inverse propagator uses one sheet-holomorphic Σ^{ren} ; $\Delta M/\Gamma$ exist only as a real/on-shell boundary split, while

poles and licensed isolated-simple-pole residues use the full self-energy. The Gram and structural weights enter intrinsic $\mathcal{G}^{\text{self}}/C_t^{\text{pole}}$. The gauge-complete $\mathcal{F}_t$ uses the full resummed propagator, a Laurent residue only on a licensed isolated-simple-pole chart, and all external vertices/interference/nonresonant/detector factors; collision/coherent strata retain full- $\mathcal{D}$ inversion or a certified complete local Laurent principal part. On the weak covered event, conditional on the true chart/nuisance tuple, the true lifetime lies in $\mathcal{T}_t(M_X)$.

4. P75-X8 freezes exhaustive routes in the finite-dimensional Euclidean/Borel space $\mathcal{Y}^{\text{ind}} = \text{ran } M_{\text{ind}}$. Only uniformly lifetime-identifiable types use $\mathcal{T}_t$; every remaining type uses gauge-complete $\mathcal{F}_t$ on $\mathcal{T}_R$, not the propagator alone. The bridge-free combined specification budget includes α_{weak} , type/domain, normalization, and nuisance failures, giving $1 - \alpha_{\text{rec}} - \alpha_{\text{spec}}$. The routes yield $\mathcal{I}_M^{\text{src}}$ then $\mathcal{I}_{m_{\text{pole}}}^{\text{phys}}$, with targeted-pole/full-support/familywise coverage preserved; optional D3 compatibility carries separate α_{bridge} .

PCR-8 may compose and audit these outputs with their premise and NotReady tags intact, but may not promote zero-radius symmetry refinements to generic robustness, infer structural mass-support identifiability from positive acceptance or pointwise Jacobian rank, identify source-mass collisions with physical complex-pole collisions, continue ordered physical-pole labels through collisions, identify the D3 source-certified benchmark band with the VJ mass domain, erase target tags, or import cross-fermion universality.

Proof. Each unconditional structural export is inherited from the cited upstream theorem. Conditional structural exports—notably H6 and the H8 certified fibrewise domain—together with protocol, pole-kernel, normalization, and inversion exports, carry their premises and status explicitly so that no claim is strengthened by handoff. Readiness does not discharge a NotReady promotion certificate. The package separates structural, protocol, empirical, and update authority, so PCR-8 can verify lawful composition without introducing a new mechanism. Any stronger statement requires the named certificate, an independent physical record, or a new target/source theorem. $\square$

Status 19.2 (PCR-8 readiness). The robustness handoff is

PCR75-HandoffReady_{child-qualified}

This denotes readiness of the typed theorem package, not readiness of every conditional empirical promotion.

20 Adversarial robustness map

Attack	Faithful burden	Disposition
Arbitrary small matrix change is same-scope	Preserve frozen source equations, support, target, and one-apex physicality	Blocked by P75-03/04
PCR-6 terminal closure by itself supplies a positive carrier radius	Exhibit a trace-transport cell, exact proper-support deletion flags, one full-support parent orbit, and explicit regularity/guard/update faces	Closure is pointwise; the positive radius is supplied separately by P75-13 on its declared cell
Degeneracy has a positive generic radius	Exhibit a ball excluding all nonzero standing traceless perturbations	False; explicit zero-radius counterperturbation
Mass splitting implies a second carrier	Produce an independently eligible complete bearer before the promotion boundary	Blocked by P75-13–15
The repeated pair in a $2 + 1$ spectrum forces zero structural support gap	Use the multiplicity-erased unordered structural support rather than labeled eigenvalue pairs	Blocked by P75-16: g_{str} is the doublet–singlet structural gap
Rank can fail inside $\eta_V < s_E$	Produce a singular V'_H in the singular-value ball	Impossible by P75-10
Flavor universality follows from rank three	Show all full-rank perturbations have equal singular values	False; anisotropic counterexample
Closure equality requires degeneracy	Break $\Delta'_{\text{act}} = \Omega'_H$ while preserving complete block isometry	Impossible by P75-12

Detector nonresolution restores degeneracy	Identify structural eigenvalue equality with overlapping confidence intervals	Blocked by P75-16/20
Appending spectral spread or $\lambda_{\min}$ defeats maximality	Produce a target-authorized structural output not factoring through the complete structural trace	Blocked by P75-17: O_{cov} is the functional-determination class of that trace, while O_{core} is maximal only relative to its pre-frozen variation-certified registry
Positive acceptance or pointwise Jacobian rank proves structural mass-support tomography	Prove a global lower-Lipschitz response on every joint signal/nuisance/control core in one common record space and positive separation of every core pair outside one declared collision-coarsening neighborhood	False without SPIC; the exempt pieces are analyzed together in that declared coarser quotient and returned as an adjacent-strata union, while physical complex poles remain a separate P75-X7 axis
Reference can be reached by small perturbation	Reach zero heavy weight with $\eta_V < s_E$	Impossible by P75-10/21
Held-out data may select robustness metric	Preserve precomparison authority while choosing metric after data	Blocked by P75-24
Child theorem is universalized	Preserve unresolved parent/source categories while applying child margins	Blocked by P75-00/18/26

20.1 Authority and inversion attacks

Attack	Faithful burden	Disposition
Every positive open interval contains a reciprocal power of two	Prove intersection with the sparse set $\{2^{-n}\}$	False; P75-X5 selects 2^{-n} inside the uniform punctured pass cell before freezing its box
Marked-child dyadic value is a physical selection	Act on a comparison class not already restricted to the marked child	Blocked by the P75-X5 selection firewall
D3 benchmark $\epsilon_{0.2}(M)$ equals VJ ϵ_{row}	Supply a typed VJ packet-to-pole-width normalization theorem	The bridge B_{norm} is NotReady
$c\tau_A^{\text{resp}} = 0.2 \text{ mm}$ is a D3 or VJ proper-lifetime prediction	Derive a full-source width, then independently fix the VJ normalization and pole kernel for VJ transfer	False: it remains the <code>ATLASReweightGridAnchored_InteriorResponseCoordinate</code> ; <code>D3FullSourceProperLifetime</code> = NotReady
Exact mass degeneracy forces a common width	Prove equality of the relevant pole imaginary parts or scalar absorptive response	P75-X7 blocks the inference; commutation alone gives modewise width eigenvalues, not equal widths
W -rest-frame length is the pp laboratory mean	First supply the D3 full-source lifetime identity, then the production-boost, angular, detector, and reconstruction convolution	The pp -response certificate is NotReady

D3 [28, 30] GeV is the VJ domain or a VJ theory-motivated priority	Supply a typed mass-domain relation into $H(I_E)$ and a separate optimization theorem for any priority claim	The SourceCertified_ATLASRweightGridAnchor_SearchBenchmark authority is established, with only 30 GeV directly anchored by ATLAS; VJ scope is PredeclaredOperationalScanSlice_NoVJMassAuthority, and D3ToVJDomainBridge = NotReady
Width eigenvalues are physical lifetimes	Prove the commuting, controlled-energy-dependence narrow-width reduction	Otherwise use chart-local complex poles of $\det \mathcal{D}^{-1} = 0$; only the commuting corollary permits $\gamma_r = 0$ and then gives infinite lifetime Blocked by P75-X8 typing
An untyped lifetime record localizes mass	Match child, branch, continuation chart or coherent stratum, response, and input scheme	
Overlapping poles may be inverted as separate lifetimes	Prove polewise lifetime identifiability throughout the response domain	Otherwise invert the full coherent-response confidence region
Profiling one nuisance point preserves coverage	Show the true nuisance and unresolved mode cannot have been discarded	False; P75-X8 uses existential projection/union
Benchmark-forming anchor may also validate the inversion	Exhibit data disjointness from benchmark formation, normalization, kernel, and nuisance formation	Blocked by the independence firewall; the response-grid-anchor authority is unchanged, while D3FullSourceResponseValidation = NotReady
Linear inversion remains valid at $5 + \kappa_C = 0$	Supply a nonzero local derivative or a controlled higher-order inverse	False; use the set-valued inverse

Criterion 20.1 (Faithful counterexample burden). A counterexample to a positive robustness theorem must preserve the exact child target, source ontology, perturbation status, norm, terminal witness family, one-apex physicality, and protocol tag of that theorem while violating its conclusion. A richer source, altered target, or different detector quotient is an update, not a same-scope defeat. A counterexample to an extension inference must additionally preserve the typed child/branch/continuation-chart-or-coherent-stratum/response/input-scheme datum, calibration partition, nuisance envelope, and declared mass domain. Dropping any of those fields changes the claim being tested.

Status 20.2 (Strongest inversion endpoint). The lawful endpoint is the conditional, coverage-preserving complex-pole inverse set on $H(I_E)$. Its theorem is proved, but the independent physical normalization, certified pole kernel and continuation atlas, normalization bridge, exhaustive compatible types, independent polewise-identifiable or coherent-response record, and numerical endpoints required for a numerical VJ result remain NotReady.

21 Conditions for revision

The core robustness theorem reopens only under an exact change at one of the following loci.

1. **Source ontology.** A new primitive heavy support, interaction, or constitutive law is admitted into the child source.
2. **Target structure.** A new observable or terminal role distinguishes heavy directions or changes bearer eligibility.

3. **Symmetry action.** A standing heavy operator fails the complete $U(3)_H$ descent assumed at the unperturbed point.
4. **Spectral gap.** The heavy cluster merges with light or other target-active support.
5. **Charged-current rank.** A lawful path reaches the singular-value boundary.
6. **Block isometry.** The complete source permits loss, absorption, or additional untracked charged-current support.
7. **Terminal transport.** The trace-transport cell is left, a proper-support deletion flag or its typing guard is lost, a second full-support carrier orbit appears, or the normal-form/update face is reached. Pointwise terminal closure alone is not a positive-radius certificate.
8. **Structural mass multiplicity.** The distinct structural mass-support gap g_{str} reaches the declared resolution boundary, including the doublet–singlet gap on a $2 + 1$ stratum. This does not determine physical complex-pole multiplicity.
9. **Structural-output trace.** A target-authorized structural output is exhibited that does not factor through $\pi_{\text{str}}\Theta_{\text{pert}}$, or the pre-frozen variation registry omits a source-authorized displacement relevant to the claimed invariant core.
10. **Protocol response.** A perturbation inside the declared structural domain is not covered by the finite response family, a fine split core loses its structural-gap floor, a carrier or explicit reference core loses coverage, the joint quotient response loses its global lower-Lipschitz bound, a required non-coarsened image separator vanishes in the common record space, or any typed record-separator, carrier/reference parameter-inverse, transported-mass, or Gram-operator guard reaches its threshold.
11. **Held-out observation.** Future data falsify one of the frozen packet or robustness inequalities.

A revision at one locus does not automatically reopen every upstream theorem. The falsifier hierarchy of Theorem 14.1 determines the minimal affected dependency level.

21.1 Promotion and numerical-endpoint conditions

The search/lifetime extension changes status only when the corresponding typed certificate is supplied or falsified:

1. **Dyadic witness formation.** New upstream evidence may revise the punctured strict-pass cell or its finite guard frontiers. No argument may instead assume that $\{2^{-n}\}$ intersects an arbitrary interval frozen beforehand.
2. **Fixed-child row-normalization constraint and child selection.** For the already marked child, a statistically independent weak-rate record may supply a jointly covered region for ϵ_{row} and every weak-response nuisance; intersecting its ϵ_{row} projection with the source-certified set gives $\mathcal{E}_X^{\text{adm}}$, as required by P75-X7/X8. This fixed-child constraint does not select the child from a broader target-independent comparison class. That stronger selection claim requires a theorem or record valid across that class. The marked witness and $\epsilon_{\text{row}} \leq 1/256$ bound alone discharge neither empirical gate.
3. **Electroweak normalization.** An input-scheme response certificate distinguishes $G_F^{(0)}$ from G_μ and maps the PCR-7 VJ packet to the complete pole-kernel normalization. A D3 coefficient is not substituted for the VJ coordinate.
4. **Complex-pole dynamics.** One renormalized sheet-holomorphic self-energy on the declared sheet, together with its pole sheet, continuation atlas, and declared uncertainty envelope, is supplied; any dispersive/absorptive split is confined to the real-energy/on-shell boundary. A separate target-complete gauge/scheme/order-certified observable-response construction combines the full resummed propagator with the appropriate vertices, interference, nonresonant terms, and detector response. Interpreting width-matrix eigenvalues as modewise pole widths requires the commuting, controlled-energy-dependence reduction of P75-X7 or an equivalent pole-level certificate. A common width or lifetime additionally requires equality of the relevant pole imaginary parts or a scalar absorptive response on that subspace.
5. **Response frame.** A proton–proton laboratory lifetime or decay-length statement requires the production-spectrum, decay-angle, acceptance, and reconstruction convolution; the W -rest-frame benchmark does not supply it.

6. **Mass-domain relation.** The D3 pair ($[28, 30]$ GeV, $c\tau_A^{\text{resp}} = 0.2$ mm) has `SourceCertified_ATLASR` `eweightGridAnchored_SearchBenchmark` authority. The response coordinate is the exact ATLASR `eweightGridAnchored_InteriorResponseCoordinate`, while `D3FullSourceProperLifetime` and `D3FullSourceResponseValidation` are `NotReady`; only 30 GeV is a direct ATLAS mass anchor. The declared-linear HEPData Figure 4a ordinate screen exceeds a factor 4.50 under the ATLAS simplified model, but is not source-level non-exclusion. A `D3ToVJDomainBridge` is nevertheless required before the benchmark can enter a typed VJ relation or benchmark-matched compatibility set. Without it the pair is only a `PredeclaredOperationalScanSlice_NoVJMassAuthority`, with no theorem-level priority or optimization claim. With the bridge's stated coverage event, the resulting set is a possibly uninformative, benchmark-derived source-coordinate compatibility/confidence set. It is not independent VJ mass-prediction authority, is not part of the core $\mathcal{I}_M^{\text{src}}$, and never by itself localizes a physical pole mass. The VJ domain remains $H(I_E)$ with numerical endpoints `NotReady` until the required bridge and response inputs are actually supplied.
7. **Independent typed response record.** The record is disjoint from benchmark formation and model formation and is matched to child, branch, simple-pole continuation chart or coherent stratum, response, and electroweak input scheme. Polewise lifetime inversion additionally requires an identifiability certificate; otherwise the full coherent-response region is inverted. Reusing the 0.2 mm response-grid coordinate cannot supply independent validation; that restriction does not demote its source-certified, grid-anchored benchmark authority.
8. **Coverage projection.** Every compatible nuisance coordinate and unresolved branch/stratum/response type is retained in the inverse union under a joint coverage event. A single best-fit profile may replace that union only after an independent uniqueness theorem.
9. **Inverse conditioning.** A linearized numerical inversion requires $|5 + \kappa_C|$ bounded away from zero on the complete allowed envelope. Otherwise the exact set-valued inverse or a certified higher-order analysis is mandatory.

The strongest presently proved extension is the conditional, coverage-preserving complex-pole inversion theorem on $H(I_E)$. The independent numerical weak normalization, VJ lifetime envelope, D3-to-VJ transfer, and numerical VJ mass localization remain `NotReady`.

22 Conclusion

PCR-7.5 determines why the heavy-sector result of PCR-6 and PCR-7 has the structure it does and how each layer can fail. The exact threefold degeneracy is protected by the full heavy $U(3)_H$ symmetry: symmetry-preserving mass perturbations shift all three heavy eigenvalues together, whereas every nonzero standing traceless component lifts the degeneracy. The lifting is quantitatively controlled by

$$\frac{3}{2}\|T\|_2 \leq \text{diam Spec}(T) \leq 2\|T\|_2.$$

Its dependency lock includes PCR-5E, PCR-6, and PCR-7 together with the declared PCR-3/PCR-4, fixed-domain-exhaustion, and SCIC authorities and their frozen transitive dependency closure. This closes the source, physicality, update, and carrier-identity provenance without reopening any upstream target or importing an unresolved parent endpoint.

The deeper carrier result is more robust. PCR-7.5 freezes a finite status-free support/mode skeleton with a finite heavy-partition decoration. On the declared trace-transport cell, every proper-support skeleton retains an exact locally constant PCR-6 deletion flag, while all full-support 3, 2 + 1, and 1 + 1 + 1 decorations map to one parent carrier orbit. The regularity-cell distance r_{reg} , normal-form/update distance d_{nf} , and explicit nonfrozen guard faces yield $r_{\text{car, cert}} > 0$. No compact candidate fibre or continuous terminal residual is assumed. On that same-scope region the perturbed realization continues to contain one terminal carrier even when the heavy spectral sector splits. This establishes the strict nonimplication

$$\text{spectral splitting} \Rightarrow \text{carrier splitting}.$$

The charged-current hierarchy is similarly stratified. On the full-rank charged-current domain, exact positive flavor universality $\Omega'_H = \delta' I_3$ with $\delta' > 0$ has zero unrestricted anisotropic radius, but rank-three heavy support persists throughout the sharp ball $\|\delta V_H\|_2 < s_E$. Most importantly, the light-heavy accounting law

$$\Delta'_{\text{act}} = \Omega'_H$$

remains exact under every complete block-isometry-preserving perturbation. Maximality has two fibrewise levels: $O_{\text{core}}^{b,X}$ is maximal relative to the pre-frozen variation-certified feature registry, while $O_{\text{cov}}^{b,X}$ is the functional-determination class of the complete joint structural trace. The joint trace-regrouping lemma used by P75-02 supplies an information-preserving factorization map, so cross-interface incidence and correlation data are retained rather than replaced by independent interface marginals. Exact laws are carried separately and the displayed heavy/CCST tuple is only a reduct. The cross-child endpoint is formed afterward, for matched ready-child realizations under matched common perturbations, as the meet of common-source reducts; fully decorated child envelopes are not identified. Exact degeneracy and exact scalar universality are excluded from the core when certified arbitrarily small standing variations destroy them, but their covariant values are not erased from O_{cov} .

The detector prediction is certified only on a declared intersection of a P75-13 trace-transport cell, the block-isometric coupling region, and a carrier-plus-reference SPIC domain; no universal protocol perturbation ball is inferred. SPIC places every structural mass-support core and an explicit coverage-certified no-vectorlike reference core in one common ordered record space, requires a global lower-Lipschitz response on the joint signal/nuisance/control coordinates, and separates their image cores outside one declared collision-coarsening neighborhood. Inside such a neighborhood the protocol uses the coarser quotient and adjacent-strata union. The coherent response coordinates are retained. The same observed record performs hypothesiswise selection whenever the attainable guard $r_{\text{rec}} < s_{\text{car,ref}}/2$ holds: on true-carrier coverage the carrier inverse contains truth and the reference inverse is empty, and conversely for a true reference. Lower-Lipschitz transport separately requires

$$\frac{2r_{\text{rec}}}{\kappa_{\text{car}}} < r_{\mathcal{Z},\text{car}}, \quad \frac{2r_{\text{rec}}}{\kappa_{\text{ref}}} < r_{\mathcal{Z},\text{ref}},$$

with r_{rec} attainable strictly above its deterministic floor. These are typed record-to-parameter inverse guards, distinct from the same-record selector and from the structural-mass and Gram guards. On the block-isometric subdomain with $\eta_V \leq \bar{\eta}_V < s_E$, the ideal pair-space bound $\|(\Delta'_{\text{act}}, \Omega'_H)\|_{\text{pair}} \geq \sqrt{6}(s_E - \bar{\eta}_V)^2$ remains a structural theorem, not a detector guard; no simultaneous distance between two nonempty counterfactual confidence regions is asserted. A split structural spectrum may remain unresolved. The condition $g_{\text{str}} > 2r_m$ gives unique true-support matching, not uniform reported-interval resolution; observed intervals resolve iff $\bar{g}_{\text{str}} > 2r_m$, while uniform design requires $r_m < g_{\text{str}}/4$. Otherwise report **ResolutionNotUniformlyCertified**. This structural partition does not determine physical poles. The SPIC gate and held-out-data status remain distinct.

The search-localization analysis establishes a positive D3 benchmark before stating the VJ firewall. PCR-5D3 certifies ([28, 30] GeV, $c\tau_A^{\text{resp}} = 0.2$ mm) as a **SourceCertified_ATLASRweightGridAnchored_SearchBenchmark**: every mass in the band has an explicit source-coherent physical realization with declared non-direct U/L passage, the complete low-mass LFU statistic obeys $8.437 < \chi_{\text{LFU}}^2 \leq 8.445 < 11.0705$, and the oscillation and CLFV records pass. These records make the predeclared pair a reproducible, motivated, admissible search benchmark; they do not derive either endpoint or select 0.2 mm uniquely.

The value 0.2 mm is exactly the **ATLASRweightGridAnchored_InteriorResponseCoordinate**, an interior member of the frozen discrete lifetime-reweighting grid whose declared span is [0.1, 1.0] mm. Its 0.1 mm distance to the nearer endpoint of that span is endpoint-distance bookkeeping only; it does not certify a continuous detector-response neighborhood. It is not a demonstrated full-source proper lifetime. Only 30 GeV, not 28 GeV, is a direct ATLAS mass anchor in the band. A frozen HEPData replay gives a declared-linear Figure 4a ordinate separation greater than a factor 4.50 across the band under the ATLAS single-Majorana, electron-only simplified model. That result is a conditional ordinate screen, not source-level non-exclusion, a likelihood recast, acceptance, or full detector-response validation. Accordingly **D3FullSourceProperLifetime** = **D3FullSourceResponseValidation** = **NotReady**. The corresponding W -rest-frame response-coordinate tag has $E_\ell^* = 34.58\text{--}35.31$ GeV and $\beta\gamma c\tau_A^{\text{resp}} = 0.231\text{--}0.252$ mm; it defines an operational prompt/near-prompt category but is not a physical mean flight length.

Separately, the VJ source supplies the symbolic mass domain $H(I_E)$, whose numerical endpoints are **NotReady**; no light-neutrino mass ratio narrows it. With no certified D3-to-VJ domain bridge or numerical overlap, the D3 pair is only a **PredeclaredOperationalScanSlice_NoVJMassAuthority** in VJ scope, with no uniqueness, optimization, or comparative-priority claim and no VJ pole-mass or pole-lifetime authority. After a D3-to-VJ domain bridge is supplied, and before the independent response record, it must freeze $\mathfrak{T}_{D3} = \text{Compat}_{D3 \rightarrow X}^r \cup \text{Compat}_{D3 \rightarrow X}^R$, export covered lifetime regions $C_{D3 \rightarrow X}^r$ and same- $\mathcal{Y}^{\text{ind}}$ full-response regions $C_{D3 \rightarrow X, \text{t}}^R$, and prohibit post-record pruning. The singleton {0.2 mm} is licensed only after both a full-source lifetime identity and an exact lifetime-coordinate bridge are certified. Let $\mathcal{E}_{D3} = \mathcal{E}_{\text{spec}} \cap \{\text{bridge pair/type/route membership}\}$; $\Pr(\mathcal{E}_{\text{spec}} \setminus \mathcal{E}_{D3} \mid \mathcal{G}_{\text{bridge}}) \leq \alpha_{\text{bridge}}$ gives $\Pr(\mathcal{E}_{D3}) \geq 1 - \alpha_{\text{spec}} - \alpha_{\text{bridge}}$. Their possibly uninformative route union is a bridge-conditional benchmark-derived source-coordinate compatibility/confidence set outside core $\mathcal{I}_M^{\text{src}}$; it is not independent mass-prediction authority and never by itself physical pole-mass localization. The VJ analysis architecture remains LNC/LNV inclusive,

flavour complete, jointly normalized in a declared electroweak input scheme, and lifetime scanned. Existing data require both an experiment-specific recast and a declared covered subset of $H(I_E)$.

Here $H(I_E)$ is compact Euclidean and the bridge relation lives in a compact Polish product. Learned relations, finite type graphs, and C^r/C^R multifunctions obey random-closed/Effros–Borel measurability. The lifetime map is continuous/Borel on Ξ_t , $\mathcal{F}_t$ is Borel on $\mathcal{U}_t^R$, and the $\mathcal{E}_{D3}$ route clauses require true-tuple domain membership before either partial map is evaluated.

The row-normalization extension closes a further authority distinction. On each fixed unperturbed VJ source, $\epsilon_{\text{row}} = \delta_E = s_E^2$ exactly. Choosing the dyadic witness before freezing the compact rational pass box repairs its existence, and the retained tester forces $n_{\text{mark}} \geq 4$, hence

$$0 < \epsilon_{\text{row}} \leq \frac{1}{256}.$$

The bounded public-data pass does not tighten that certified ceiling. Conditional on applicability of the declared light-response model, the universal-response HFLAV coupling-ratio Jacobian along the common- ϵ direction vanishes, while the frozen covariance test gives $\chi_4^2 = 7.0193$ and $p = 0.1349$. Thus HFLAV has status `ConditionalLightResponseCompatibleAt95` and supplies no common-deficit magnitude; full VJ compatibility remains `NotReady`. The direct/indirect m_W screen sets the unreported cross-covariance to zero only as a screening assumption and yields $\epsilon_{95}^{\text{scr}} = 1.62 \times 10^{-3}$; this is not an independence certificate, and the calculation lacks the finite-mass HNL electroweak, cross-covariance, and template-response certificates required for coverage on $H(I_E)$. The predeclared stop rule therefore retains $1/256$ and leaves $\epsilon_{95}^{\text{weak}}$, $\mathcal{E}_X^{\text{weak}}$, and $\mathcal{E}_X^{\text{adm}}$ `NotReady`.

For the fixed child, X7 intersects a certified outer source enclosure $\mathcal{E}_X^{\text{src}} \subseteq (0, 1/256]$, containing the true ϵ_X , with $\mathcal{E}_X^{\text{weak}} = \text{pr}_\epsilon \mathcal{C}_X^{\text{weak}}$, the coverage-preserving epsilon projection of a jointly covered weak-record region over the compatible mass, branch, electroweak scheme, muon normalization, open channels, response, and theory nuisances. The weak coverage holds almost surely conditional on any design data. X7 proceeds only if $\mathcal{E}_X^{\text{adm}} = \mathcal{E}_X^{\text{src}} \cap \mathcal{E}_X^{\text{weak}} \neq \emptyset$; an empty intersection is `FixedChildNormalizationIncompatibility`, not a lifetime curve. This fixed-child check does not select the child; target-independent child selection requires the broader comparison class.

A VJ lifetime is defined by chart-local complex-energy zeros of $\mathcal{D}^{-1} = zI - M_X^{(0)} - \Sigma^{\text{ren}}(z)$, with one sheet-holomorphic self-energy, not by width-matrix eigenvalues alone. The $\Delta M/\Gamma$ split exists only on the declared real/on-shell boundary; the pole equation and any licensed isolated-simple-pole residue use the full Σ^{ren} , with no separate complex continuation of either boundary piece. The normalization bridge transports the full Gram and structural mass-support weights into the intrinsic self-energy/pole kernel. If licensed, $\Gamma = \epsilon \mathcal{G}^{\text{self}}$ is a boundary factorization and $\mathcal{C}_t^{\text{pole}}$ contains only self-energy channel/flavour/mode, threshold/energy, shifted-pole, and pole-derivative effects. The gauge-complete production–propagation–decay map $\mathcal{F}_t$ is built from the full resummed propagator, using a Laurent residue only on a licensed isolated-simple-pole chart, together with external vertices, interference, required nonresonant terms, and detector response; collision/coherent strata retain full- $\mathcal{D}$ inversion or a certified complete local Laurent principal part. No external response factor enters the intrinsic pole width. On the weak covered event, conditional on the true chart and nuisance tuple, the true lifetime belongs to the standalone envelope $\mathcal{T}_t(M_X)$.

Before the independent record, the exhaustive admitted universe is partitioned as $\mathfrak{T}_{\text{adm}} = \mathfrak{T}_\tau \dot{\cup} \mathfrak{T}_R$. The lifetime route requires uniform identifiability over its entire routed domain; any pre-frozen identifiable subtype sends its complement to $\mathfrak{T}_R$, and otherwise the whole type uses the full-response route. The common finite-dimensional space $\mathcal{Y}^{\text{ind}} = \text{ran } M_{\text{ind}}$, with its Euclidean/Borel structure, retains only genuinely observation-defined coordinates; unavailable ones use a hypothesis-independent zero-weight/unconstrained rule and enter neither distance, conditioning, nor inversion. Parameter-uniform or conditional-on-design coverage first yields $\mathcal{I}_M^{\text{src}}$. Mapping accepted witnesses through the pole kernel yields $\mathcal{I}_{m_{\text{pole}}}^{\text{phys}}$. A lifetime chart covers its targeted pole; several such poles require familywise coverage, while the $\mathcal{F}_t$ route covers the complete unordered support. The VJ-packet-to-pole-width normalization bridge and D3-to-VJ domain bridge are deliberately separate. At present the numerical VJ lifetime envelope, domain transfer, source-coordinate localization, and physical pole-mass localization remain `NotReady`.

The bridge-free combined well-specification event includes true-type admission, routed-domain validity, normalization-intersection membership, nuisance membership, and the weak-region event; α_{weak} is charged inside α_{spec} . Accordingly the target-qualified core level is $1 - \alpha_{\text{rec}} - \alpha_{\text{spec}}$, with $\alpha_{\text{spec}} = 0$ only if all components are deterministic. Optional D3 compatibility instead carries its separate conditional α_{bridge} .

The resulting endpoint is

$$\begin{aligned}
& \text{NestedRobustnessClosure}_{\text{VJ}} \\
& \oplus \text{RobustPredictionLock}_{\text{CCST}}^{\text{SPIC}} \\
& \oplus \text{TargetQualifiedRowBound}_{1/256} \\
& \oplus \text{ConditionalComplexPoleEnvelope}_{H(I_E)} \\
& \oplus \text{FrozenStructuralPrediction} \\
& \oplus \text{DetectorEvaluationConditional}_{\text{InstantiatedSPIC+HeldOutData}} \\
& \oplus \text{X7X8InputsNotReady}
\end{aligned}$$

The detector term denotes same-record carrier/reference model/core selection and requires both an instantiated SPIC certificate and held-out data; held-out data alone are not the missing step. Likewise, the bridges, target-complete pole kernel, independent normalization, exhaustive route universe, and coverage procedures are specification/status gates, not supplied certificates. PCR-8 may now close the global dependency chain with a precise account of which conclusions are exact, which have positive radii, which are symmetry-protected zero-radius refinements, which detector claims are SPIC-qualified, and which mass/lifetime conclusions remain conditional or NotReady.

A Notation and status calculus

Symbol/status	Meaning
$\mathcal{B}_{7.5}^{\text{ready}}$	The two PCR-5E/PCR-6/PCR-7 ready vectorlike-joint child scopes.
$X \in \text{PhysMod}_{E,\text{VJ}}^b$	An accepted joint physical realization in child b .
$K_H = H_\lambda^2 I_3$	Unperturbed heavy mass-squared operator.
$\delta K_H = a I_3 + T$	Central plus traceless Hermitian heavy perturbation.
$\rho_K = \ T\ _2$	Symmetry-breaking heavy mass-squared perturbation norm.
$V_H = -s_E U$	Unperturbed heavy charged-current block.
$\epsilon_{\text{row}} = \delta_E = s_E^2$	Electron-row light charged-current normalization deficit on a fixed unperturbed VJ source; the marked-child target gives $0 < \epsilon_{\text{row}} \leq 1/256$, not an independently selected weak-rate value.
n_{mark}	Dyadic witness index chosen in the uniform punctured pass cell before the compact rational response box is frozen; $n_{\text{mark}} \geq 4$.
$\epsilon_{0.2}(M)$	PCR-5D3 benchmark coefficient realizing the declared response-chart identity $c\tau_A^{\text{resp}} = 0.2$ mm on the source-certified realization continuum; this is not a full-source proper lifetime.
$c\tau_A^{\text{resp}} = 0.2$ mm	ATLASReweightGridAnchored_InteriorResponseCoordinate; it is an exact discrete grid member whose distance to the lower endpoint of the declared grid span is 0.1 mm. This distance is not a continuous response radius; D3FullSourceProperLifetime and D3FullSourceResponseValidation are NotReady.
$H(I_E)$	Source-qualified symbolic VJ mass domain; numerical endpoints are NotReady.
$[28, 30]$ GeV	Source-continuum mass coordinate of the D3 SourceCertified_ATLASReweightGridAnchored_SearchBenchmark; only 30 GeV is a direct ATLAS mass anchor. In VJ scope it is a PredictedOperationalScanSlice_NoVJMassAuthority, pending a typed domain bridge.
$\mathcal{D}_{X,b}^{-1}(z; M, \epsilon, \theta)$	Target-complete renormalized VJ inverse propagator whose lower-half-plane zeros define physical unstable-state poles; currently NotReady numerically.
$\mathcal{Z}_{X,b}(M; \epsilon, \theta)$	Unordered multiset of source-connected relevant-sheet complex-energy poles.

Symbol/status	Meaning
$\Xi_t, \mathcal{T}_t(M)$	Certified simple-pole continuation domain and its chart-qualified proper-lifetime envelope; collisions use the unordered multiset and coherent response instead.
B_{norm}	VJ-packet-to-pole-width normalization bridge, including the declared electroweak input-scheme map.
$\mathfrak{D}_{D3 \rightarrow X}$	D3-to-VJ domain relation; separate from width normalization and currently NotReady.
$\eta_V = \ \delta V_H\ _2$	Charged-current perturbation norm.
$\Omega'_H = V'_H V_H^\dagger$	Perturbed inclusive heavy charged-current Gram matrix.
$\Delta'_{\text{act}} = I - N' N'^\dagger$	Perturbed active charged-current deficit.
g_{LH}	Light-heavy spectral gap.
$\mathcal{S}_b(X), K_{\min}^b(X)$	Finite source-generator set and its deletion-essential kernel required by every complete same-apex terminal candidate.
$\text{sk}_0, \mathcal{A}_b^{\text{dec}}(X)$	Status-free support/mode skeleton and its finite heavy-partition decoration.
$\mathcal{A}_b^{\text{prop, supp}}(X)$	Decorated skeletons omitting at least one deletion-essential generator; full-support decorations instead map to one parent carrier orbit.
$C_X, r_{\text{reg}}, d_{\text{nf}}$	Certified trace-transport cell, distance to its regularity boundary, and distance to the normal-form/update face.
$q_a \equiv 1, \gamma_{a,k}, s_P$	Exact transported deletion flag for a proper-support skeleton, explicit witness-typing face slacks, and explicit nonfrozen parent-guard slack.
$r_{\text{car, cert}}^b(X)$	Half-minimum of the regularity, normal-form/update, parent-guard, and proper-witness face radii on the declared cell.
$\partial \mathcal{P}_{\text{term}}$	Explicit union of the regularity boundary, normal-form/update face, parent-guard failure, deletion-witness loss, and witness-typing faces.
$\mathcal{U}_{\text{car}}$	Connected component of the complement of $\partial \mathcal{P}_{\text{term}}$ containing the accepted realization.
g_{str}	Minimum separation between distinct structural heavy mass-support points; not applicable on the one-point degenerate support and not a physical complex-pole gap.
$\mathcal{O}_{\text{core}}^{b,X}$	Fibrewise invariant structural core, maximal relative to the pre-frozen variation-certified feature registry.
$\mathcal{O}_{\text{cov}}^{b,X}$	Functional-determination class of the complete fibrewise structural trace; the displayed heavy/CCST tuple is only a reduct.
$\kappa_{\pi,p}$	Global lower-Lipschitz response constant on one joint signal/nuisance/control SPIC core.
ϵ_{op}	Simultaneous operator-norm confidence radius for detector-level Gram certification.
$\mathcal{R}_{7.5}(X)$	Feature-specific robustness vector.
Skin	Presentation change with no standing-bearing effect.
SameScopeTensor	Standing-bearing deformation inside the fixed source/target ontology.
StandingBreakingTensor	Same-scope deformation that changes a standing symmetry or internal sector structure.
SourceUpdate	New primitive source support or law.
TargetUpdate	New role, witness, target, or comparison class.
ProtocolUpdate	Structural target remains meaningful but the frozen detector protocol no longer covers the point.
NotReady	A named unresolved theorem or typing burden.

B Perturbation source normal form

A perturbation normal form over a fixed source hypergraph H_X is

$$\text{NF}_{\text{pert}}(X) = (H_X, \delta\theta, \delta\mathcal{E}, \delta\mathcal{R}, \delta\text{Match}, \delta\text{Cont}, \delta\text{Bdy}, \delta\text{Pop}, \text{Status}),$$

where $\delta\theta$ changes existing coefficient values, $\delta\mathcal{E}$ changes existing equation values without changing equation type, and $\delta\mathcal{R}$ changes existing relation values without adding relation arity. The remaining entries record matching, continuation, boundary, population, and status.

Proposition B.1 (Normal-form decision order). *The perturbation status is decided in the following order:*

$$\begin{aligned} &\text{typing} \rightarrow \text{source formation} \rightarrow \text{target preservation} \\ &\quad \rightarrow \text{activity reduction} \rightarrow \text{one-apex physicality} \\ &\quad \rightarrow \text{terminal standing} \rightarrow \text{protocol applicability}. \end{aligned}$$

A failure at an earlier stage cannot be repaired by a later stage.

Proof. This is the inherited source-first and no-repair order of PCR-3, PCR-4, PCR-5E, and the fixed-domain theorem. Later evaluation presupposes earlier well-typed formation, so downstream success cannot supply missing upstream standing. $\square$

Submitted change	Normal-form status	Reason
Heavy common mass shift	SameScope tensor, if positive and inside threshold domain	Existing coefficient changes; support unchanged.
Traceless heavy mass matrix	Standing-breaking tensor, if allowed by frozen source equations	Existing heavy operator gains anisotropic standing.
Small charged-current matrix perturbation	SameScope tensor inside source/isometry constraints	Existing coupling incidence changes.
New heavy field	SourceUpdate	New primitive bearer support.
New detector observable	TargetUpdate or ProtocolUpdate	Witness/record family changes.
Basis rotation in heavy eigenspace	Skin	Complete witness trace invariant.
Structural mass coordinate leaves parent phase space	ProtocolUpdate or TargetUpdate	Frozen source/analyzer support no longer applicable.
Undefined perturbation norm	NotReady	Robustness statement lacks a comparison metric.

C Matrix perturbation lemmas

This appendix collects the external finite-dimensional matrix results used in the quantitative estimates. They do not determine the AASC perturbation scope; they apply only after the source and target classes are fixed.

Lemma C.1 (Weyl eigenvalue bound). *For Hermitian A, E , ordered eigenvalues satisfy*

$$|\lambda_i(A + E) - \lambda_i(A)| \leq \|E\|_2.$$

Lemma C.2 (Singular-value perturbation). *For arbitrary A, E ,*

$$|\sigma_i(A + E) - \sigma_i(A)| \leq \|E\|_2.$$

Lemma C.3 (Davis–Kahan projector estimate). *If two spectral clusters of a Hermitian A are separated by gap $g > 0$ and $\|E\|_2 < g/2$, then the corresponding perturbed projector P' exists with the same rank and*

$$\|P' - P\|_2 \leq \frac{2\|E\|_2}{g}.$$

Lemma C.4 (Gram perturbation). For $V_0 = -sU$ and $V' = V_0 + E$,

$$\|V'V'^\dagger - s^2I\|_2 \leq 2s\|E\|_2 + \|E\|_2^2.$$

Proof. Expand

$$V'V'^\dagger - V_0V_0^\dagger = V_0E^\dagger + EV_0^\dagger + EE^\dagger$$

and apply submultiplicativity. $\square$

Lemma C.5 (Positive square-root spread transport). If $c > \rho > 0$ and $t_{\max} - t_{\min} = D$ with $|t_i| \leq \rho$, then

$$\frac{D}{2\sqrt{c+\rho}} \leq \sqrt{c+t_{\max}} - \sqrt{c+t_{\min}} \leq \frac{D}{2\sqrt{c-\rho}}.$$

Authoritative references include [3], [5], [6], and [13].

D Finite terminal skeleton and trace-transport ledger

The carrier certificate is finite at the level that the upstream source theorems actually make finite. PCR-6 Definition 18.1 forms the within-realization candidate class before eligibility or identity comparison; P6-04 and PCR-5E E-S1/E-S2 give a finite source and incidence presentation [1, 10]. They do not assert that the full candidate class is finite, that every candidate fibre is compact, or that a real-valued terminal missing-work residual is continuous.

Let $\mathcal{S}_b(X)$ be the finite deletion-essential support and irreducible-relation generator set. Write $K_{\min}^b(X) \subseteq \mathcal{S}_b(X)$ for the complete deletion-essential generator kernel of the accepted parent, namely the generators that every terminal-role-complete same-apex candidate must contain. Every source-authorized candidate first maps, before any status is evaluated, to the partition-forgetful core

$$\text{sk}_0(A) = (\text{supp}_{\mathcal{S}}(A), \text{mode}(A)),$$

and then to the finite partition decoration

$$\text{sk}(A) \in 2^{\mathcal{S}_b(X)} \times \{\text{Node, Aggregate, Relation, Mixed, Parent}\} \times \{\perp, 3, 2+1, 1+1+1\}.$$

The finite image is $\mathcal{A}_b^{\text{dec}}(X)$. Continuous projector orientations remain inside a decorated-skeleton fibre and play no role in the proof. The full-support parent subset

$$\mathcal{P}_X^{\text{dec}} = \{P_{X,3}, P_{X,2+1}, P_{X,1+1+1}\}$$

has one partition-forgetful support/mode core, and every member maps to the same terminal carrier orbit $[P_X]_{\text{term}}$. Define

$$\mathcal{A}_b^{\text{prop, supp}}(X) = \{a \in \mathcal{A}_b^{\text{dec}}(X) : K_{\min}^b(X) \not\subseteq \text{supp}_{\mathcal{S}}(a)\}.$$

Changed full-support standing data define another realization or an update.

D.1 Deletion-essential obstruction ledger

Support/obligation	Deletion or failure effect	Transported certificate
Light sectors S_1, S_2, S_3	Loses relative phase, propagation, or boundary response	Exact omission flag on the fixed support cell.
Heavy aggregate P_H	Loses D3 heavy response and D5 matching/protection incidence	Rank-three aggregate is retained by P75-09; a split subprojector remains proper.
Light-heavy incidence	Loses charged-current closure and source matching	Active incidence is retained by the P75-10 rank cell.

Support/obligation	Deletion or failure effect	Transported certificate
Charged-current boundary	Loses preparation/detection reconstruction	Exact boundary-support omission.
D4 population/background support	Loses accepted child population/transport work	Exact target-support omission.
D5 action/protector support	Loses branch-specific charge standing	Exact branch-qualified omission.
Common-cone relation	Marginal witnesses cannot coexist on one apex	Exact one-apex compatibility obstruction.
Matching/continuation	Loses future use and threshold identity	Exact continuation-support obstruction.

Choose for every $a \in \mathcal{A}_b^{\text{prop, supp}}(X)$ one omitted generator g_a and its PCR-6 P6-S2B/G witness. On the certified cell of Lemma 7.3, define

$$q_a(\delta) = \begin{cases} 1, & \text{the transported witness excludes every candidate over } a, \\ 0, & \text{the witness is lost or ceases to be typed.} \end{cases}$$

The trace-transport lemma proves

$$q_a(\delta) \equiv 1, \quad L_a^{\text{del}} = 0 \quad (a \in \mathcal{A}_b^{\text{prop, supp}}(X)).$$

This exact locally constant flag replaces the unsupported minimum of invented continuous residuals over a candidate fibre.

D.2 Regularity and explicit faces

Let C_X be the closed normal-form/activity/response cell of Lemma 7.3, with $0 \in \text{relint } C_X$, and put

$$\mathfrak{F}_{\text{reg}}^{b,X} = \mathfrak{P}^{b,X} \setminus \text{relint } C_X, \quad r_{\text{reg}}^b(X) = \text{dist}(0, \mathfrak{F}_{\text{reg}}^{b,X}) > 0.$$

Let $\mathfrak{F}_{\text{nf}}^{b,X}$ be the closed normal-form or update face and $d_{\text{nf}}(X) = \text{dist}(0, \mathfrak{F}_{\text{nf}}^{b,X}) > 0$.

For each nonfrozen open parent guard use its distance-to-complement slack and let s_P be their finite minimum, with Lipschitz constant L_P . For any nonfrozen guard needed to type the chosen witness for a , write $\gamma_{a,k} > 0$ and $L_{a,k}$ for its distance slack and Lipschitz constant. Define

$$r_a^{\text{face}}(X) = \min \left(\left\{ \frac{q_a(0)}{L_a^{\text{del}}} \right\} \cup \left\{ \frac{\gamma_{a,k}(0)}{L_{a,k}} : k \in \mathcal{G}_a \right\} \right),$$

using $u/0 = +\infty$ for every positive constant slack u , including a constant parent slack, and $+\infty$ for an empty family of nonfrozen guards. The constant deletion flag therefore introduces no fictitious finite scale; its only falsifier is loss of the trace-transport cell or of an explicitly listed typing guard.

Proposition D.1 (Auditable trace-transport carrier certificate). *At an accepted ready-child point in the declared strict-margin trace-transport cell,*

$$r_{\text{reg}}^b(X) > 0, \quad d_{\text{nf}}(X) > 0, \quad q_a(0) = 1 \quad (a \in \mathcal{A}_b^{\text{prop, supp}}(X)),$$

and every displayed nonfrozen slack is positive. With absent guard terms omitted,

$$r_{\text{car, cert}}^b(X) = \frac{1}{2} \min \left\{ r_{\text{reg}}^b(X), d_{\text{nf}}(X), \frac{s_P(0)}{L_P}, \min_{a \in \mathcal{A}_b^{\text{prop, supp}}(X)} r_a^{\text{face}}(X) \right\} > 0.$$

This ball avoids the explicit locus

$$\begin{aligned} \partial \mathcal{P}_{\text{term}}^b(X) &= \mathfrak{F}_{\text{reg}}^{b,X} \cup \mathfrak{F}_{\text{nf}}^{b,X} \cup \{\delta \in \text{relint } C_X : s_P(\delta) \leq 0\} \\ &\cup \bigcup_{a \in \mathcal{A}_b^{\text{prop, supp}}(X)} \left(\{\delta \in \text{relint } C_X : q_a(\delta) = 0\} \cup \bigcup_{k \in \mathcal{G}_a} \{\delta \in \text{relint } C_X : \gamma_{a,k}(\delta) \leq 0\} \right). \end{aligned}$$

Consequently

$$r_{\text{car}}^b(X) = \text{dist}(0, \partial \mathcal{P}_{\text{term}}^b(X)) \geq r_{\text{car, cert}}^b(X) > 0.$$

If a proposed chart does not supply the stated closed cell, fixed generator identification, or transported deletion witnesses, this proposition is conditional/NotReady on that chart rather than an unconditional extrapolation of PCR-6.

Proof. PCR-5E E-S4A supplies the strict solved response/routing cell, while P75-09 and P75-10 give the heavy aggregate and active-incidence neighbourhoods. Intersecting them inside the relative exact-equation chart and choosing a closed subcell gives $r_{\text{reg}} > 0$. The normal-form/update face is closed and avoided. Distance-to-complement slacks are Lipschitz, so the displayed half-minimum keeps every nonfrozen guard positive.

The remaining proper-support obstruction is exact: by Lemma 7.3, $q_a = 1$ throughout the cell. A full-support candidate belongs to a partition-decorated parent skeleton that maps to the same parent orbit, or leaves the comparison as an update. Thus no proper-support skeleton becomes terminal-complete in the displayed ball, while the parent remains live. This proves the separation from $\partial \mathcal{P}_{\text{term}}^b(X)$ without candidate-fibre compactness or a continuous terminal residual hypothesis. $\square$

For a compact family K , let Z_K be the zero section in a common finite-dimensional perturbation bundle and let $\mathfrak{F}_K$ be the closed union of the regularity, normal-form, parent-guard, and proper-witness faces. If $Z_K \cap \mathfrak{F}_K = \emptyset$, compact/closed separation gives

$$r_{\text{car, cert}}^b(K) = \frac{1}{2} \text{dist}(Z_K, \mathfrak{F}_K) > 0.$$

No compactness of a terminal-candidate fibre is asserted or needed.

Receiving a distinct heavy eigenvalue changes only the finite partition decoration of the same full-support parent orbit and supplies none of the omitted deletion-essential work. A second terminal carrier therefore requires loss of a transported obstruction, failure of an explicit typing guard, or escape through the declared regularity/update boundary; spectral splitting alone is insufficient.

E Perturbed protocol normal form

The robustness protocol is

$$\text{Prot}_{7.5} = (J_Q, J_H^{\text{rob}}, \text{Src}, \text{Ana}, \text{Cal}, \text{Bkg}, \text{Rec}_{d/q/m}, \alpha, \text{Exposure}, \text{Falsifier}).$$

It preserves the PCR-7 source/analyzer geometry and enlarges only the predeclared structural heavy-mass-support and response-bin cover. Completeness is always relative to this finite binning and response-kernel class.

For a structural mass partition π , let $P_a \in \text{End}(E_H)$, $a = 1, \dots, k_\pi$, be the distinct heavy spectral projectors, with $k_\pi = 1, 2, 3$, and let

$$G_3 = U(3), \quad G_{2+1} = U(2) \times U(1), \quad G_{1+1+1} = U(1)^3.$$

Heavy spectral projectors act on E_H , whereas $\Omega'_H = V'_H V_H^{\prime\prime\prime}$ acts on the active flavor space, so those operators are never multiplied directly. The mass-support-resolved active-space weights are

$$W_a := V'_H P_a V_H^{\prime\prime\prime} \in \text{Herm}_{\geq 0}(E_{\text{act}}).$$

For each declared analysis bin j , let $D_\pi^{(j)}(\vartheta) \in \text{End}(E_H)$ be the full complex production–propagation–decay protocol kernel at nuisance point ϑ . The entire nuisance-indexed family is frozen protocol metadata and is not a physical coordinate in Ξ_π , a function reconstructed from held-out data, or a computation of the P75-X7 complex propagator poles z_t . At a joint point $(\xi, \vartheta, c) \in \mathcal{Z}_{\pi, p}$, the response evaluator forms

$$\mathcal{A}_\pi^{(j)}(\xi, \vartheta) := V'_H D_\pi^{(j)}(\vartheta) V_H^{\prime\prime\prime} = \sum_{a,b} V'_H P_a D_\pi^{(j)}(\vartheta) P_b V_H^{\prime\prime\prime}.$$

Thus overlap-bin interference is retained by the full matrix response rather than approximated by separate positive component rates. The calibrated real and imaginary quadratures of $\mathcal{A}_\pi^{(j)}(\xi, \vartheta)$, together with the rate kernels formed from it, are evaluated inside $\mathcal{R}_{\pi, p}$.

At the frozen finite binning, define the unquotiented lifted datum and its physical quotient exactly by

$$\widetilde{\Xi}_\pi = \left(\{m_a, P_a, W_a\}_{a=1}^{k_\pi}, V'_H, d(\Delta'_{\text{act}}), q(\Delta'_{\text{act}}), d(\Omega'_H), q(\Omega'_H) \right), \quad \boxed{\Xi_\pi := \widetilde{\Xi}_\pi / G_\pi}.$$

The group acts on the physical datum by

$$P_a \mapsto g P_a g^\dagger, \quad V'_H \mapsto V'_H g^\dagger.$$

The frozen metadata family is required to be equivariant, $D_\pi^{(j)} \mapsto g D_\pi^{(j)} g^\dagger$; consequently W_a , the evaluated $\mathcal{A}_\pi^{(j)}$, and the d/q reduct are invariant. On the exactly mass-degenerate stratum only the quotient aggregate is labeled. Mass-support status `Deg3` does not imply $D_3^{(j)} \propto I_3$ or $\mathcal{A}_3^{(j)} \propto I_3$; a common scalar response kernel is a separate certified condition. These structural data do not determine physical complex-pole multiplicity: conditional P75-X7 is the sole continuation route, and the empirical VJ pole record remains `NotReady`.

Let $K_{\pi,p} \subset \Xi_\pi$ be a compact physical core, $\Theta_{\pi,p}$ the complete declared nuisance domain, and $C_{\pi,p}$ the calibration/background control domain. The response is fitted on the augmented space

$$\mathcal{Z}_{\pi,p} := K_{\pi,p} \times \Theta_{\pi,p} \times C_{\pi,p}, \quad \mathcal{R}_{\pi,p} : \mathcal{Z}_{\pi,p} \rightarrow \mathbb{R}^{N_{\text{prot}}}.$$

The map $\mathcal{R}_{\pi,p}$ evaluates the frozen families $D_\pi^{(j)}(\vartheta)$ and $\mathcal{A}_\pi^{(j)}(\xi, \vartheta)$ at the joint physical/nuisance/control point. It reconstructs ξ , nuisance, and control coordinates subject to SPIC; it does not reconstruct the frozen kernel functions themselves. The common ordered codomain $\mathbb{R}^{N_{\text{prot}}}$, the product metric $d_{\mathcal{Z}}$, and all coordinate scalings are frozen with the protocol. The record ordering contains all declared observed-bin coordinates, calibrated real and imaginary quadratures, nonnegative rate summaries, and calibration/background controls. Every retained coordinate is a genuinely defined observed summary under every hypothesis, and every $\mathcal{R}_{\pi,p}$ predicts that same summary. There is no model-predicted padding. A physically observed bin with a zero rate under one model remains an ordinary observed coordinate carrying that model's zero prediction.

One predeclared hypothesis-independent availability projection M_{obs} is frozen before held-out data, with

$$\mathcal{R}_{\pi,p}^{\text{obs}} := M_{\text{obs}} \mathcal{R}_{\pi,p}.$$

An unavailable coordinate is either zero-weighted by M_{obs} or quotiented as unconstrained, identically under every hypothesis; it contributes to neither image distance, lower-Lipschitz conditioning, nor inversion. Core labels, hypothesis labels, structural tags, routing masks, and core-specific shortening or reordering are forbidden in the data vector. All comparison and inversion operations use the common effective observed-summary space $\text{ran } M_{\text{obs}}$. The joint confidence set $C_{\pi,p}^{\text{joint}} \subset \mathcal{Z}_{\pi,p}$ is formed before physical projection, and the reported signal set is

$$C_{\pi,p}^{\text{phys}} = \text{pr}_{\Xi_\pi} C_{\pi,p}^{\text{joint}}.$$

Freezing a nuisance coordinate at its best fit before projection is not a coverage-preserving substitute.

Let the no-vectorlike reference be represented by an explicit compact augmented core

$$\mathcal{Z}_{\text{ref},p} := K_{\text{ref},p} \times \Theta_{\text{ref},p} \times C_{\text{ref},p}, \quad \mathcal{R}_{\text{ref},p} : \mathcal{Z}_{\text{ref},p} \rightarrow \mathbb{R}^{N_{\text{prot}}},$$

with $\mathcal{R}_{\text{ref},p}^{\text{obs}} := M_{\text{obs}} \mathcal{R}_{\text{ref},p}$. Here $K_{\text{ref},p}$ has physical zero heavy amplitude. Its response uses the same ordered observed-summary coordinates, calibration/background class, nuisance/control convention, hypothesis-independent availability projection M_{obs} , and held-out split as the carrier cores. A coverage-certified joint reference region $C_{\text{ref},p}^{\text{joint}} \subset \mathcal{Z}_{\text{ref},p}$ is constructed and projected rather than treating the reference as an exact point.

E.1 SPIC requirements

A *StratifiedProtocolIdentifiabilityConditioningCertificate* (SPIC) contains all of the following.

1. **Declared response class.** The finite mass/energy binning, complex kernel family, nuisance set, control samples, common ordered observed-summary space, hypothesis-independent availability projection, and retained interference quadratures are frozen before held-out data. Every retained coordinate must be defined and observed under every hypothesis; unavailable coordinates are zero-weighted or quotiented out of comparison and inversion. A core or routing label may not be encoded in the data vector or image separator. Material sub-bin structure outside this class is `ProtocolUpdate`.

2. **Separated fine cores.** Every $2 + 1$ or $1 + 1 + 1$ fine-stratum core obeys

$$g_{\text{str}} \geq g_{\pi, \min} > 0.$$

If a fine core is declared uniformly resolved at transported mass-coordinate radius r_m , its predeclared floor also satisfies $g_{\pi, \min} > 4r_m$. The condition $g_{\text{str}} > 2r_m$ supplies only unique matching to true support points.

3. **Global within-core conditioning.** For one recorded $\kappa_{\pi, p} > 0$,

$$\|\mathcal{R}_{\pi, p}^{\text{obs}}(z) - \mathcal{R}_{\pi, p}^{\text{obs}}(z')\|_2 \geq \kappa_{\pi, p} d_{\mathcal{Z}}(z, z') \quad \text{for all } z, z' \in \mathcal{Z}_{\pi, p}.$$

A full-rank Jacobian at each point does not, by itself, satisfy this condition. The reference core has its own predeclared product metric $d_{\mathcal{Z}, \text{ref}}$ and independent conditioning bound

$$\|\mathcal{R}_{\text{ref}, p}^{\text{obs}}(z) - \mathcal{R}_{\text{ref}, p}^{\text{obs}}(z')\|_2 \geq \kappa_{\text{ref}, p} d_{\mathcal{Z}, \text{ref}}(z, z'), \quad z, z' \in \mathcal{Z}_{\text{ref}, p}, \quad \kappa_{\text{ref}, p} > 0.$$

Carrier-core conditioning does not imply this reference-core inequality.

4. **Typed inter-core separation.** For every pair of cores not jointly contained in one explicitly declared collision-coarsening neighborhood, their complete response images in the common $\mathbb{R}^{N_{\text{prot}}}$ obey

$$\text{dist}(\mathcal{R}_{\pi, p}^{\text{obs}}(\mathcal{Z}_{\pi, p}), \mathcal{R}_{\pi', p'}^{\text{obs}}(\mathcal{Z}_{\pi', p'})) \geq s_{(\pi, p), (\pi', p')} > 0.$$

The exemption applies only to restricted pieces analyzed together in that same coarser quotient; generic adjacency is not an exemption.

5. **Collision disposition.** A neighborhood of the fine-core boundary $g_{\text{str}} \downarrow 0$ is analyzed in the adjacent coarser quotient. The output is a coarser physical confidence set plus the set-valued union of compatible adjacent fine strata. Ordered mass-support/eigenvector labels are not continued through the collision.
6. **Acceptance and analyzer response.** The production/analyzer acceptance has a positive minimum, while the calibrated signed/complex response satisfies the preceding conditioning bounds. Acceptance is not identifiability and complex interference coefficients are not required to be positive.
7. **Joint nuisance coverage.** Event, calibration, background, and nuisance coordinates enter one joint confidence construction; only afterward is the set projected to the physical quotient.
8. **Reference-core coverage and separation.** The no-vectorlike $\mathcal{Z}_{\text{ref}, p}$ has its own hypothesiswise coverage-certified joint region in the same record space and obeys the displayed lower-Lipschitz bound with $\kappa_{\text{ref}, p} > 0$. Every carrier core has the recorded positive carrier/reference image separator

$$s_{(\pi, p), (\text{ref}, p)} := \text{dist}(\mathcal{R}_{\pi, p}^{\text{obs}}(\mathcal{Z}_{\pi, p}), \mathcal{R}_{\text{ref}, p}^{\text{obs}}(\mathcal{Z}_{\text{ref}, p})) > 0$$

. A global lower-Lipschitz certificate on a disjoint carrier/reference union does not replace this separator. From one common record, the guard $r_{\text{rec}} < s_{(\pi, p), (\text{ref}, p)}/2$ makes the carrier and reference inverse sets mutually exclusive: on the true carrier coverage event the carrier set contains truth and the reference set is empty, while on the true reference coverage event the roles reverse. This is hypothesiswise model/core selection, not simultaneous coverage of two nonempty counterfactual regions. Carrier-only conditioning may not be exported as the aggregate SPIC required by P75-21.

9. **Sampling-law certificate.** The protocol freezes either independent retained trials or an explicit martingale, mixing, or effective-sample-size condition with a simultaneous tail bound satisfying $\epsilon_{\text{rec}}^{\text{stat}}(n) \downarrow 0$. Bounded event/control summaries alone do not imply concentration under arbitrary dependence.
10. **Common inverse-set geometry.** The protocol freezes one $\widehat{Y}_n \in \text{ran } M_{\text{obs}}$. For every declared carrier or reference core label a , write $\mathcal{Z}_a$ and $\mathcal{R}_a^{\text{obs}}$ for its augmented core and response. Its true-point event obeys the uniform hypothesiswise guarantee

$$\mathcal{E}_a(z_a) := \{\|\widehat{Y}_n - \mathcal{R}_a^{\text{obs}}(z_a)\|_2 \leq r_{\text{rec}}(n)\}, \quad \inf_a \inf_{z_a \in \mathcal{Z}_a} \Pr(\mathcal{E}_a(z_a)) \geq 1 - \alpha.$$

Its inverse sets are

$$\mathcal{C}_{\pi, p}^{\text{joint}}(\widehat{Y}_n) = \{z \in \mathcal{Z}_{\pi, p} : \|\mathcal{R}_{\pi, p}^{\text{obs}}(z) - \widehat{Y}_n\|_2 \leq r_{\text{rec}}(n)\},$$

and the identical definition with (π, p) replaced by (ref, p) . The common record ball, not an assumed parameter ball, supplies the factor of two in inverse confidence-set diameters and the half-separator selection rule. No probability statement treats mutually exclusive hypotheses as simultaneously true.

Field	Requirement
Parent support J_Q	Strictly above every perturbed heavy mass-support coordinate plus tagged charged-lepton and resolution margin.
Heavy support J_H^{rob}	Compact union containing all structural mass-support coordinates in the certified perturbation domain.
Source/analyzer cells	Positive minimum preparation/acceptance plus conditioned signed and complex response.
Physical quotient	Independent structural mass-support coordinates; $W_a = V'_H P_a V_H^{\dagger}$; and the d/q reduct. Nuisance-indexed kernel functions are not coordinates of Ξ_π .
Kernel metadata	The frozen family $D_\pi^{(j)}(\vartheta)$ and derived coherent response are evaluated inside $\mathcal{R}_{\pi,p}$, retain interference, are not reconstructed from held-out data, and are not an X7 pole computation.
Record space	One ordered observed-summary $\mathbb{R}^{N_{\text{prot}}}$ for every carrier and reference core, including bin, quadrature, rate, and control coordinates. Every retained coordinate is observed under every hypothesis; one hypothesis-independent availability projection zero-weights or quotients unavailable coordinates out of distance, conditioning, and inversion. A physically observed zero-rate prediction is ordinary model output, never padding.
Identifiability	Global carrier-core and independent reference-core lower-Lipschitz bounds, plus positive image separation for every pair not contained in one declared collision-coarsening neighborhood; the exception is restricted to pieces analyzed in that same coarse quotient.
Parameter transport	Record radius r_{rec} gives carrier/reference parameter diameters at most $2r_{\text{rec}}/\kappa_{\pi,p}$ and $2r_{\text{rec}}/\kappa_{\text{ref},p}$; finite-exposure targets require the analogous strict deterministic-floor guards.
Fine split cores	$g_{\text{str}} \geq g_{\pi,\min} > 0$; a uniform reported-resolution claim additionally requires $g_{\pi,\min} > 4r_m$; collision neighborhoods use the coarser quotient.
Nuisance/control	Joint confidence on $K_{\pi,p} \times \Theta_{\pi,p} \times C_{\pi,p}$, followed by physical projection.
Reference	Explicit compact zero-heavy-amplitude core with its own metric, $\kappa_{\text{ref},p} > 0$, the same global record space and design, a hypothesiswise coverage-certified joint region, and a required positive carrier/reference image separator for every carrier core.
Confidence	Uniform hypothesiswise coverage $1 - \alpha$, frozen before held-out records; it does not assert simultaneous truth of carrier and reference hypotheses.
Inverse sets	One common $\widehat{Y}_n \in \text{ran } M_{\text{obs}}$; carrier and reference joint sets are the preimages of its radius- $r_{\text{rec}}(n)$ ball under their observed responses. Below a carrier/reference half-separator, the true-hypothesis set contains truth and the incompatible set is empty.
Sampling law	Independent retained trials or a predeclared martingale/mixing/effective-sample-size condition with an explicit simultaneous tail bound; boundedness alone is insufficient.
Resolution	$g_{\text{str}} > 2r_m$ gives unique true-support matching, $\widehat{g}_{\text{str}} > 2r_m$ gives realized record resolution, and $g_{\text{str}} > 4r_m$ gives uniform reported-interval disjointness. Below the last threshold report ResolutionNotUniformlyCertified , or SplitButUnresolved when the realized intervals overlap. Complex-pole multiplicity remains a separate X7 axis.
Update	Uncovered mass support, zero acceptance, material sub-bin structure, or a changed response class produce ProtocolUpdate .

Proposition E.1 (Finite robust exposure above the deterministic floor). *On a compact SPIC-certified domain with positive minimum acceptance, carrier constants $\kappa_{\pi,p} > 0$, independent reference constants $\kappa_{\text{ref},p} > 0$, typed*

inter-core separators, coverage-certified reference cores, and the frozen sampling-law certificate, write

$$r_{\text{rec}}(n) = \epsilon_{\text{rec}}^{\text{det}} + \epsilon_{\text{rec}}^{\text{stat}}(n), \quad \epsilon_{\text{rec}}^{\text{stat}}(n) \downarrow 0.$$

For every predeclared record-space target $r_{\text{rec},\star} > \epsilon_{\text{rec}}^{\text{det}}$, finite exposure attains $r_{\text{rec}}(n) < r_{\text{rec},\star}$ at uniform hypothesis-wise coverage $1 - \alpha$; this target lies in $\text{ran } M_{\text{obs}}$, not in a parameter space.

For separately predeclared parameter-diameter targets $r_{\mathcal{Z},\pi,p}$ and $r_{\mathcal{Z},\text{ref},p}$, every nonempty joint inverse region obeys its corresponding bound

$$\text{diam}_{d_{\mathcal{Z}}} C_{\pi,p}^{\text{joint}}(\widehat{Y}_n) \leq \frac{2r_{\text{rec}}(n)}{\kappa_{\pi,p}}, \quad \text{diam}_{d_{\mathcal{Z},\text{ref}}} C_{\text{ref},p}^{\text{joint}}(\widehat{Y}_n) \leq \frac{2r_{\text{rec}}(n)}{\kappa_{\text{ref},p}}.$$

Finite exposure attains these parameter targets only under

$$\frac{2\epsilon_{\text{rec}}^{\text{det}}}{\kappa_{\pi,p}} < r_{\mathcal{Z},\pi,p}, \quad \frac{2\epsilon_{\text{rec}}^{\text{det}}}{\kappa_{\text{ref},p}} < r_{\mathcal{Z},\text{ref},p},$$

and then supplies n satisfying the same inequalities with $r_{\text{rec}}(n)$ in place of $\epsilon_{\text{rec}}^{\text{det}}$. Noncollision classification further requires an attainable record radius below the applicable typed half-separator. For a carrier/reference pair this means that, on the true carrier coverage event, the carrier inverse contains truth and the reference inverse is empty, while on the true reference coverage event the roles reverse. Uniform reported-interval fine resolution requires the separately transported mass radius below $g_{\text{str}}/4$. The physical confidence set for whichever hypothesis is true is obtained by projecting its joint set, and collision neighborhoods retain the compatible adjacent-strata union.

Proof. The frozen record family consists of finitely many bounded event and control summaries, including calibrated differences and real/imaginary interference quadratures in one common ordered record space. Under the frozen sampling-law certificate, the applicable independent-trial or declared martingale/mixing/effective-sample-size tail bound and simultaneous union control within each true-hypothesis law shrink only $\epsilon_{\text{rec}}^{\text{stat}}(n)$, not the certified deterministic floor. Boundedness alone would not supply this step. Thus every record-space target strictly above that floor is attainable at finite exposure. Two candidates in one record ball differ by at most $2r_{\text{rec}}(n)$; the carrier and independent reference lower-Lipschitz bounds yield the two parameter-diameter inequalities. Their strict deterministic-floor guards are therefore necessary for finite-exposure attainment of the declared parameter targets. If two separated cores both had nonempty inverse sets, one point from each would put their image distance at most $2r_{\text{rec}}(n)$, contradicting the half-separator guard. On the true-hypothesis coverage event its inverse contains truth, hence is nonempty and forces the incompatible set to be empty. Projection preserves that true hypothesis's physical coverage, not simultaneous counterfactual coverage. At a declared collision the separator exemption applies only to the restricted pieces analyzed in the same coarser quotient, where set-valued adjacent-strata projection replaces a single fine-stratum inverse. $\square$

The typed parameter-inversion guards are

$$\left(\frac{2r_{\text{rec}}}{\kappa_{\pi,p}} < r_{\mathcal{Z},\pi,p}, \quad \frac{2r_{\text{rec}}}{\kappa_{\text{ref},p}} < r_{\mathcal{Z},\text{ref},p} \right),$$

with the corresponding strict deterministic-floor guards obtained by replacing r_{rec} with $\epsilon_{\text{rec}}^{\text{det}}$. These parameter-space targets are not record, mass-support, or operator-Gram radii; the ideal P75-21 pair-space gap is not a detector-radius target at all.

The detector export is the typed guard vector

$$\left(r_{\text{rec}} < \frac{1}{2} s_{(\pi,p),(\pi',p')}, \quad r_m < \frac{1}{4} g_{\text{str}}, \quad 2s_E \bar{\eta}_V + \bar{\eta}_V^2 + \epsilon_{\text{op}} < \frac{s_E^2}{3} \right).$$

The first entry is evaluated only for pairs requiring typed separation, including every carrier/reference pair, and r_{rec} must be attainable above $\epsilon_{\text{rec}}^{\text{det}}$. It implements the hypothesiswise same-record selector above. The third entry uses the predeclared uniform protocol-domain bound $\bar{\eta}_V$, not an unknown actual perturbation norm. The ideal P75-21 pair-space gap remains separate and supplies no fourth detector guard. These three guards live in different metrics and are not replaced by a single scalar tolerance.

F Support-manuscript index

ID	Support manuscript/source	Use
P75-S01	Constraint-Induced Physical Carrier Realization: Detailed ARC Plan	PCR-7.5 placement, PCR-8 boundary, prediction-lock and capstone discipline.
P75-S02	Physical Nature and Minimal Identity of the Neutrino Carrier	One terminal carrier, 3+1 sector core, complete heavy $U(3)_H$ descent, terminal promotion framework.
P75-S03	Observable Projection and Experimental Prediction Lock for the Neutrino Carrier	CCST packet, finite nonlinear protocol, reference separation, held-out firewall.
P75-S04	Joint Standing-Carrier Closure for the Neutrino Sector	Ready child source categories, strict-margin source region, parent boundary.
P75-S05	Symmetry and Charge Extensions of the Neutrino Standing-Carrier Space	Vectorlike Dirac source, charge/protector structure, spectral charge uniformity.
P75-S06	Successor-Target Quantitative Active Closure of the Neutrino Standing-Carrier Space	Heavy response, process tester, matching, D3-1.3 target.
P75-S07	Heavy-State, Active-Closure, and CLFV Extension	Bundled exact PCR-5D3 source for the prompt-response map, declared $c\tau_A^{\text{resp}} = 0.2$ mm response-chart coordinate, exact membership in the discrete ATLAS grid spanning $[0.1, 1.0]$ mm, no continuous response-neighborhood authority, and full-source-lifetime/production/decay separation.
P75-S08	Cosmological Population and Transport Extension	Population and source-guard semantics.
P75-S09	Lepton-Number-Violating Target Extension	Conjugation/LNV standing and one-apex physicality.
P75-S10	Direct Kinematic-Mass Target Extension	Absolute-scale and fixed-base source authorization.
P75-S11	Functorial Target Extension of Standing-Carrier Quotients	Target update, restriction, fibres, path and failure provenance.
P75-S12	Exact Standing-Carrier Stratification	B15 matching/continuation and source atlas.
P75-S13	Standing-Carrier Upper Closure	Seven-channel packet and source/carrier distinction.
P75-S14	Compatible Role-Occupancy Closure	Internal factorization versus terminal projection.
P75-S15	Standing Carrier Identity Closure	Complete continuation signatures and identity uniqueness.
P75-S16	Role-Occupancy Closure under Admissibility	Eligibility, compatibility, live residue, assignment orbit.
P75-S17	A Fixed-Domain Exhaustion Theorem	Scope, fail-closed status, anchor/tensor/skin exhaustion.
P75-S18	Source-First Exhaustion of the Physical Carrier Universe	Source perturbation grammar and activity reduction.
P75-S19	Simultaneous Physical Constraint Collapse	One-apex residual and common-cone conditions.

ID	Support manuscript/source	Use
P75-S20	Mass-Matrix, Eigenstructure, and Ratio Rigidity	Spectral standing and matrix/basis discipline.
P75-S21	Mixing-Angle and Phase Standing	Invariant mismatch and boundary reconstruction.
P75-S22	Neutrino Ordering Prediction-Lock Closure	Prediction-lock and exact negative endpoint discipline.
P75-S23	The Higgs Bridge Carrier	Premetric carrier-first dependency.
P75-S24	The Higgs Domain Endpoint	Projection and no-backflow discipline.
P75-S25	Neutrino Carrier Target-Extension Mini-ARC Skeleton	D-series sequencing and endpoint neutrality.
P75-S26	Mini-ARC Refinement Addendum	Standing-coordinate and common-generator distinctions.
P75-S27	FCGS Mini-ARC Detailed Skeleton	Cross-fermion universality boundary.
E1	ATLAS prompt HNL search, EPJC 86 (2026) 153	Published 8–65 GeV LNV search: direct mass grid includes 30 but not 28 GeV; the lifetime-reweight grid includes the interior 0.2 mm coordinate; every selected event contains a same-flavour, same-charge pair, and three-same-flavour signal events are rejected.
E2	CMS trilepton HNL search, JHEP 06 (2024) 123	Dirac/Majorana trilepton coverage from 10 GeV to 1.5 TeV, hadronic-tau channels, trigger and flavour-category context.
E3	ATLAS displaced HNL search, JHEP 07 (2025) 196	Long-lived Dirac/Majorana coverage below 14.5 GeV and displaced-vertex comparison boundary.
E4	ATLAS W -boson mass and width, EPJC 84 (2024) 1309	Numerical W -rest-frame kinematic localization.
E5	ATLAS prompt-HNL HEPData, Figure 4a	Observed electron-mixing ordinates at the tabulated 20 and 30 GeV mass points; PCR-7.5 freezes straight-line display interpolation as a conditional ordinate screen, not a likelihood recast or source-level non-exclusion.
M1	Bhatia, <i>Matrix Analysis</i>	Spectral, singular-value and commutant results.
M2	Stewart and Sun, <i>Matrix Perturbation Theory</i>	Singular-value and subspace perturbation.
M3	Davis–Kahan theorem	Spectral projector stability.
M4	Horn and Johnson, <i>Matrix Analysis</i>	Gershgorin, matrix norms and spectral inequalities.

G Theorem dependency index

ID	Local result	Immediate dependencies
P75-00	Import/no-backflow lock	PCR-5E, PCR-6, PCR-7, ARC, the declared PCR-3/PCR-4, fixed-domain-exhaustion, and SCIC authorities, and their frozen transitive dependency closure
P75-01	Robustness target	P75-00, fixed-domain theorem
P75-02	Semantic perturbation class and information-preserving joint-trace regrouping/factorization	P75-01; PCR-3/5E/6/7 interface completeness; complete cross-interface incidence/correlation record and the explicit lossless regrouping maps
P75-03	Source perturbation grammar	P75-02, PCR-3/4/5E
P75-04	Status classifier	P75-01–03, fixed-domain exhaustion
P75-05	Central/traceless decomposition	PCR-6 heavy action, matrix commutant
P75-06	Symmetry-preserving rigidity	P75-05
P75-07	Degeneracy lifting	P75-05/06
P75-08	Spectral spread	P75-07, min-max theory
P75-09	Heavy subspace stability	P75-03, Weyl, Davis–Kahan
P75-10	Charged-current rank	PCR-7 source, singular-value perturbation
P75-11	Universality rigidity	P75-10, polar decomposition
P75-12	Closure transport	PCR-7 block isometry
P75-13	Trace-transport carrier radius	P75-03/04/09/10; PCR-5E E-S1/E-S2 finite incidence and E-S4A strict cell; PCR-6 Definition 18.1, P6-04, P6-S2B/G deletion witnesses, and full-support inheritance
P75-14	One-carrier stability	P75-13 trace-transport cell, exact proper-support deletion flags, and single full-support parent orbit
P75-15	Splitting without promotion	P75-07, P75-14
P75-16	Structural mass-support resolution	P75-07–09; $g_{\text{str}} > 2r_m$ for true-support matching, $\widehat{g}_{\text{str}} > 2r_m$ for observed interval resolution, and $r_m < g_{\text{str}}/4$ for a uniform design guarantee
P75-17	Fibrewise invariant core/complete covariant envelope	P75-01–03, P75-05–15, P75-02 information-preserving complete-joint-trace regrouping/factorization including cross-interface relations, pre-frozen variation registry, inherited PCR-7 structural reduct; no P75-18–21 premise
P75-18	Cross-child common reduct	P75-17, PCR-5E common-source forgetful maps, matched ready-child realizations, and matched common perturbations; decorated child envelopes remain separate
P75-19	Stratified mass-support/response coverage conditional on SPIC	PCR-7 protocol grammar, P75-09/10/16, one common record space, coherent response normal form, within-core global lower-Lipschitz bounds, all non-coarsened core separators, explicit reference core, collision coarsening, joint nuisance/control coverage, typed deterministic floors, same-record model/core selector $r_{\text{rec}} < s_{\text{car,ref}}/2$, and carrier/reference record-to-parameter inverse-radius guards
P75-20	Factorized detector disposition	P75-10/16/19, SPIC where detector reconstruction is claimed, response-kernel status, and Gram/Gershgorin bounds including ϵ_{op}

ID	Local result	Immediate dependencies
P75-21	Ideal gap and certified-record model/core selection	Ideal pair-space gap: P75-10/12. Detector selection: P75-19/20, carrier-plus-reference SPIC, attainable $r_{\text{rec}} < s_{\text{car,ref}}/2$, truth-containing inverse for the true hypothesis and empty competing inverse, plus separate parameter-inverse, mass, and Gram guards
P75-22	Robustness vector	P75-06–21
P75-23	Falsifier hierarchy	P75-06 central-stratum conditional, P75-07 noncentral split classification, P75-17/22
P75-24	Held-out firewall	PCR-7 empirical partition, P75-22/23
P75-25	Capstone	P75-00–24, P75-X1–X8, and P75-X2A
P75-26	PCR-8 handoff	P75-25, ARC boundary
P75-X1	D3 benchmark and separated VJ mass-domain authority	PCR-5E source-qualified $H(I_E)$, PCR-5D3 source-certified realization continuum and ATLAS-reweight-grid-anchored [28, 30] GeV benchmark, no D3-to-VJ domain bridge
P75-X2	D3 source-certified response-grid tag	P75-X1, two-body W -rest-frame kinematics, ATLAS m_W , exact discrete-grid membership and endpoint-distance bookkeeping; no continuous detector-response radius and no full-source lifetime identity
P75-X2A	Predeclared D3 benchmark-admissibility certificate	PCR-5D3 non-direct-constraint realization continuum, ATLAS direct 30 GeV mass anchor and lifetime-reweight grid, HEPData Figure 4a ordinate replay, P75-X2 kinematic tag; endpoint selection, full response, and non-exclusion
P75-X3	D3 benchmark/VJ search architecture	NotReady
P75-X4	Resolution-adaptive authority refinement	P75-X1, P75-X2, P75-X2A, PCR-7 packet, P75-13–21 robustness hierarchy, current ATLAS/CMS analyses; no P75-X7 premise
P75-X5	Row-normalization identity, dyadic-existence repair, and marked-child bound	P75-X1–X3, detector mass resolution, independent-constraint route, P75-24 empirical firewall
P75-X6	D3 response-grid-coordinate authority	PCR-5E E-S4A strict-pass/guard construction, PCR-7 charged-current block and empirical partition; bounded weak-record disposition is ConditionalLightResponseCompatibleAt95 only, full VJ compatibility and $\epsilon_{95}^{\text{weak}}$ NotReady, with zero direct/indirect m_W cross-covariance only a screening assumption and not an independence certificate
		PCR-5D3 source realization and declared response map, exact interior membership in the frozen discrete ATLAS lifetime-reweighting grid, endpoint-distance bookkeeping that supplies no continuous detector-response radius, full-source lifetime/response NotReady firewalls, and separated VJ nonprediction firewall; P75-X1, P75-X2, P75-X2A

ID	Local result	Immediate dependencies
P75-X7	Conditional chart-qualified complex-pole lifetime envelope	P75-X5/X6 firewalls; certified outer $\mathcal{E}_X^{\text{src}} \ni \epsilon_X$, coverage-preserving projection $\mathcal{E}_X^{\text{weak}}$ of a jointly covered weak region, nonempty $\mathcal{E}_X^{\text{adm}}$, and $\alpha_{\text{weak}} \subset \alpha_{\text{spec}}$; one sheet-holomorphic Σ^{ren} , boundary-only $\Delta M/\Gamma$, intrinsic $\mathcal{G}^{\text{self}}/C_t^{\text{pole}}$ factors, separate gauge-complete full-resummed-propagator $\mathcal{F}_t$, isolated-simple-pole-only residue expansion, standalone $\mathcal{T}_t$ inclusion on the weak covered event, and B_{norm}
P75-X8	Source-coordinate inversion and physical pole-mass projection on $H(I_E)$	P75-X7; exhaustive disjoint $\mathfrak{T}_\tau \dot{\cup} \mathfrak{T}_R$, uniform lifetime identifiability, gauge-complete full- $\mathcal{D}$ $\mathcal{F}_t$ on $\mathfrak{T}_R$, finite-dimensional Euclidean/Borel $\mathcal{Y}^{\text{ind}} = \text{ran } M_{\text{ind}}$, and bridge-free combined α_{spec} ; multiple lifetime charts require familywise control. Optional D3 compatibility prefreeszes $\mathfrak{T}_{D3} = \text{Compat}^\tau \dot{\cup} \text{Compat}^R$, uses covered C^τ/C^R , and is an $\mathcal{E}_{D3}/\alpha_{\text{bridge}}$ -qualified calibration set outside core $\mathcal{I}_M^{\text{src}}$

H Supplementary formal records

The project archive contains formal representations of the perturbation target, source grammar, spectral atlas, terminal trace-transport certificate, robustness vector, invariant/covariant packet pair, conditioned protocol extension, separated D3/VJ search authorities, complex-pole lifetime route, falsifier hierarchy, and PCR-8 handoff. These records support provenance and reproducibility; they do not replace the semantic or mathematical proofs.

Record	Content
pcr75_target_certificate.json	Ready child tags, perturbation scope, update boundaries and empirical status.
pcr75_perturbation_atlas.json	Source normal forms, status classes and trace fields.
pcr75_spectral_stability.json	Central/traceless decomposition, 3, 2 + 1, and 1 + 1 + 1 structural mass-support partitions, distinct-support gap, spread, and projector bounds; no physical complex-pole claim.
pcr75_carrier_stability.json	Status-free finite support/mode skeleton, heavy-partition decorations, single full-support parent orbit, transported deletion flags, explicit trace/guard/update faces, and positive one-carrier radius.
pcr75_robust_packet.json	Fibrewise registry-relative invariant core $\mathcal{O}_{\text{core}}$ on $\mathcal{D}_{\text{str}}^{(b)}(X)$, complete-trace covariant envelope $\mathcal{O}_{\text{cov}}$, exact law packet, certified variation witnesses, and a cross-child common reduct only for matched ready-child realizations under matched common perturbations.
pcr75_protocol_extension.json	SPIC common record space, frozen sampling law and deterministic-floor attainability, same-record carrier/reference selector $r_{\text{rec}} < s_{\text{car,ref}}/2$, hypothesiswise truth-containing/competing-empty inverses, record-to-parameter inverse guards, $r_m < g_{\text{str}}/4$ uniform mass-resolution guard, true-support and observed-gap distinctions, collision coarsening, nuisance/control coverage, and a separate Gram guard. On block isometry with $\eta_V \leq \bar{\eta}_V < s_E$, the ideal pair-space gap is structural, independent of SPIC, and supplies no finite-record guard.

Record	Content
pcr75_search_localization.json	D3 SourceCertified_ATLASRweightGridAnchored_SearchBenchmark, ATLASRweightGridAnchored_InteriorResponseCoordinate, direct 30 GeV mass anchor, declared-linear HEPData Figure 4a conditional ordinate screen, symbolic VJ domain $H(I_E)$, operational comparison slice, trigger/selection logic, and typed domain-bridge boundary.
pcr75_weak_deficit_lifetime.json	Certified outer source enclosure, coverage-preserving epsilon projection of a jointly covered weak region with $\alpha_{\text{weak}} \subset \alpha_{\text{spec}}$, admissible intersection/incompatibility, fixed-child selection firewall, one sheet-holomorphic self-energy with boundary-only dispersive/absorptive split, intrinsic $\mathcal{G}^{\text{self}}/\mathcal{C}^{\text{pole}}$ factors, standalone lifetime-envelope inclusion, separate gauge-complete full-resummed-propagator $\mathcal{F}_t$ with isolated-simple-pole-only residue expansion and full- $\mathcal{D}$ /complete-principal-part coherent inversion, finite-dimensional observed space, exhaustive X8 routes, source-coordinate inverse, pole-mass projection, conditional/familywise coverage, and optional pre-record D3 lifetime/full-response bridge routes with $\mathcal{E}_{D3}/\alpha_{\text{bridge}}$.
pcr75_robustness_vector.json	Feature-specific structural radii, class-conditional exactness, and certified-domain protocol guard vector without an inferred protocol perturbation radius.
pcr75_falsifier_hierarchy.json	P75-23 core first-failure localization plus separately attributed P75-X1–X8 and P75-X2A authority and inversion attacks.
pcr75_pcr8_handoff.json	P75-H1–H10 downstream interface.
theorem_registry.json	P75-00–P75-26 and P75-X1–P75-X8 plus P75-X2A inventory and source dependencies.

These records support a conditional source-coordinate inverse $\mathcal{I}_M^{\text{src}}$ on $H(I_E)$. Its accepted witnesses map separately through the target-complete complex-pole kernel to $\mathcal{I}_{m_{\text{pole}}}^{\text{phys}}$. A lifetime-route chart covers its targeted pole; complete support from several such charts needs simultaneous or familywise control. The full-response route covers the complete unordered support. Numerical VJ domain endpoints, lifetime, source-coordinate localization, and physical pole-mass localization remain NotReady. Independently, D3 establishes the SourceCertified_ATLASRweightGridAnchored_SearchBenchmark on [28, 30] GeV at $c\tau_A^{\text{resp}} = 0.2$ mm. The response coordinate is an exact interior ATLAS reweight-grid point and only 30 GeV is a direct ATLAS mass anchor. The declared-linear HEPData Figure 4a screen is greater than a factor 4.50 under the ATLAS simplified model, not source-level non-exclusion; $\text{D3FullSourceProperLifetime} = \text{D3FullSourceResponseValidation} = \text{NotReady}$. In VJ scope the pair is a PredeclaredOperationalScanSlice_NoVJMassAuthority, with no priority or optimization claim. After its domain bridge, that benchmark can additionally contribute a possibly uninformative bridge-conditional source-coordinate compatibility/confidence set. The set is the predeclared union of lifetime-region and same- $\mathcal{V}^{\text{ind}}$ full-response compatibility, is qualified by $\mathcal{E}_{D3}/\alpha_{\text{bridge}}$, lies outside core $\mathcal{I}_M^{\text{src}}$, and is neither independent VJ mass authority nor physical pole-mass localization.

I Claim and nonclaim register

Claim	Status	Boundary
One-carrier stability	Proved on the declared $\mathcal{U}_{\text{car}}$	Exact transported proper-support deletion flags and a single full-support parent orbit inside an explicit trace-transport cell; uncertified charts and unresolved parents are not covered.
Exact heavy degeneracy	Proved on central stratum	Zero unrestricted breaking radius.

Claim	Status	Boundary
Heavy splitting under $T \neq 0$	Proved	Does not imply carrier splitting.
Rank-three robustness	Proved for $\eta_V < s_E$	Sharp operator-norm radius.
Flavor universality	Proved on the full-rank equal-singular-value stratum	Positive universality means $\Omega'_H = \delta' I_3$ with $\delta' > 0$; it is not generic in the rank ball.
$\Delta'_{\text{act}} = \Omega'_H$	Exact on complete block-isometry class	Failure means deeper source/target change.
Maximal structural information	Proved fibrewise on the certified structural domain	O_{core} is maximal relative to its pre-frozen variation registry; O_{cov} is the complete structural-trace class modulo functional determination; cross-child comparison uses only the common reduct for matched ready-child realizations under matched common perturbations.
Ideal carrier/reference separation	Proved for $\eta_V < s_E$ with block isometry	Algebraic pair-metric gap; no detector inverse is used.
Detector-record carrier/reference selection	Proved conditional on aggregate SPIC	One common record space, global within-core conditioning, explicit coverage-certified reference core, collision coarsening, joint nuisance/control coverage, and attainable $r_{\text{rec}} < s_{\text{car,ref}}/2$. On true-carrier coverage the carrier inverse contains truth and the reference inverse is empty, and conversely for a true reference. The ideal pair-space gap remains structural; there is no simultaneous counterfactual nonempty-region distance or detector pair-space guard.
Row-normalization identity	Proved on each fixed unperturbed VJ source	$\epsilon_{\text{row}} = \delta_E = s_E^2$; a measured weak-rate interpretation also requires an EW input-scheme response.
Marked dyadic coefficient	Existence repaired; child-qualified upper bound proved	$\epsilon_{\text{row}} \leq 1/256$, but the exact mark and an independently physical normalization selection remain unexposed.
Public weak-record pass	Conditional light-response compatibility; common normalization not identified	Conditional on applicability of the published light-response model, HFLAV gives $\chi_4^2 = 7.0193$, $p = 0.1349$, with zero common- ϵ Jacobian; full VJ compatibility is NotReady. The separate m_W screen gives $\epsilon_{95}^{\text{scr}} = 1.62 \times 10^{-3}$ after setting unreported direct/indirect cross-covariance to zero as a screening assumption, not an independence certificate. It lacks an X7-ready finite-mass response certificate, so the source ceiling is retained.

Claim	Status	Boundary
D3 search benchmark	SourceCertified_ATLASR eweightGridAnchored_Se archBenchmark	The pair ([28, 30] GeV, $c\tau_A^{\text{resp}} = 0.2$ mm) has an explicit D3 source-coherent physical realization at every mass, with the declared non-direct oscillation/LFU/CLFV/matching passage. The coordinate is the ATLASReweightGridAnchored_In teriorResponseCoordinate, and only 30 GeV is a direct ATLAS mass anchor. A declared-linear HEPData Figure 4a screen clears by a factor greater than 4.50 under the ATLAS simplified model, but is not source-level non-exclusion. D3FullSourceProperLifetime and D3FullSourceResponseValidati on are NotReady; VJ use is only a PredeclaredOperationalScanSl ice_NoVJMassAuthority, with no priority or optimization claim.
VJ complex-pole lifetime envelope	Conditional theorem	Requires certified outer $\mathcal{E}_X^{\text{src}} \ni \epsilon_X$, the coverage-preserving epsilon projection $\mathcal{E}_X^{\text{weak}}$ of a jointly covered weak region, nonempty $\mathcal{E}_X^{\text{adm}}$, and α_{weak} inside α_{spec} ; one sheet-holomorphic Σ^{ren} with boundary-only $\Delta M/\Gamma$; intrinsic $\mathcal{G}^{\text{self}}/C^{\text{pole}}$, separate gauge-complete full-resummed-propagator $\mathcal{F}_t$ with isolated-simple-pole-only residue expansion and full- $\mathcal{D}$ /complete-principal-part coherent inversion, and true-lifetime inclusion in $\mathcal{T}_t$ on the weak covered event conditional on the true chart/nuisance tuple; all remain NotReady.
Typed source-mass-parameter localization	Conditional/NotReady on $H(I_E)$	$\mathcal{I}_M^{\text{src}}$ projects pre-record exhaustive disjoint routes through the finite-dimensional Euclidean/Borel space $\mathcal{Y}^{\text{ind}} = \text{ran } M_{\text{ind}}$ under parameter-uniform or conditional-on-design coverage; $\mathfrak{T}_\tau$ requires uniform identifiability, every complement lies in $\mathfrak{T}_R$, and the bridge-free well-specification failure α_{spec} enters coverage.
Physical pole-mass localization	Conditional/NotReady	$\mathcal{I}_{m_{\text{pole}}}^{\text{phys}}$ is the separate complex-pole-kernel image of accepted source-coordinate witnesses. A lifetime-route chart covers its targeted pole; the full-response route covers the complete unordered support, and multiple lifetime-route poles require simultaneous/familywise coverage.

Claim	Status	Boundary
D3-to-VJ benchmark transfer	NotReady and optional	Before the response record the bridge freezes $\mathfrak{T}_{D3} = \text{Compat}^r \dot{\cup} \text{Compat}^R$, exports covered $C_{D3 \rightarrow X}^r$ and same- $\mathcal{Y}^{\text{ind}}$ regions $C_{D3 \rightarrow X, t}^R$, and forbids post-record pruning. The singleton $\{0.2 \text{ mm}\}$ is allowed only after both a full-source lifetime identity and an exact lifetime-coordinate bridge are certified. With $\mathcal{E}_{D3} = \mathcal{E}_{\text{spec}} \cap \{\text{bridge pair/type/route membership}\}$, $\Pr(\mathcal{E}_{\text{spec}} \setminus \mathcal{E}_{D3} \mid \mathcal{G}_{\text{bridge}}) \leq \alpha_{\text{bridge}}$ implies $\Pr(\mathcal{E}_{D3}) \geq 1 - \alpha_{\text{spec}} - \alpha_{\text{bridge}}$. The possibly uninformative union is benchmark-derived, lies outside core $\mathcal{I}_M^{\text{src}}$, and creates neither independent VJ mass authority nor physical pole-mass localization; none of those limits demotes the benchmark itself.
Measured heavy splitting	Not claimed	Future held-out record.
Universal neutrino robustness	Not claimed	Only two ready children.
Cross-fermion robustness	Not claimed	FCGS target.
Statistical confidence in theorem	Not applicable	Confidence applies to empirical instantiation.

J Falsifier lattice

The robustness comparison contains independently scoped implication branches. In particular, exact structural mass degeneracy and charged-current universality are separate coordinates on the unrestricted perturbation target. No implication between them is used. The positive-universality branch is instead

$$\Omega'_H = \delta' I_3, \quad \delta' > 0 \implies \text{Rank3} \implies \text{NonzeroHeavyWeight},$$

whereas P75-06/P75-07 separately characterize preservation and lifting of the structural mass degeneracy.

The block-isometry closure branch is

$$\text{BlockIsometry} \implies \Delta'_{\text{act}} = \Omega'_H.$$

Equality of these two matrices does not by itself imply separation from the no-vectorlike reference: both may vanish. The ideal structural reference gap has the stronger antecedent

$$\text{BlockIsometry} \wedge \{\eta_V < s_E\} \implies \text{IdealPairSpaceReferenceGap}.$$

This is still not a detector statement. With

$$s_{\text{car,ref}} := s_{(\pi, p), (\text{ref}, p)},$$

the detector branch is separately conditioned by

$$\text{AggregateSPIC} \wedge \{s_{\text{car,ref}} > 0\} \wedge \left\{ r_{\text{rec}} < \frac{1}{2} s_{\text{car,ref}} \right\} \implies \text{HypothesiswiseSameRecordSelection}$$

on the applicable true-hypothesis coverage event. The ideal pair-space gap does not replace the observed-record separator or the SPIC premises.

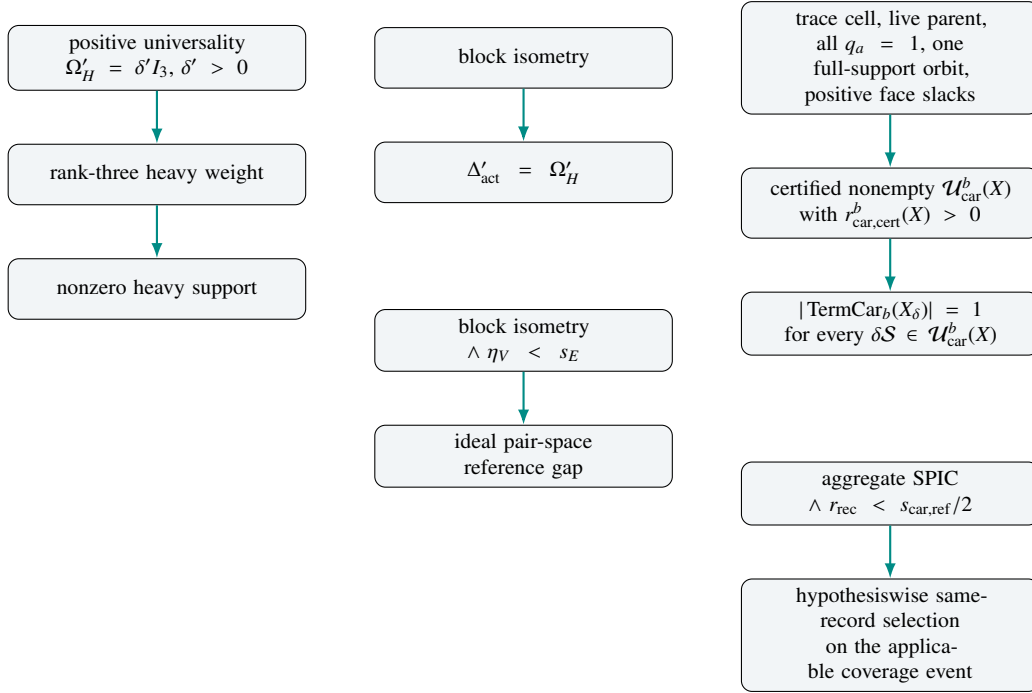
Carrier persistence has its own complete antecedent:

$$\begin{aligned} & \text{TraceCell} \wedge \text{ParentLive} \wedge (\forall a \in \mathcal{A}^{\text{prop, supp}} : q_a = 1) \wedge \text{OneFullSupportOrbit} \wedge \text{PositiveFaceSlacks} \\ & \implies r_{\text{car, cert}}^b(X) > 0, \quad \mathcal{U}_{\text{car}}^b(X) \neq \emptyset, \end{aligned}$$

followed by

$$\delta S \in \mathcal{U}_{\text{car}}^b(X) \implies |\text{TermCar}_b(X_\delta)| = 1.$$

One carrier at a single point is not, by itself, a persistence theorem. Within $\mathcal{U}_{\text{car}}^b(X)$, P75-15 additionally proves that structural spectral splitting does not promote a second carrier.



A future result is localized to the branch whose complete antecedent it preserves and whose conclusion it defeats. No implication may be inferred between branches from visual placement. Leaving an antecedent's certified domain challenges that premise or invokes the corresponding update status; it does not falsify a conditional downstream theorem.

K PCR-8 handoff schema

ID	Export
P75-H1	Exact child target and source identifiers; upstream no-backflow map.
P75-H2	Perturbation trace, source grammar and status classifier.
P75-H3	Heavy $U(3)_H$ action, central/traceless decomposition and multiplicity atlas.
P75-H4	Spectral spread, structural mass-coordinate spread, and heavy-projector stability bounds; no physical pole-mass spread is inferred.
P75-H5	Charged-current rank radius, full-rank positive universality stratum $\Omega'_H = \delta'I_3$ with $\delta' > 0$, and closure identity.
P75-H6	Status-free finite support/mode skeleton, finite heavy-partition decoration, exact proper-support deletion flags, one full-support parent orbit, and certified carrier radius only on declared trace-transport cells; absent chart witnesses remain conditional/NotReady, and compact-family uniformity requires the separately declared common perturbation bundle.
P75-H7	Structural mass support plus separate mass-support, response, reference, and physical-pole dispositions.
P75-H8	Fibrewise registry-relative invariant core and complete-trace covariant envelope at fixed child/source scope; cross-child output exists only for matched ready-child realizations under matched common perturbations and is then the common-source reduct, while decorated child envelopes remain separate.

P75-H9	Frozen SPIC sampling law, deterministic floors and attainability; same-record carrier/reference selector $r_{\text{rec}} < s_{\text{car,ref}}/2$ with truth-containing/competing-empty inverses; typed inverse guards $2r_{\text{rec}}/\kappa_{\text{car}} < r_{\mathcal{Z},\text{car}}$ and $2r_{\text{rec}}/\kappa_{\text{ref}} < r_{\mathcal{Z},\text{ref}}$; $r_m < g_{\text{str}}/4$ for uniform mass resolution, $g_{\text{str}} > 2r_m$ only for true-support matching, and $\widehat{g}_{\text{str}} > 2r_m$ for observed interval resolution; separate structural-mass and Gram guards. On block isometry with $\eta_V \leq \bar{\eta}_V < s_E$, the ideal pair-space gap is structural, not a detector guard.
P75-H10	Robustness vector, falsifier hierarchy, empirical firewall and parent boundary.

L Compact theorem synopsis

ID	Theorem	Principal output
P75-00	Import/no-backflow	Ready child scope and upstream preservation, including the declared PCR-3/PCR-4, fixed-domain-exhaustion, and SCIC authorities and their frozen transitive dependency closure.
P75-01	Target formation	Candidate-neutral robustness target.
P75-02	Semantic perturbations	Complete joint perturbation trace with information-preserving regrouping/factorization and retained cross-interface relations.
P75-03	Source grammar	Finite source-normal-form completeness.
P75-04	Status classifier	Skin/tensor/update/NotReady exhaustion.
P75-05	Heavy decomposition	$\delta K = aI + T$, commutant classification.
P75-06	Symmetry rigidity	Central perturbations preserve degeneracy.
P75-07	Degeneracy lifting	$T \neq 0 \Rightarrow 2 + 1$ or $1 + 1 + 1$.
P75-08	Spectral spread	$3\rho/2 \leq \text{diam} \leq 2\rho$.
P75-09	Projector stability	Rank-three heavy aggregate and Davis–Kahan bound.
P75-10	Rank stability	Sharp radius $\eta_V < s_E$.
P75-11	Universality	On the full-rank charged-current domain, equal singular values iff $\Omega'_H = \delta' I_3$ for some $\delta' > 0$.
P75-12	Closure	$\Delta'_{\text{act}} = \Omega'_H$ on block-isometry class.
P75-13	Terminal trace transport	Status-free finite support skeleton, partition-decorated single parent orbit, exact deletion flags, explicit faces, and positive certified radius on declared cells.
P75-14	One carrier	$ \text{TermCar} = 1$ throughout the P75-13 trace-transport region.
P75-15	Nonpromotion	Spectral splitting need not split carrier.
P75-16	Structural mass-support regimes	$g_{\text{str}} > 2r_m$ matches true supports; observed intervals resolve iff $\widehat{g}_{\text{str}} > 2r_m$; uniform design requires $r_m < g_{\text{str}}/4$, else ResolutionNotUniformlyCertified.
P75-17	Fibrewise maximal structural information	Registry-relative invariant core $\mathcal{O}_{\text{core}}$, complete-joint-trace covariant envelope $\mathcal{O}_{\text{cov}}$ supported by the lossless regrouping certificate, exact law packet, and named heavy/CCST reduct.

ID	Theorem	Principal output
P75-18	Cross-child common reduct	For matched ready-child realizations and matched common perturbations, the common-source forgetful cospan and invariant meet; decorated child envelopes remain separate.
P75-19	Protocol coverage	Finite stratified mass-support/coherent-response protocol with global conditioning, non-coarsened separators, reference core, collision coarsening, nuisance/control coverage, typed floors, and carrier/reference record-to-parameter inverse-radius guards.
P75-20	Confidence transport	Separate structural mass-support, response-kernel, and reference statuses, explicitly excluding physical-pole status; ϵ_{op} -qualified detector rank certification.
P75-21	Ideal gap and record selection	On block isometry with $\eta_V \leq \bar{\eta}_V < s_E$, the ideal pair-space gap is structural; detector inference instead uses attainable $r_{\text{rec}} < s_{\text{car,ref}}/2$ for hypothesiswise same-record selection, with the true inverse truth-containing and the competing inverse empty, plus separate parameter-inverse, mass, and Gram guards.
P75-22	Robustness vector	Feature-specific positive/zero/class-exact radii.
P75-23	Falsifier hierarchy	First-failure localization; a split exits/tests the P75-06 central stratum and is classified by P75-07 rather than contradicting P75-06's conditional implication.
P75-24	Empirical firewall	Frozen held-out robustness prediction.
P75-25	Capstone	NestedRobustnessClosure _{VJ} .
P75-26	PCR-8 handoff	P75-H1–H10 ready.
P75-X1	Benchmark/domain authority	([28, 30] GeV, $c\tau_A^{\text{resp}} = 0.2$ mm) is the D3 source-certified, ATLAS-reweight-grid-anchored benchmark; $H(I_E)$ remains symbolic and separately typed.
P75-X2	D3 response-grid benchmark	Approximate W -rest-frame energy/boost and response-coordinate images motivate the category; they are not a physical mean flight, VJ pole, or pp -lab prediction.
P75-X2A	Predeclared D3 benchmark admissibility	The complete [28, 30] GeV band has source-coherent realizations and uniform non-direct margins; 0.2 mm is an exact ATLAS reweight-grid point, 30 GeV a direct mass anchor, and the HEPData ordinate screen passes. Endpoint selection, full response, and non-exclusion remain NotReady.
P75-X3	Separated search architecture	D3 source-certified near-prompt benchmark lane; VJ flavour-complete LNC/LNV-inclusive mass/lifetime scan on declared subsets of $H(I_E)$.

ID	Theorem	Principal output
P75-X4	Authority refinement	Resolution-adaptive benchmark cells; VJ narrowing only by independent constraint/source theorem, with bridge required for D3-to-VJ transfer.
P75-X5	Row-normalization authority	Exact $\epsilon_{\text{row}} = \delta_E = s_E^2$, repaired dyadic existence, and marked-child bound $\epsilon_{\text{row}} \leq 1/256$. HFLAV gives only conditional light-response compatibility at 95 percent; full VJ compatibility and $\epsilon_{95}^{\text{weak}}$ remain NotReady. Zero direct/indirect m_W cross-covariance is only a screening assumption, not an independence certificate.
P75-X6	Response-grid-coordinate authority	$c\tau_A^{\text{resp}} = 0.2$ mm is an exact interior member of the frozen discrete ATLAS reweighting grid. Its 0.1 mm distance to the nearer endpoint of the declared grid span is not a continuous detector-response radius; full-source proper lifetime, full response validation, and VJ pole lifetime are NotReady.
P75-X7	Lifetime envelope	Conditional envelope from outer $\mathcal{E}_X^{\text{src}}$, coverage-preserving projected joint-weak $\mathcal{E}_X^{\text{weak}}$, nonempty $\mathcal{E}_X^{\text{adm}}$, one sheet-holomorphic Σ^{ren} with boundary-only $\Delta M/\Gamma$, intrinsic $\mathcal{G}^{\text{self}}/C^{\text{pole}}$, separate gauge-complete full-resummed-propagator $\mathcal{F}_t$ with isolated-simple-pole-only residue expansion, and true-lifetime inclusion in $\mathcal{T}_t$ on the covered event.
P75-X8	Source and pole-mass localization	Exhaustive routes in finite-dimensional Euclidean/Borel $\mathcal{Y}^{\text{ind}} = \text{ran } M_{\text{ind}}$ use gauge-complete full- $\mathcal{D}$ $\mathcal{F}_t$ on $\mathfrak{T}_R$ and bridge-free combined α_{spec} , yielding $\mathcal{I}_M^{\text{src}}$ then $\mathcal{I}_{m_{\text{pole}}}^{\text{phys}}$; optional D3-to-VJ compatibility prefrees lifetime/full-response routes and gives an $\mathcal{E}_{D3}/\alpha_{\text{bridge}}$ -qualified benchmark-derived set outside core $\mathcal{I}_M^{\text{src}}$.

M Representative perturbations and negative controls

M.1 Central mass motion

$$K'_H = (H_\lambda^2 + a)I_3, \quad V'_H = V_H.$$

All three structural mass-squared eigenvalues move together. The structural heavy support remains degenerate, universal, and rank three. No physical complex-pole motion or multiplicity follows without P75-X7. One-carrier identity also persists when the path remains inside the declared P75-13 trace-transport cell and avoids every regularity, witness-typing, guard, and update face.

M.2 Two-plus-one spectral splitting

$$T = \rho \text{diag}(1, -1/2, -1/2), \quad \delta V_H = 0.$$

The structural mass-support partition becomes 1 + 2 with the sharp minimum spread $3\rho/2$. This is not a physical complex-pole claim. Closure, rank, and universality of the unchanged charged-current Gram matrix persist.

One-carrier identity persists only while this path remains inside the declared P75-13 trace-transport cell and avoids its explicit faces.

M.3 Three-way spectral splitting

$$T = \rho \operatorname{diag}(1, 0, -1), \quad \delta V_H = 0.$$

The structural mass-support partition becomes $1 + 1 + 1$ with maximal spread 2ρ , without fixing the physical complex-pole multiset. The same deeper conclusions persist at their own scopes; in particular, the carrier conclusion requires membership in the declared P75-13 cell.

M.4 Anisotropic charged-current weight

$$\delta K_H = 0, \quad V'_H = \operatorname{diag}(s_E + \epsilon, s_E - \epsilon, s_E)U.$$

For $0 < |\epsilon| < s_E$, the Gram matrix is positive definite but not scalar. Rank persists. Closure persists only when N' is co-varied so that the complete light-plus-heavy block remains isometric. One-carrier identity persists only inside the declared P75-13 trace-transport cell; flavor universality fails.

M.5 Rank-boundary control

At $\|\delta V_H\|_2 = s_E$, a rank-one singular direction can be canceled. This is the exact boundary of the rank theorem. Crossing it need not yet produce multiple carriers, but it leaves the PCR-7 rank-three packet and reaches a deeper falsifier.

M.6 Nonphysical negative control

An arbitrary variation of N with V_H fixed that violates

$$NN^\dagger + V_H V_H^\dagger = I_3$$

is not a same-scope physical perturbation. It cannot be used as a counterexample to the closure theorem; it is a source/target failure or update.

N Release and finality gates

Gate	Condition
RG-1	PCR-6 and PCR-7 publication endpoints imported with exact child tags and no parent leakage.
RG-2	Robustness target frozen before perturbation classification.
RG-3	Semantic and source-first perturbation completeness established.
RG-4	Total skin/same-scope/breaking/update/protocol status classifier established.
RG-5	Central/traceless heavy decomposition and exact commutant theorem proved.
RG-6	Spectral splitting and sharp spread bounds proved.
RG-7	Light-heavy subspace and charged-current rank margins proved.
RG-8	Block-isometry closure transported exactly.
RG-9	On each declared trace-transport cell, a status-free finite support skeleton, heavy-partition decorations, one full-support parent orbit, exact proper-support deletion flags, the r_{reg} cell-boundary term, explicit guard/update faces, and a positive one-carrier radius are proved; an uncertified chart is conditional/NotReady.
RG-10	Structural mass-support status is separated from detector resolution and physical complex-pole status, with the $2 + 1$ gap taken only between distinct eigenvalues of $(K'_H)^{1/2}$.

RG-11	The invariant core is proved maximal relative to a pre-frozen variation-certified feature registry; the covariant packet is the functional-determination class of the complete structural trace, with laws separate and cross-child output only for matched ready-child realizations under matched common perturbations, as a common-source reduct.
RG-12	Robust protocol coverage and same-record core selection are conditional on a common ordered record space, hypothesiswise uniform coverage, global within-core lower-Lipschitz bounds, all non-coarsened core separators including the reference, structural collision coarsening, joint nuisance/control projection, and separate record-separator, parameter-inverse, transported-mass, and Gram guards with typed deterministic floors. On block isometry with $\eta_V \leq \bar{\eta}_V < s_E$, the ideal pair-space gap is structural rather than a detector-confidence guard; no protocol perturbation radius is inferred.
RG-13	Feature-specific radius vector and zero-radius results proved with metrics/classes fixed.
RG-14	Falsifier hierarchy and held-out firewall frozen.
RG-15	PCR-8 handoff complete and branch qualified.
RG-16	Dyadic existence is proved by selecting 2^{-n} in the punctured pass neighborhood before freezing the interval; no density claim is used.
RG-17	Marked-child row normalization and its $1/256$ bound are separated from independently physical selection and from an observed weak-rate deficit.
RG-18	The D3 pair ([28, 30] GeV, $c\tau_A^{\text{resp}} = 0.2$ mm) is certified as a SourceCertified_ATLASReweightGridAnchored_SearchBenchmark; 0.2 mm is the ATLASReweightGridAnchored_InteriorResponseCoordinate, only 30 GeV is a direct ATLAS mass anchor, and D3FullSourceProperLifetime = D3FullSourceResponseValidation = NotReady.
RG-19	The declared-linear HEPData Figure 4a ordinate screen exceeds a factor 4.50 under the ATLAS simplified model but is not source-level non-exclusion. The D3 band is separated from the symbolic VJ domain $H(I_E)$; its VJ status is PredeclaredOperationalScanSlice_NoVJMassAuthority, with no priority or optimization claim, while numerical VJ endpoints and the D3-to-VJ domain bridge remain NotReady.
RG-20	VJ lifetimes are defined by chart-local full complex poles on a declared sheet; ordered labels are not transported through collisions, and a width-eigenvalue formula is used only under a certified commuting, controlled-energy-dependence reduction.
RG-21	Independent records are child/branch/continuation-chart-or-coherent-stratum/response/input-scheme typed, disjoint from benchmark formation, and inverted through either an identifiable polewise lifetime region or the full coherent-response region with coverage-preserving nuisance projection.
RG-22	The local inverse condition number is used only where $ 5 + \kappa_C $ is bounded away from zero; singular loci route to exact set-valued or higher-order inversion.
RG-23	Strongest extension endpoint is the conditional complex-pole inverse set on $H(I_E)$; numerical VJ lifetime and mass localization remain NotReady.

O External mathematical authority and applicability

The paper uses the following standard results only after source/target typing is complete.

Result	Use	PCR-7.5 applicability condition
Weyl eigenvalue inequalities	cluster motion and spectral support	Hermitian mass-squared perturbation
Davis–Kahan $\sin-\Theta$ theorem	heavy projector stability	positive pre-perturbation light-heavy gap

Singular-value perturbation	charged-current rank stability	operator-norm perturbation of V_H
Distance to singular matrices	exact rank radius s_E	square full-rank heavy block
Polar decomposition	scalar-Gram characterization	square charged-current block
Hoeffding inequality	finite record confidence	independent bounded categorical trials
Gershgorin theorem	optional rank certification from record intervals	Hermitian positive-semidefinite Gram estimate
Lower-Lipschitz inverse stability	global response inversion and confidence transport on each frozen compact core	SPIC-certified lower-Lipschitz bound on the joint signal/nuisance/control space plus typed separation of every non-coarsened core pair, including the reference, in one common record space
Analytic matrix-function and pole theory	basis-invariant unstable-state lifetimes	declared renormalized inverse propagator, fixed sheet and scheme, and chart-local isolated lower-half-plane poles

No standard matrix theorem decides whether a variation is a lawful physical perturbation, whether a split sector becomes a carrier, or whether a detector record belongs to the target. Those are local PCR-7.5 burdens. Compactness, positive acceptance, and pointwise full-rank Jacobians also do not imply global response injectivity; the SPIC common-record-space, lower-Lipschitz, all-non-coarsened-core separation, explicit-reference, collision-coarsening, joint-nuisance, and typed-error certificate is an independent hypothesis.

P Full endpoint matrix by perturbation class

Perturbation class	Carrier one	$\Delta = \Omega$	Rank 3	Universal	Degenerate
Lawful heavy basis rotation	exact	exact	exact	exact	exact
Central mass shift	margin-limited	exact	exact	exact	exact
Standing traceless mass tensor	margin-limited	exact	exact	exact if V fixed	no
Isotropic charged-current change	margin-limited	exact	if $\eta < s_E$	if $\eta < s_E$ (then $\delta' > 0$)	mass-dependent
Anisotropic charged-current change	margin-limited	exact	if $\eta < s_E$	generally no	mass-dependent
Full mass perturbation below half-gap	margin-limited	source-dependent	coupling-dependent	no general claim	no general claim
Source topology change	SourceUpdate	no same-scope claim	no same-scope claim	no same-scope claim	no same-scope claim
Target-role change	TargetUpdate	target-relative	target-relative	target-relative	target-relative
Protocol-only loss	structural theorem intact	structural theorem intact	structural theorem intact	structural theorem intact	structural theorem intact

P.1 Authority and inference endpoint matrix

Object	Strongest current endpoint	Blocked promotion or failure boundary
Terminal carrier	Positive radius from the trace-transport cell, exact proper-support deletion flags, single full-support parent orbit, and explicit face slacks	Escape from the cell, witness-typing loss, or a normal-form/update face is the first carrier-level falsifier; pointwise PCR-6 closure alone proves no radius
2 + 1 structural mass-support partition	Doublet-singlet gap uses distinct eigenvalues of $(K'_H)^{1/2}$	This is not a physical complex-pole gap; the repeated doublet creates no spurious zero structural gap
Robust packet	Fibrewise registry-relative invariant core O_{core} and complete-trace covariant envelope O_{cov} modulo functional determination	A stronger invariant claim requires a pre-certified omitted variation factor; a stronger covariant claim requires an output not factoring through $\pi_{\text{str}}\Theta_{\text{pert}}$

Stratified mass-support/response protocol	Finite hypothesiswise coverage and same-record core selection in one common record space on globally lower-Lipschitz SPIC cores, with all non-coarsened core separators including the reference, collision coarsening, joint nuisance/control coverage, and separate record-separator, carrier/reference parameter-inverse, transported-mass, and Gram-operator guards. On block isometry with $\eta_V \leq \bar{\eta}_V < s_E$, the ideal pair-space gap is structural and supplies no detector guard	Positive acceptance or local Jacobian rank without those certificates is <code>ProtocolUpdate</code> ; physical complex-pole claims require P75-X7
Marked row normalization	$0 < \epsilon_{\text{row}} \leq 1/256$ on the repaired marked-child/target scope	No independently physical value or confidence set is selected
Public weak-record reconstruction	<code>ConditionalLightResponseCompatible</code> At95: $\chi^2_A = 7.0193$, $p = 0.1349$, conditional on applicability of the published light-neutrino response	Common ϵ is ratio-null; full VJ compatibility is <code>NotReady</code> . The 1.62×10^{-3} zero-cross-covariance screen is not an independence certificate and is not promoted, so 1/256 remains the ceiling
D3 $c\tau_A^{\text{resp}} = 0.2$ mm point	<code>ATLASRweightGridAnchored_InteriorResponseCoordinate</code> , an exact discrete grid member with 0.1 mm distance to the lower endpoint of the declared grid span	Operational prompt/near-prompt response-coordinate tag only; no continuous response radius follows, and <code>D3FullSourceProperLifetime</code> and <code>D3FullSourceResponseValidation</code> are <code>NotReady</code>
D3 [28, 30] GeV band	<code>SourceCertified_ATLASRweightGridAnchored_SearchBenchmark</code> , with only 30 GeV a direct ATLAS mass anchor	Declared-linear HEPData Figure 4a screen greater than factor 4.50 under the ATLAS simplified model, not source-level non-exclusion; VJ use is <code>PredeclaredOperationalScanSlice_NoVJMassAuthority</code> , with no priority or optimization claim
VJ mass domain VJ lifetime dynamics	Symbolic source-qualified domain $H(I_E)$ Conditional chart-qualified complex-pole envelope from $\det \mathcal{D}^{-1} = 0$	Numerical endpoints <code>NotReady</code> Width eigenvalues suffice only under the commuting, controlled-energy-dependence reduction; collisions use unordered poles and coherent response
VJ mass localization	Conditional coverage-preserving union of polewise-identifiable and coherent-response projections on $H(I_E)$	Numerical result needs independent normalization, complete pole kernel, exhaustive compatible types, independent response record, and joint nuisance envelope
Local scalar inversion	First-order condition number where $ 5 + \kappa_C \geq \kappa_{\min} > 0$	At or across $5 + \kappa_C = 0$, use exact set-valued or certified higher-order inversion
W-rest-frame response-coordinate image	Kinematic image of $\beta\gamma c\tau_A^{\text{resp}}$, not a physical mean decay length	A pp laboratory mean requires a full-source proper lifetime plus boost, angular, acceptance, and reconstruction convolution

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